MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **DESIGN CALCULATION**
6000 VILLAGE:CENTRAL COMMUNITY CENTER
DESIGN OF BUILDING STRUCTURE

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
Typ.	:	Typical
HCS	:	Hollow Core Slab
Fc'	:	Concrete Compressive Strength
Fy	:	Yield Strength of Reinforcing Steel
ϕ	:	Strength Reduction Factor
SBC	:	Saudi Building Code
SASO	:	Saudi Standards, Metrology and Quality Organization
ACI	:	American Concrete Institute
ASCE	:	American Society of Civil Engineers
PCI	:	Precast/Prestressed Concrete Institute

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1. INTENT

This Document is intended to produce the technical needs in Engineering including particular reference to design justification & Engineering validation works. The work described below is related to Design of Central Community Center Building in Village Area.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The Scope of this document is limited to Complete Analysis & Structural Design Verification of Central Community Center Building only.

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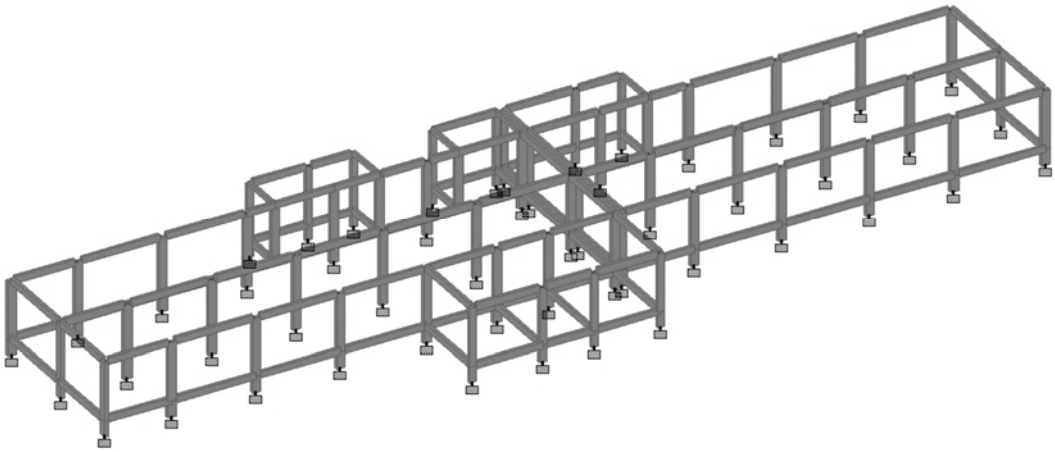
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4. DESIGN METHODOLOGY & ASSUMPTIONS

This Building Is a Single Story Frame Structure with Foundation Being Cast in Site. The Building Shall Have Insulated Block Walls and Cast in Site Framing Beams to support the HCS (Hollow Core Slab)

The Superstructure for this Accommodation Building was modeled using STAAD software. The Analysis of Frames (Cast-In-Situ Columns & Beams) has been performed meeting design requirements & also keeping in view the Architectural requirements. The loads were applied as per SBC/ASCE/ACI standards & analysis was done to generate the Column Reactions at the Base which are assumed to be fixed. The Foundation design has been done by importing Column Base Reactions into Foundation Design Software.



STAAD 3D MODEL

4.1. Codes

[1] SBC-301-2007-Structural Loading & Forces.

[2] SBC-303-2007-Structural Soil & Foundations.

[3] SBC-304-2007-Structural Concrete Structures.

[4] SASO SSA 2 – Steel Bars for the Reinforcement of Concrete.

[5] SASO SSA 224 – Steel Fabric for Reinforcement of Concrete

[6] ASCE-7-2005 Minimum Design Loads for Buildings and other Structures.

[7] ACI 318-08 Building Code Requirements.

[8] PCI –MNL-120-04 –Design Handbook, 6th edition.

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4.2. Geotechnical investigations

Foundations have been designed based on recommendations of the Soil Investigation Report, Doc # MD-216-3000-EG-CV-RPT-1008.

4.3. Material Properties

Compressive Strength of Concrete at 28th days

fc' = 35 MPa (5.00 Ksi) for beams, columns, suspended slabs.

fc' = 35 MPa (5.00 Ksi) for foundations, grade beams, columns.

Yield Strength of Steel

fy = 414 MPa (60 Ksi) for all reinforcing steel

5. DESIGN LOADS

5.1. Dead Load

5.1.1. Roof Dead Loads

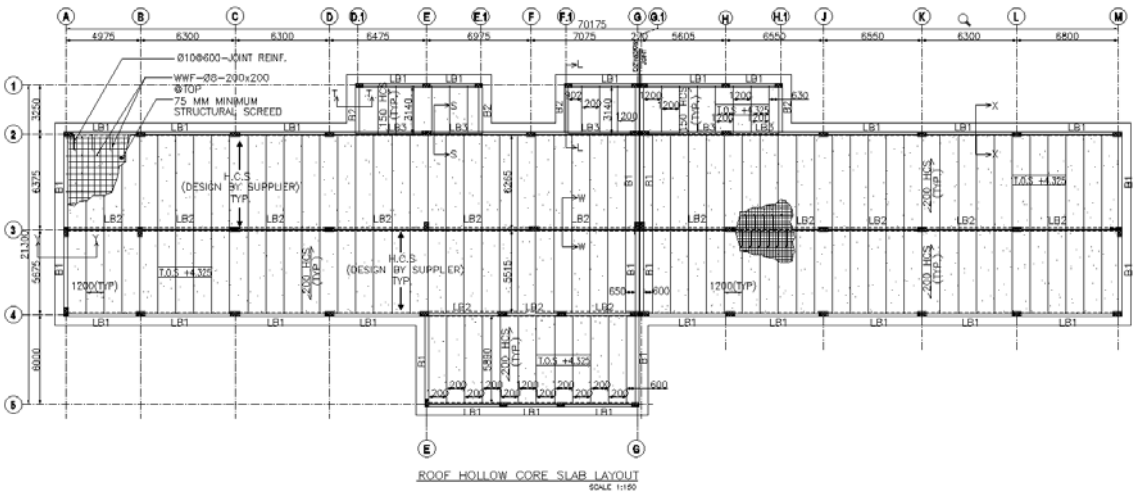
- Provision for Electrical Lighting = 0.05 kN/m²

Ceiling

Suspended steel channel system = 0.10 kN/m²

Acoustical Fiber Tile (16mm) = 0.05 kN/m²

Sub-total = 0.2 kN/m²



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- Hollow Core Slab (PCI 6th Edition)

(HCS-200 Topped) b/w Grid 2 & 3, Grid 3 & 4, Grid 4 & 5 = **3.10 KN/m2**

(HCS-150 Topped) b/w Grid 1 & 2 = **2.52 KN/m2**

Prestressed Hollowcore Floor										
Overall Structural Depth (mm)	**Self Wt- KN/M2**	**No. of Cores in Unit**	**Spans indicated below allow for Characteristic Service Load (Live Load) Plus Self Weight Plus 1.5Kn/m² for Finishes Characteristic Service Loads Kn/m²**							
			0.75	1.50	2.00	2.50	3.00	4.00	5.00	10.00
			Clear Span in metres							
100	2.38	0	5.39	5.19	5.08	4.91	4.72	4.40	4.14	3.30
150	2.52	11	7.40	7.40	7.40	7.40	7.28	6.80	6.40	5.12
150 (Dense	2.94	11	7.40	7.40	7.40	7.35	7.08	6.64	6.27	5.06
200	3.10	11	9.40	9.40	9.40	9.40	9.26	8.69	8.21	6.65
250	3.43	6	10.80	10.80	10.80	10.80	10.80	10.80	10.56	8.60
300	3.88	6	13.50	13.50	13.50	13.50	13.18	12.44	11.81	9.70
350	4.23	6	15.00	15.00	15.00	15.00	14.89	14.08	13.39	11.06
400	4.57	6	16.00	16.00	16.00	16.00	15.97	15.13	14.42	11.98
450	4.92	6	18.00	18.00	18.00	17.70	17.19	16.33	15.58	13.02
500	5.27	6	18.00	18.00	18.00	18.00	18.00	17.13	16.38	13.76

- Built-in Water Proofing System

100mm thk Polystyrene Foam

0.01 kN/m² per 25.4mm thickness x (100/25.4mm) = 0.039 kN/m²

Water proofing membrane

Bituminous, smooth surface = 0.07 kN/m²

90 mm thick average cement foam concrete screed Lightweight

Concrete 0.38 kN/m² per 25mm thickness x (90/25mm) = 1.368 kN/m²

75 mm Concrete screed above HCS = 21 x 0.075 =1.6 kN/m²

Sub-total = **3.1 kN/m²**

Total Load b/w Grid 2 & 3, Grid 3 & 4, Grid 4 & 5 = **6.4 kN/m²**

Total Load b/w Grid 1 & 2 = **5.82 kN/m²**

UDL on External Beams along Grid Line-4 = 5.675 x 6.4/2 =**18.16 Kn/m**

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UDL on External Beams along Grid Line-2 = 6.375 x 6.4/2 =**20.4 Kn/m**

UDL on External Beams along Grid Line-5 = 6.000 x 6.4/2 =**19.2 Kn/m**

UDL on External Beams along Grid Line-1 = 3.250 x 5.82/2 =**9.45 Kn/m**

UDL on Internal Beams along Grid Line-3 = 20.4+18.16 =**38.56 Kn/m**

UDL on Internal Beams along Grid Line-4, B/w E&G = 18.16+19.2=**37.36 Kn/m**

UDL on Internal Beams along Grid Line-2, B/w D&F, G&J = **29.85 Kn/m**

Moment on all the External Beams due to Cantilever Portion of slab and Screed

= (0.15x24+ (0.05+0.2)/2x20)*0.6*0.6/2 = **1 knm/m**

5.1.2. Ground Floor Walls

- 250 thk. Outer CMU

16 x 0.25 x 3.5m clear height = **14.0 kN/m**

5.2. Live Load

5.2.1. Roof Live Loads

Live loads were taken from Project Design Criteria, Doc-MD-216-1000-EG-ST-CRT-1001, Clause 4.2.2..

Ordinary Flat = **0.50 kN/m²**

UDL on External Beams along Grid Line-4 = 5.675 x 0.50/2 =**1.418 Kn/m**

UDL on External Beams along Grid Line-2 = 6.375 x 0.50/2 =**1.59 Kn/m**

UDL on External Beams along Grid Line-5 = 6.000 x 0.50/2 =**1.50 Kn/m**

UDL on External Beams along Grid Line-1 = 3.250 x 0.50/2 =**0.812 Kn/m**

UDL on Internal Beams along Grid Line-3 = 1.4+1.6 =**3.0 Kn/m**

UDL on Internal Beams along Grid Line-4, B/w E&G = 1.42+1.50=**2.92 Kn/m**

UDL on Internal Beams along Grid Line-2, B/w D&F, G&J = **2.40 Kn/m**

Moment on all the External Beams due to Live Load on

Cantilever Portion of slab = 1*0.6*0.6/2 = **0.18 kN/m /m**

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5.3. Wind Load

Wind loads were automatically generated in STAAD as per ASCE 7-05. The parameters used for the wind load generation are as follows (Refer STRUCTURAL DESIGN CRITERIA NON PROCESS BUILDINGS Doc No MD-216-0000-EG-ST-CRT-0001)

All structures and foundations (except for buried structures) designed and shall be constructed to resist the wind effects determined in accordance with Section-7.2, SBC-301-Analytical Procedure.

5.4. Wind Load (Analytical Verification)

WIND LOADING (SBC-301)

Using the directional design method

Tedds calculation version 2.0.20



Building data

Type of roof;	Flat
Length of building;	b = 70000 mm
Width of building;	d = 13000mm
Height to eaves;	H = 4325 mm
Mean height;	h = 4325 mm

General wind load requirements

Basic wind speed;	V = **49** m/s (or) 177 kmph (Refer Clause-4.2.6, Doc- MD-216-1000-EG-ST- CRT-1001)
Risk category;	II(As per SBC-301, Table 1.6-1)
Velocity pressure exponent coeff;	K_d = 0.85(As per SBC-301, Table 6.4-1)
Exposure category ;	D(As per SBC-301, Table 6.4.2.3)
Enclosure classification ;	Partially enclosed buildings(As per SBC-301, Section-6.2)

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Internal pressure coef +ve; GC_{pi_p} = 0.55(As per SBC-301, Figure-7.2-1)

Internal pressure coef –ve; GC_{pi_n} = -0.55(As per SBC-301, Figure-7.2-1)

Gust effect factor; G_f = 0.85(As per SBC-301, Section-7.2.7.1)

Importance Factor; 1(As per SBC-301, Section-9.5)

Topography

Topography factor not significant; K_{zt} = 1.0(As per SBC-301, Section-6.4.3.2)

Velocity pressure equation; $q = 0.0473x10^{-3}Kn/m^2 \times K_z \times K_{zt} \times K_d \times V^2$ xI

V in Km/hr(As per SBC-301, Section-7.2.8)

Velocity pressures table

z (mm)	K_z (SBC-301,Table 7.2-2)	q_z (kN/m^2)
4325	1.03	1.29
4325	1.03	1.29

Pressures and forces

Net pressure; $p = q \times G_f \times C_{pe} - q_i \times GC_{pi}$;

Net force; $F_w = p \times A_{ref}$;

Walls load case 1 - Wind 0, GC_{pi} 0.55, - c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	4.33	0.80	1.29	0.17	302.76	50.72
B	4.33	-0.50	1.29	-1.26	302.75	-380.36
C	4.33	-0.70	1.29	-1.48	56.23	-82.96
D	4.33	-0.70	1.29	-1.48	56.23	-82.96

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Walls load case 2 - Wind 0, GC_{pi} -0.55, + c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	4.33	0.80	1.29	1.58	302.76	479.85
B	4.33	-0.50	1.29	0.16	302.75	48.76
C	4.33	-0.70	1.29	-0.06	56.23	-3.26
D	4.33	-0.70	1.29	-0.06	56.23	-3.26

Walls load case 3 - Wind 90, GC_{pi} 0.55, - c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	4.33	0.80	1.29	0.17	56.23	9.42
B	4.33	-0.20	1.29	-0.93	56.23	-52.16
C	4.33	-0.70	1.29	-1.48	302.75	-446.68
D	4.33	-0.70	1.29	-1.48	302.75	-446.68

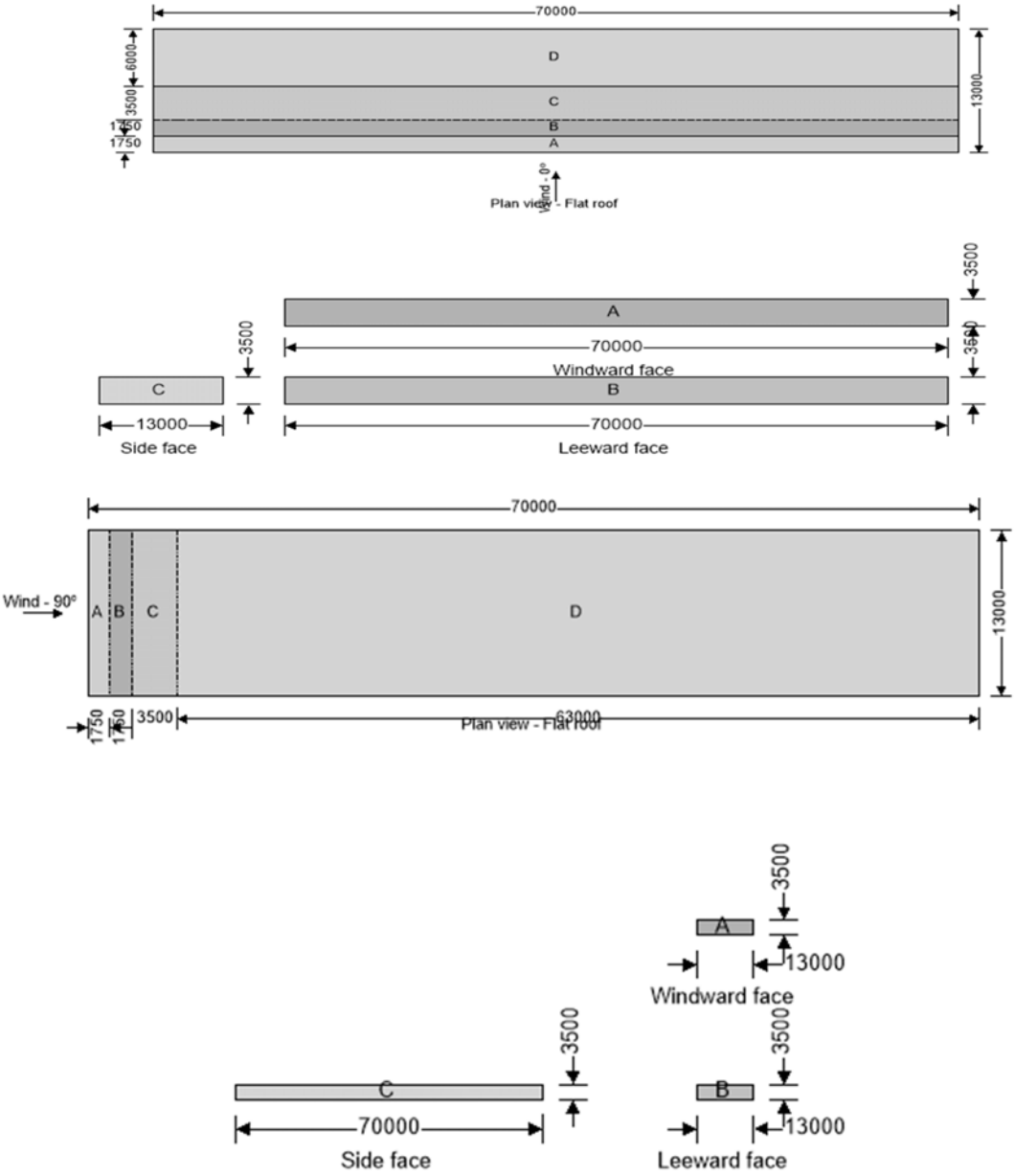
Walls load case 4 - Wind 90, GC_{pi} -0.55, + c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	4.33	0.80	1.29	1.58	56.23	89.12
B	4.33	-0.20	1.29	0.49	56.23	27.53
C	4.33	-0.70	1.29	-0.06	302.75	-17.56
D	4.33	-0.70	1.29	-0.06	302.75	-17.56

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So from the above analysis, the Wind Loads generated in STAAD are in accordance with SBC-301-2007-Structural Loading & Forces.

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5.5. Seismic Load

Earthquake loads were automatically generated in STAAD as per IBC 2003. The basic parameters affecting the generation of seismic loads is shown below:

Importance Factor, I = 1.0(As per SBC-301, Section-9.5)

Design Spectral Response Acceleration (Refer Clause-4.2.7b, MD-216-1000-EG-ST- CRT-1001)

Ss = 0.1g (Spectral Response Acceleration 0.2 second)

S1 = 0.02g (Spectral Response Acceleration 1 second)

S_{D_1} = 2/3*Fv*S1 = 2/3 * 2.4 * 0.02 = 0.032 (As per SBC-301, Section-9.4.4)

S_{DS} = 2/3*Fa*S1 = 2/3 * 1.60 * 0.10 = 0.10 (As per SBC-301, Section-9.4.4)

Where S_{D_1} Damped Design Spectral Response Acceleration@1 second

S_{DS} Damped Design Spectral Response Acceleration@short period

Site Class = D

Site Coefficient, Fa = 1.60 (As per SBC-301, Table 9.4.3a)

Site Coefficient, Fv = 2.40 (As per SBC-301, Table 9.4.3b)

Building period Coefficient, Ct=0.04 (As per SBC-301, Table 10.9.3.2)

For Ordinary reinforced concrete moment frame,

Response Modification Factor, Rx = 2.5 (As per SBC-301, Table 10.9.3.2)

Response Modification Factor, Rz = 2.5 (As per SBC-301, Table 10.2)

System Over Strength Factor, Omega = 3 (As per SBC-301, Table 10.2)

Deflection Amplification factor, Cd = 2.5 (As per SBC-301, Table 10.2)

5.6. Thermal/Temperature Loads (Self-restraining and Thermal Expansion and Contraction Loads)

An ambient temperature ranges from a maximum of 50° C to a low of 0° C. In direct exposure to sunlight, the temperature shall be assumed to rise a minimum of 15° C above ambient. As a minimum, the temperature gradient for thermal loads shall be +25° C and -15° C with the most severe condition governing the design.

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6. DESIGN COMBINATIONS

Service Loads (Allowable Stress Design) (SBC-301, Section 2.4.1)

D

D + T

D +Lr

D +0.75Lr

D + (W or 0.7E)

D + 0.75W + 0.75Lr

0.6D + W

0.6D + 0.7E

Factored loads (Strength Design) (SBC-301, Section 2.3.2)

1.4D

1.4D+1.4T

1.2(D + T) + 0.5Lr

1.2D + 1.6Lr + 0.8W

1.2D + 1.6W + 0.5Lr

1.2D + 1.0E

0.9D + 1.6W

0.9D + 1.0E

Where:

D = Dead Load

Lr = Live Load

W = Wind Load

E = Seismic Load

T = Temperature Load

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7. CRACKED SECTIONS

In lieu of a more accurate analysis of the effects of cracking of concrete, provisions of SBC-304-Section 10.11 was used.

For Beams:	I = 0.35Ig
For Columns:	I = 0.70Ig
For Walls:	
Un-cracked	I = 0.70Ig
Cracked	I = 0.35Ig
For Slabs	I = 0.25Ig

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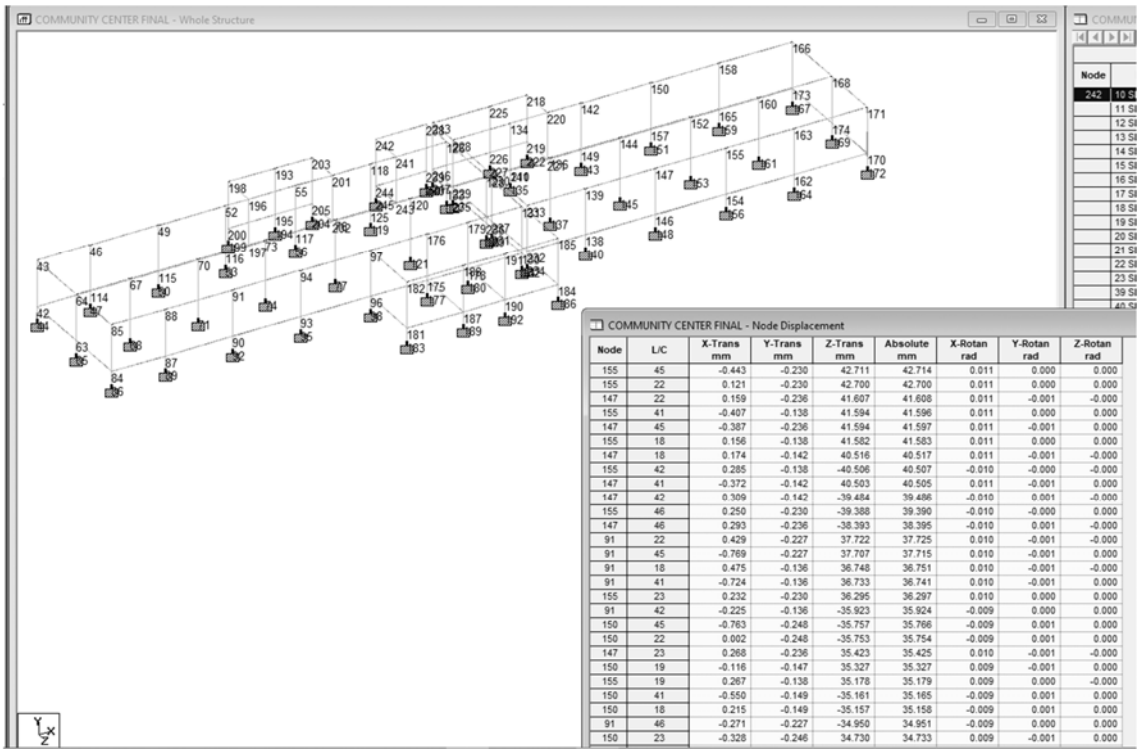
8. DEFLECTIONS/DRIFT CHECK

8.1. Maximum Allowable Deflection

L/240: for member load with live load plus dead load

L/600: for members supporting masonry under total loads

Where, L = span of member between support



Node	L/C	X-Trans mm	Y-Trans mm	Z-Trans mm	Absolute mm	X-Rotan rad	Y-Rotan rad	Z-Rotan rad
155	45	-0.443	-0.230	42.711	42.714	0.011	0.000	0.000
155	22	0.121	-0.230	42.700	42.700	0.011	0.000	0.000
147	22	0.159	-0.236	41.607	41.608	0.011	-0.001	-0.000
155	41	-0.407	-0.138	41.594	41.596	0.011	0.000	0.000
147	45	-0.387	-0.236	41.594	41.597	0.011	-0.001	0.000
155	18	0.156	-0.138	41.582	41.583	0.011	0.000	0.000
147	18	0.174	-0.142	40.516	40.517	0.011	-0.001	-0.000
155	42	0.285	-0.138	-40.506	40.507	-0.010	-0.000	-0.000
147	41	-0.372	-0.142	40.503	40.505	0.011	-0.001	0.000
147	42	0.309	-0.142	-39.484	39.486	-0.010	0.001	-0.000
155	46	0.250	-0.230	-39.388	39.390	-0.010	-0.000	0.000
147	46	0.293	-0.236	-38.393	38.395	-0.010	0.001	-0.000
91	22	0.429	-0.227	37.722	37.725	0.010	-0.001	-0.000
91	45	-0.769	-0.227	37.707	37.715	0.010	-0.001	0.000
91	18	0.475	-0.136	36.748	36.751	0.010	-0.001	-0.000
91	41	-0.724	-0.136	36.733	36.741	0.010	-0.001	0.000
155	23	0.232	-0.230	36.295	36.297	0.010	0.000	0.000
91	42	-0.225	-0.136	-35.923	35.924	-0.009	0.000	0.000
150	45	-0.763	-0.248	-35.757	35.766	-0.009	0.001	0.000
150	22	0.002	-0.248	-35.753	35.754	-0.009	0.001	0.000
147	23	0.268	-0.236	35.423	35.425	0.010	-0.001	-0.000
150	19	-0.116	-0.147	35.327	35.327	0.009	-0.001	0.000
155	19	0.267	-0.138	35.178	35.179	0.009	0.000	-0.000
150	41	-0.550	-0.149	-35.161	35.165	-0.009	0.001	0.000
150	18	0.215	-0.149	-35.157	35.158	-0.009	0.001	0.000
91	46	-0.271	-0.227	-34.950	34.951	-0.009	0.000	0.000
150	23	-0.328	-0.246	34.730	34.733	0.009	-0.001	0.000

Maximum attained Drift as per Analysis from above figure = 42.70 mm

Allowable Drift as per Table-10.12, SBC-301-2007 is H/100 = 4325/100 = 43.25mm

Hence the Structure is safe considering drift check. Also the HCS panels once placed with Joint Reinforcement & screed over HCS will stabilize the Structure furtherer more with diaphragm action …

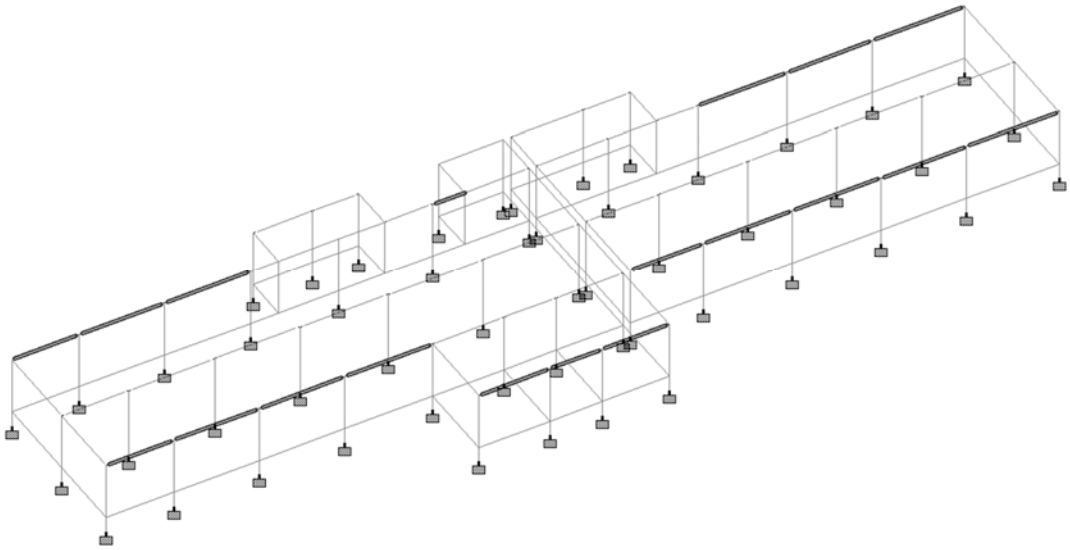
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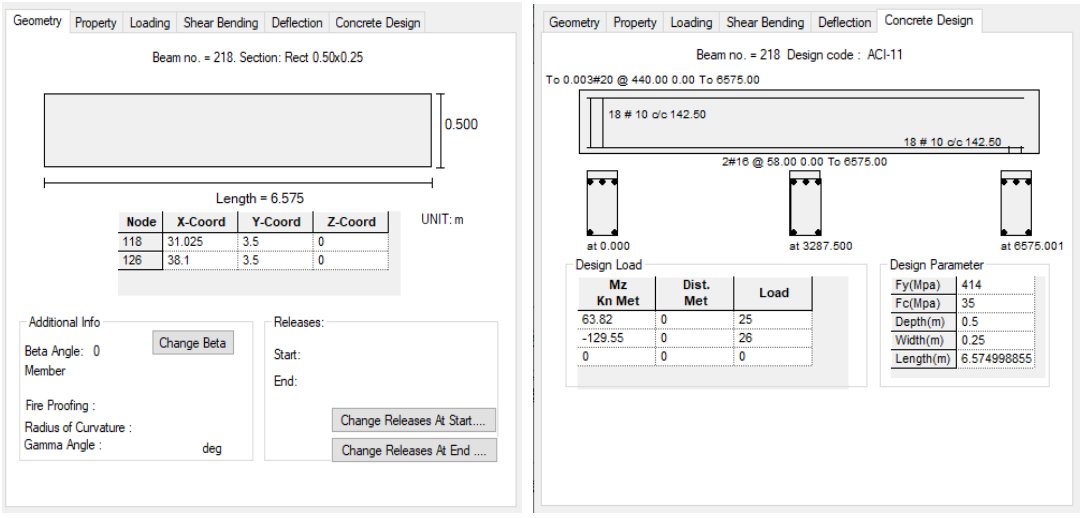
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9. CONCRETE ROOF BEAM DESIGN

9.1. Design of Ledger Beam(LB1)





Design of Ledger Beam (LB1)

Maximum Moment required from Analysis = **129 Knm (Hogging)**

Therefore Ast_{req} =

$$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

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Where b=250mm, Fy=414 Mpa, d=500-40-10-20/2=440mm, Fc'=35 Mpa & ϕ =0.9

Ast1.req = **830 mm²**

Ast.min= (As per clause-10.5-ACI-318)

$$A_{s,\,min} = \frac{0.25\sqrt{f'_c}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **471mm²**

Maximum Moment required from Analysis = **63 Knm (Sagging)**

Ast2.req = **410 mm²**

However Provide 3-20 dia -Full Straight Bars Top & 3-20 dia-Full Straight Bottom.

Ast1.pro**>>>>>>(Ast.min, Ast1.req)**

Ast2.pro**>>>>>>(Ast.min, Ast2.req)**

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9.1.1. Staad Output For LB1

```
            ACI 318-11   BEAM  NO.   218   DESIGN RESULTS
            ==========================================

 LEN -  6575. MM  FY -  414.  FC -  35.  MPA, SIZE -  250. X  500. MMS

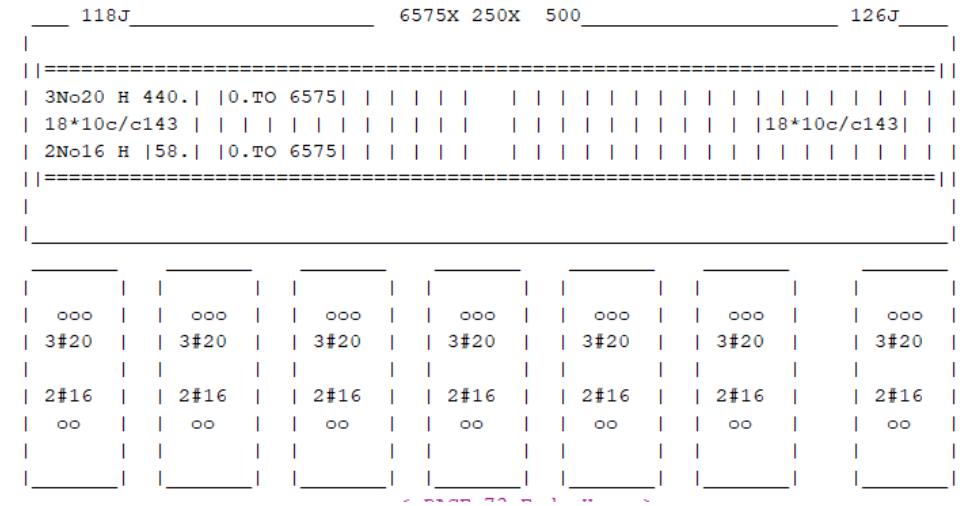
 LEVEL     HEIGHT      BAR INFO        FROM          TO          ANCHOR
             (MM)                        (MM)         (MM)       STA  END
 ____________________________________________________________________

   1         58.     2 - 16MM          0.          6575.        YES  YES
   2        440.     3 - 20MM          0.          6575.        YES  YES

     B E A M  N O.   218 D E S I G N  R E S U L T S - SHEAR


AT START SUPPORT - Vu=   71.44 KNS  Vc=  111.53 KNS  Vs=    0.00 KNS
Tu=    7.52 KN-MET  Tc=    4.2 KN-MET  Ts=   10.0 KN-MET  LOAD    28
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 2348. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   2.48 SQ.CM.

AT END   SUPPORT - Vu=   63.70 KNS  Vc=  128.72 KNS  Vs=    0.00 KNS
Tu=    6.70 KN-MET  Tc=    4.2 KN-MET  Ts=    8.9 KN-MET  LOAD    49
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 2348. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   2.21 SQ.CM.
```

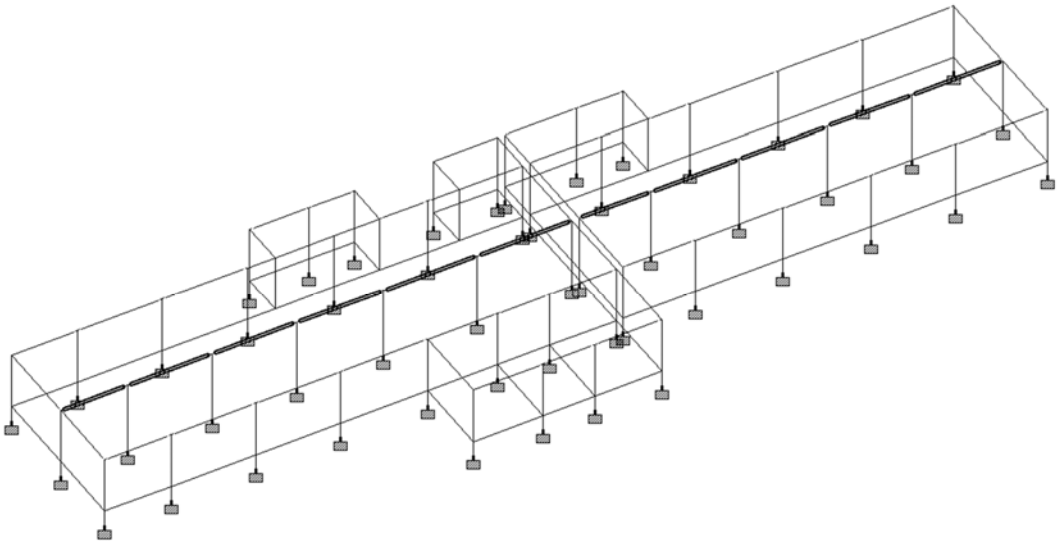


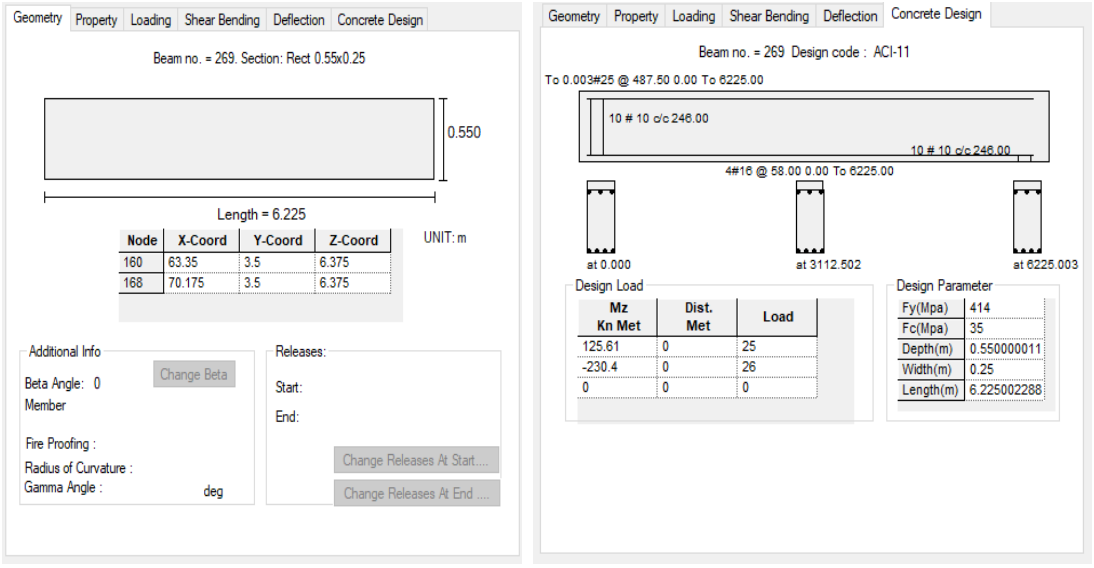
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9.2. Design of Ledger Beam(LB2)





Design of Ledger Beam (LB2)

Provide 3-25 dia-Full Straight Bars Top & 3-25 dia-Full Straight Bottom.

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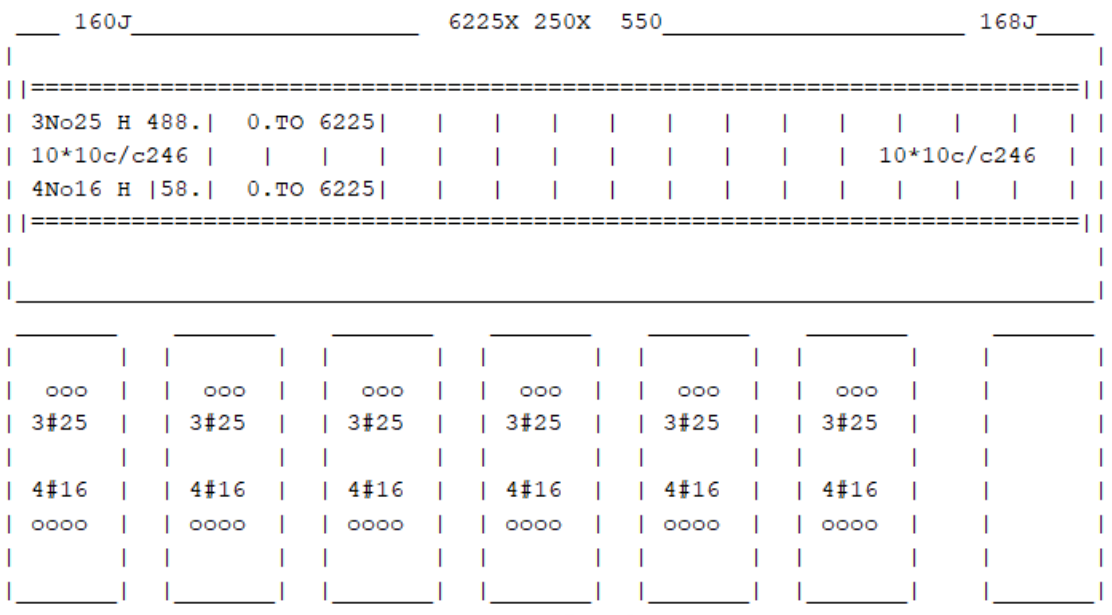
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9.2.1. Staad Output For LB2

```
                B E A M  N O.   269 D E S I G N  R E S U L T S - SHEAR


    AT START SUPPORT - Vu=  144.59 KNS  Vc=  137.89 KNS  Vs=   54.90 KNS
    Tu=    0.01 KN-MET  Tc=    4.7 KN-MET  Ts=    0.0 KN-MET  LOAD    25
    NO STIRRUPS ARE REQUIRED FOR TORSION.
    REINFORCEMENT IS REQUIRED FOR SHEAR.
    PROVIDE 10 MM 2-LEGGED STIRRUPS AT 246. MM  C/C FOR 2073. MM

    AT END   SUPPORT - Vu=   72.57 KNS  Vc=  138.68 KNS  Vs=    0.00 KNS
    Tu=    0.00 KN-MET  Tc=    4.6 KN-MET  Ts=    0.0 KN-MET  LOAD     3
    NO STIRRUPS ARE REQUIRED FOR TORSION.
    REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
    PROVIDE 10 MM 2-LEGGED STIRRUPS AT 246. MM  C/C FOR 2073. MM
```

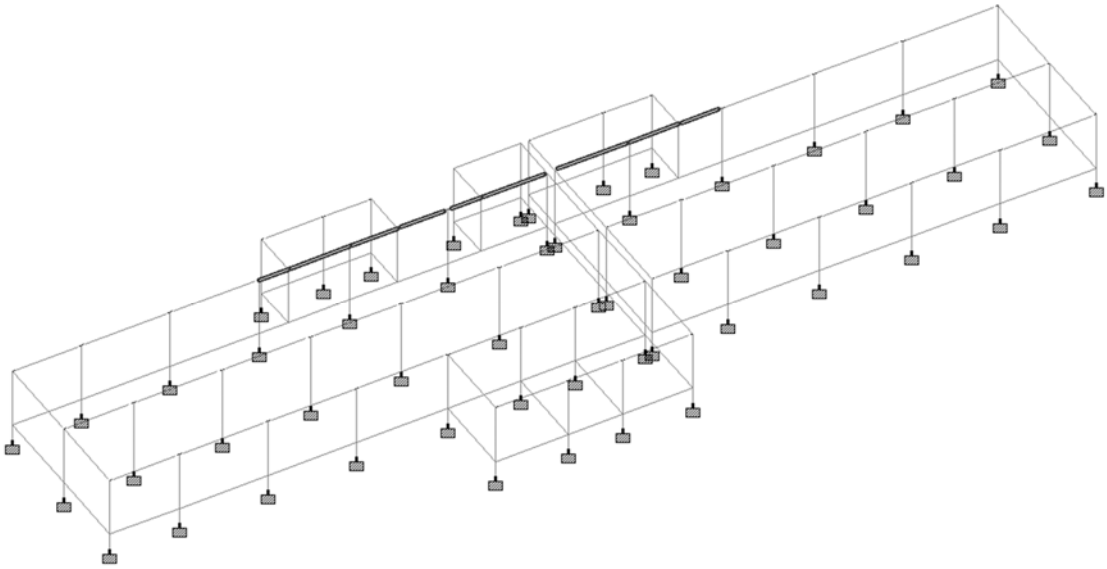


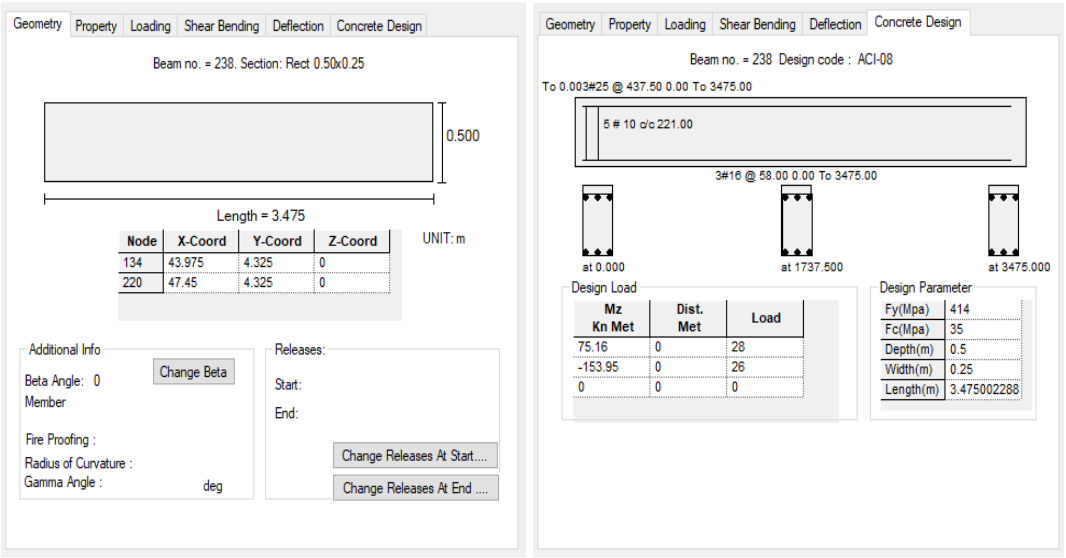
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9.3. Design of Ledger Beam(LB3)





Design of Ledger Beam (LB3)

Provide 3-25 dia-Full Straight Bars Top & 3-25 dia-Full Straight Bottom.

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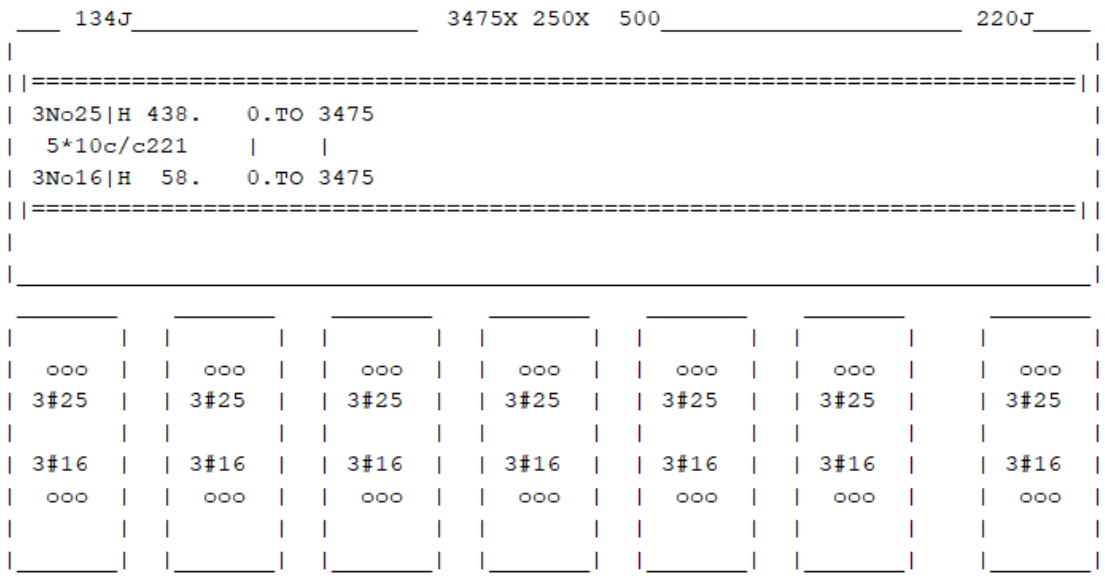
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9.3.1. Staad Output For LB3

```
            B E A M   N O.    238 D E S I G N   R E S U L T S - SHEAR


   AT START SUPPORT - Vu=    67.19 KNS   Vc=  116.05 KNS   Vs=     0.00 KNS
   Tu=     0.52 KN-MET   Tc=     4.0 KN-MET   Ts=     0.0 KN-MET   LOAD      3
   NO STIRRUPS ARE REQUIRED FOR TORSION.
   REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
   PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM  C/C FOR  798. MM

   AT END    SUPPORT - Vu=    14.88 KNS   Vc=  104.59 KNS   Vs=     0.00 KNS
   Tu=     0.52 KN-MET   Tc=     4.0 KN-MET   Ts=     0.0 KN-MET   LOAD      3
                      STIRRUPS ARE NOT REQUIRED.
```

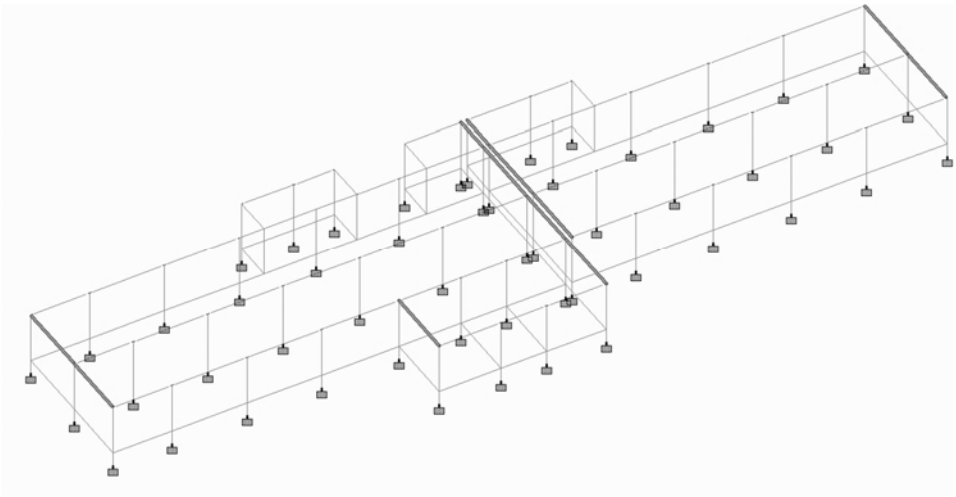


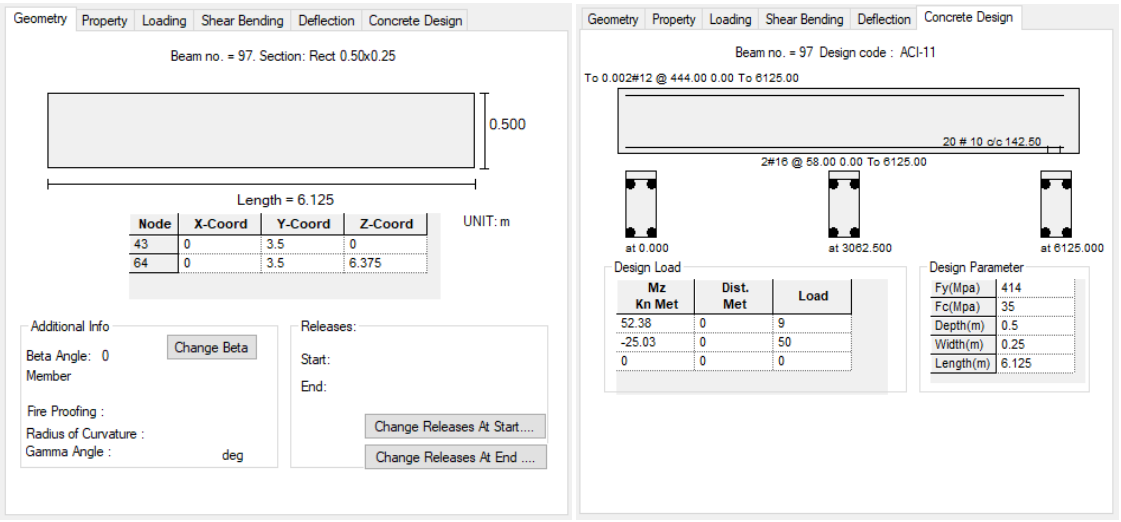
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9.4. Design of Beam-(B1/B3)





Design of Beam (B1/B3)

However Provide minimum 3-16 dia-Full Straight Bars Top & Bottom.

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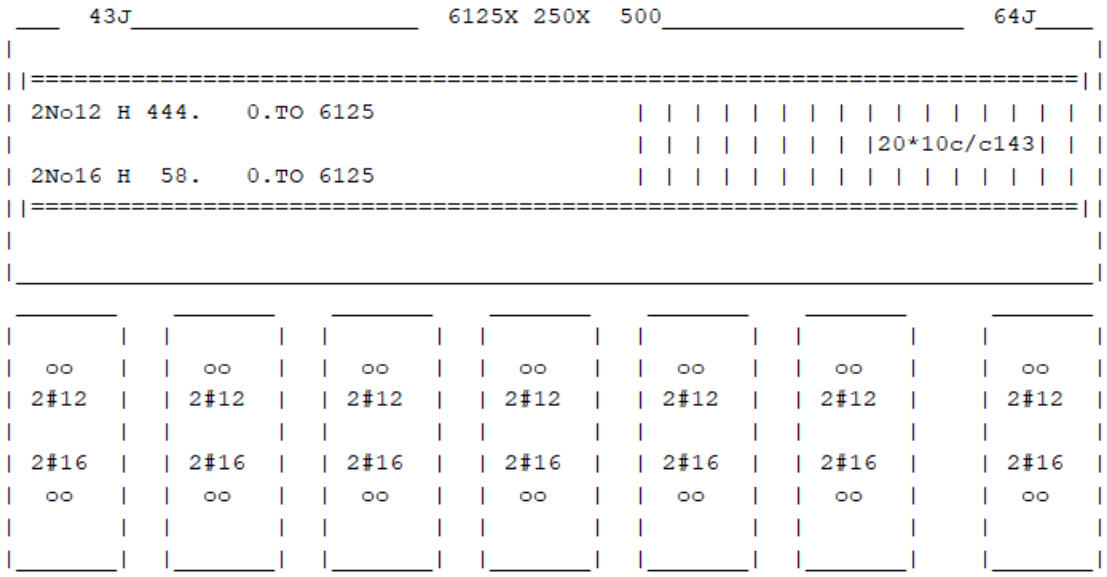
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9.4.1. Staad Output For B1/B3

```
        B E A M  N O.    97 D E S I G N  R E S U L T S - SHEAR


AT START SUPPORT - Vu=    7.84 KNS  Vc=  108.13 KNS  Vs=    0.00 KNS
Tu=     1.36 KN-MET  Tc=     3.9 KN-MET  Ts=    0.0 KN-MET  LOAD     3
                  STIRRUPS ARE NOT REQUIRED.
AT END   SUPPORT - Vu=   10.65 KNS  Vc=  108.71 KNS  Vs=    0.00 KNS
Tu=     5.71 KN-MET  Tc=     3.9 KN-MET  Ts=    7.6 KN-MET  LOAD    25
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 2622. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   1.88 SQ.CM.
```

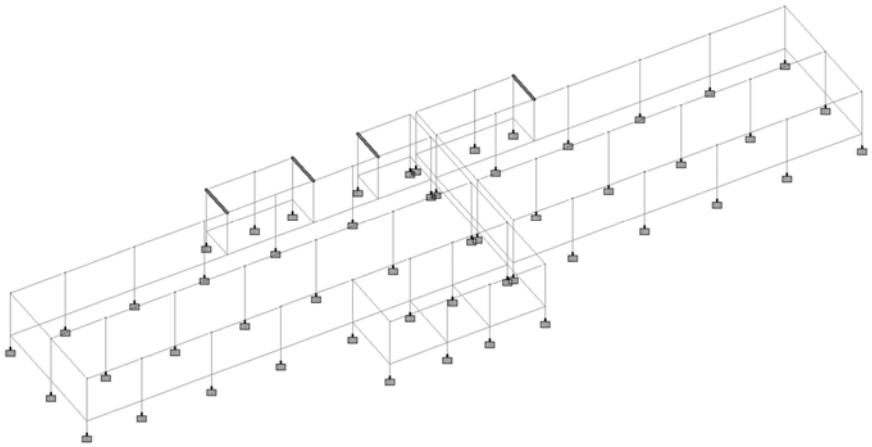


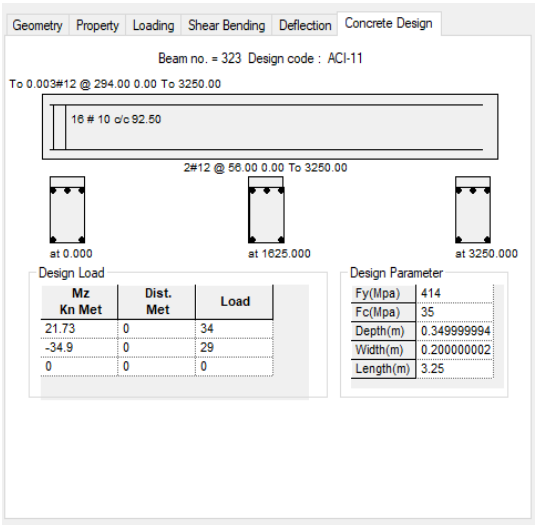
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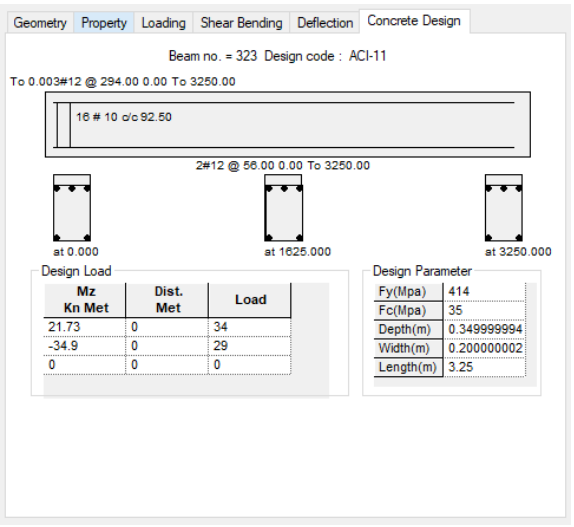
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9.5. Design of Beam-(B2)







However Provide minimum 3-12 dia-Full Straight Bars Top & Bottom.

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9.5.1. Staad Output For B2

```
            ACI 318-11   BEAM  NO.   323   DESIGN RESULTS
            ==============================================

 LEN -  3250. MM  FY -  414.  FC -  35.  MPA, SIZE -  200. X  350. MMS

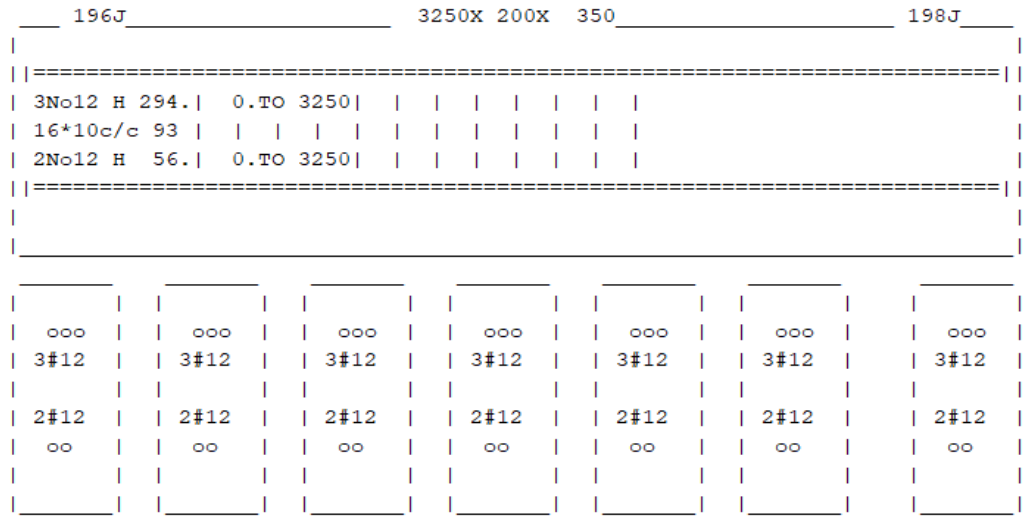
 LEVEL     HEIGHT      BAR INFO        FROM          TO          ANCHOR
             (MM)                        (MM)         (MM)       STA  END
 ____________________________________________________________________________


   1         56.     2 - 12MM          0.          3250.        YES  YES
   2        294.     3 - 12MM          0.          3250.        YES  YES

      B E A M  N O.   323 D E S I G N  R E S U L T S - SHEAR

 AT START SUPPORT - Vu=    1.24 KNS  Vc=   55.25 KNS  Vs=    0.00 KNS
 Tu=    3.78 KN-MET  Tc=    1.7 KN-MET  Ts=    5.0 KN-MET  LOAD    25
 STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT  93. MM  C/C FOR 1335. MM
 ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   1.85 SQ.CM.

 AT END   SUPPORT - Vu=    5.26 KNS  Vc=   56.03 KNS  Vs=    0.00 KNS
 Tu=    0.00 KN-MET  Tc=    1.7 KN-MET  Ts=    0.0 KN-MET  LOAD     3
                  STIRRUPS ARE NOT REQUIRED.
```



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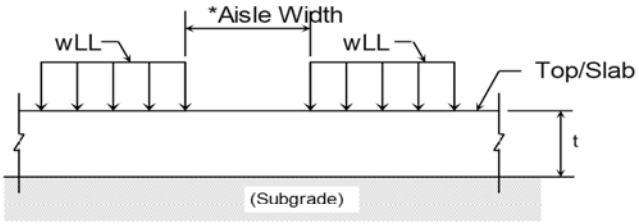
10. CONCRETE GRADE SLAB DESIGN

CONCRETE SLAB ON GRADE ANALYSIS

For Slab Subjected to Stationary Uniformly Distributed Live Loads

Input Data:

Slab Thickness, t =	3.937	in.	100.000	mm		
Concrete Strength, f 'c =	5000	psi	35.000	kpa		
Subgrade Modulus, k =	39.7866	pci	10800	Kn/m3	(AS PER SECTION-7.7-SOIL REPORT)	
Factor of Safety, FS =	1.500		1.500			
Uniform Live Load, wLL =	104.45	psf	5.00	kpa		



Concrete Slab on Grade with Uniform Loads

Results:

Design Parameters:

Modulus of Rupture, MR =	636.40	psi	4387.79	kpa	MR = 9*SQRT(f 'c)
Allow. Bending Stress, Fb =	424.26	psi	2925.19	kpa	Fb = MR/FS
Modulus of Elasticity, Ec =	4030509	psi	27789329	kpa	Ec = 57000*SQRT(f 'c)
Poisson's Ratio, μ =	0.15		0.15		μ = 0.15 (assumed for concrete)
Radius of Stiffness, Lr =	26.94	in.	684.37	mm	Lr = (Ec*t^3/(12*(1-μ^2)*k))^0.25
Critical Aisle Width, Wcr =	4.96	ft.	1511.77	mm	Wcr = (2.209*Lr)/12

Stationary Uniformly Distributed Live Loads:

wLL(allow) =	682.05	psf	wLL(allow) = 257.876*Fb*SQRT(k*t/Ec)
	32.65	kpa	**wLL(allow) >= wLL, O.K.**

Provide Minimum Reinforcement of 0.14%@both faces = 0.14%(h) = 140 mm2/M

Table 24.4.3.2—Minimum ratios of deformed shrinkage and temperature reinforcement area to gross concrete area

Reinforcement type	f_y, psi	Minimum reinforcement ratio	
Deformed bars	< 60,000	0.0020	
Deformed bars or welded wire reinforcement	≥ 60,000	Greater of:	$\frac{0.0018 \times 60,000}{f_y}$
			0.0014

Provide WWF-6mm Dia@200 c-c Both Ways-TOP

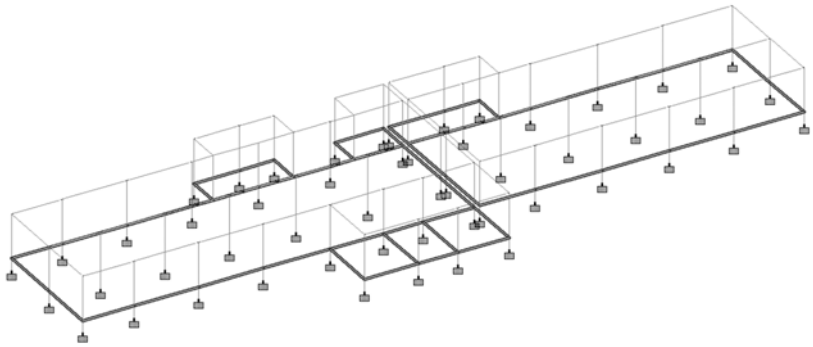
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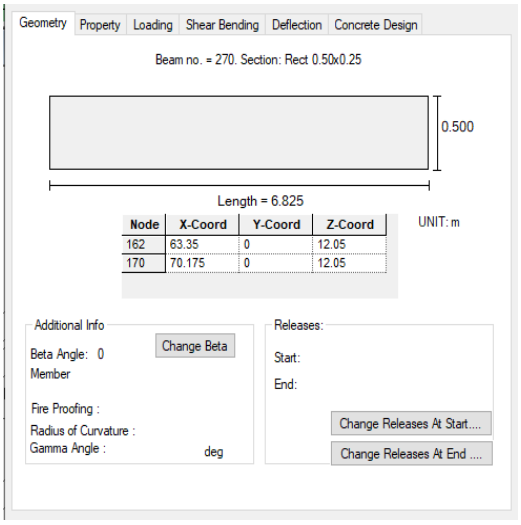
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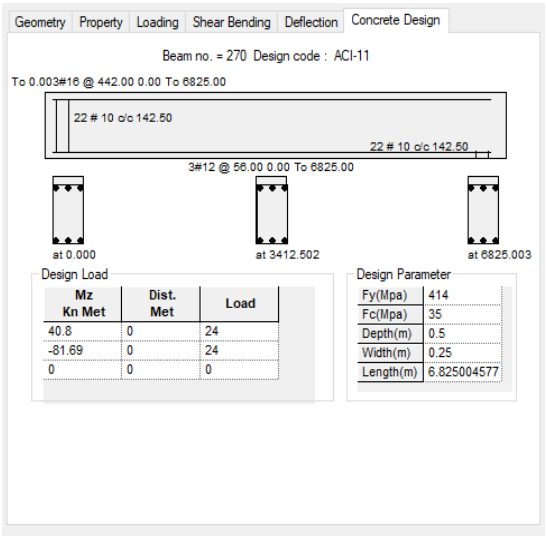
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11. DESIGN OF GRADE BEAM-(GB1)

11.1. Design of Beam-(GB1)







Provide minimum 3-16 dia-Full Straight Bars Top & Bottom.

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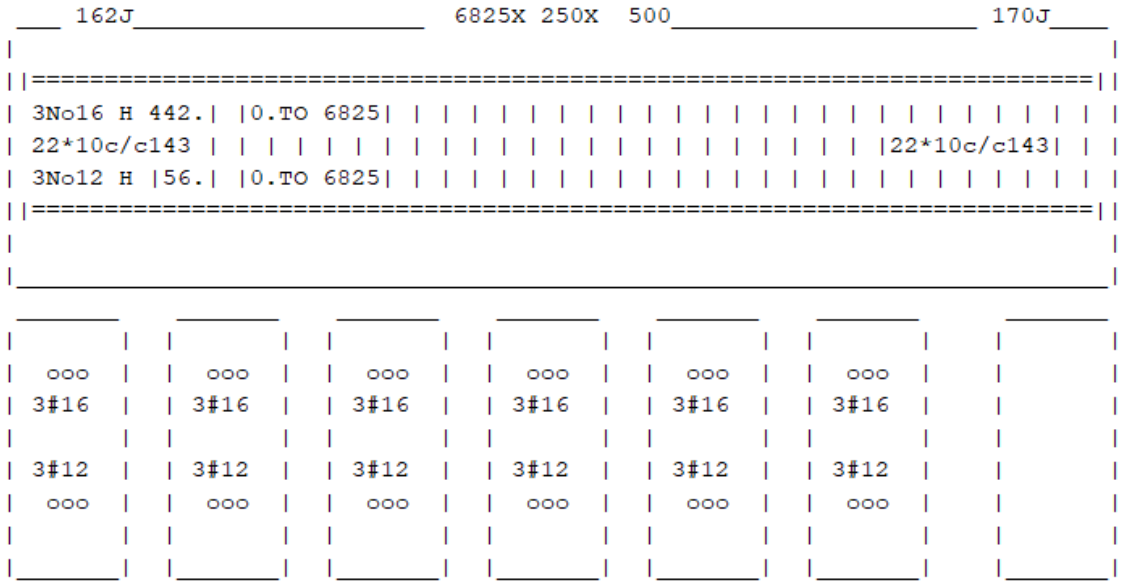
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11.1.1. Staad Output for GB1

```
        B E A M  N O.   270 D E S I G N  R E S U L T S - SHEAR


AT START SUPPORT - Vu=   53.80 KNS  Vc=  107.64 KNS  Vs=    0.00 KNS
Tu=    4.12 KN-MET  Tc=    3.8 KN-MET  Ts=    5.5 KN-MET  LOAD    49
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 2973. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   1.36 SQ.CM.

AT END   SUPPORT - Vu=   52.82 KNS  Vc=  107.64 KNS  Vs=    0.00 KNS
Tu=    4.12 KN-MET  Tc=    3.8 KN-MET  Ts=    5.5 KN-MET  LOAD    49
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 2973. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   1.36 SQ.CM.
```



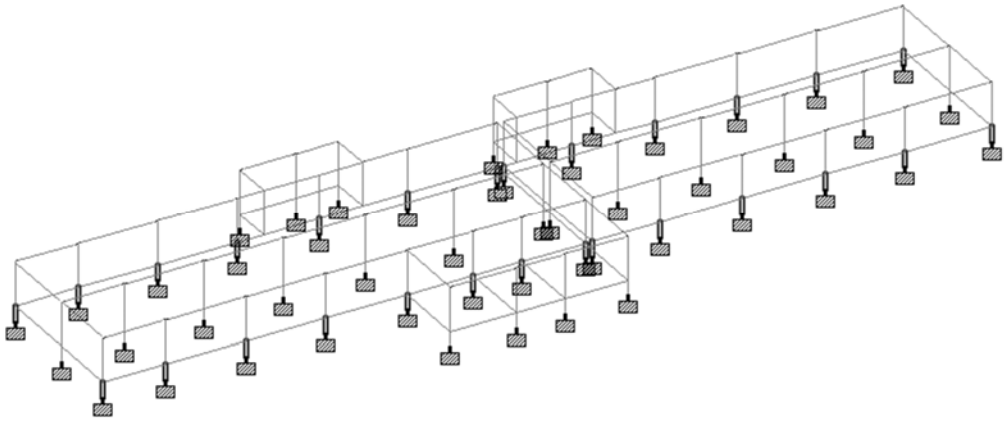
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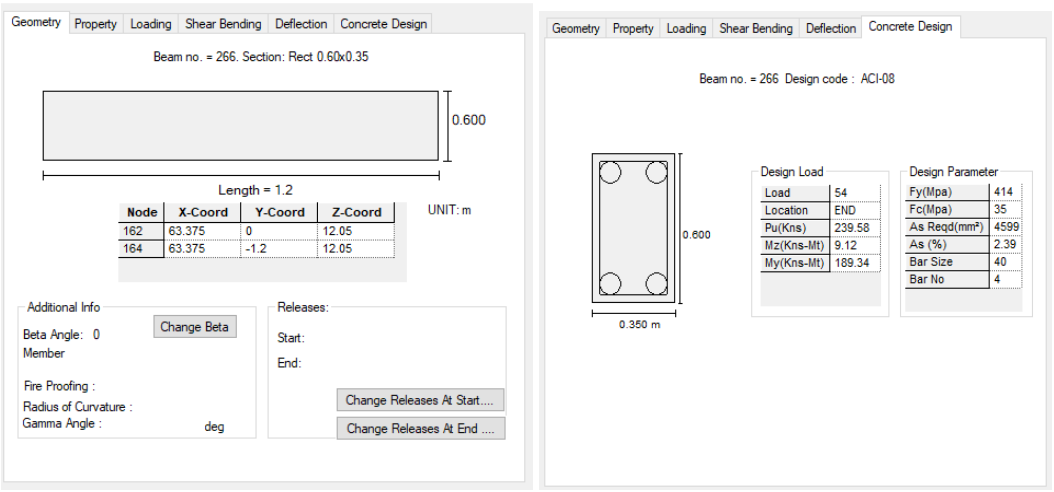
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12. CONCRETE COLUMN DESIGN

12.1. Design of Column-(CR)





Column Design CR

Provide equivalent 10 No's-Dia 25 equivalent Vertical Reinforcement in the Column Neck.

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12.1.1. Staad Output For Column CR

```
        COLUMN  NO.    266  DESIGN PER ACI 318-08 - AXIAL + BENDING


--------------------------------< PAGE 29 Ends Here >----------------------------
   STAAD SPACE                                                 -- PAGE NO.    30

  FY - 413.7   FC -   35.0 MPA,   RECT SIZE - 350.0 X 600.0 MMS, TIED
          AREA OF STEEL REQUIRED =   4599.0   SQ. MM

  BAR CONFIGURATION        REINF PCT.    LOAD    LOCATION    PHI
  ------------------------------------------------------------

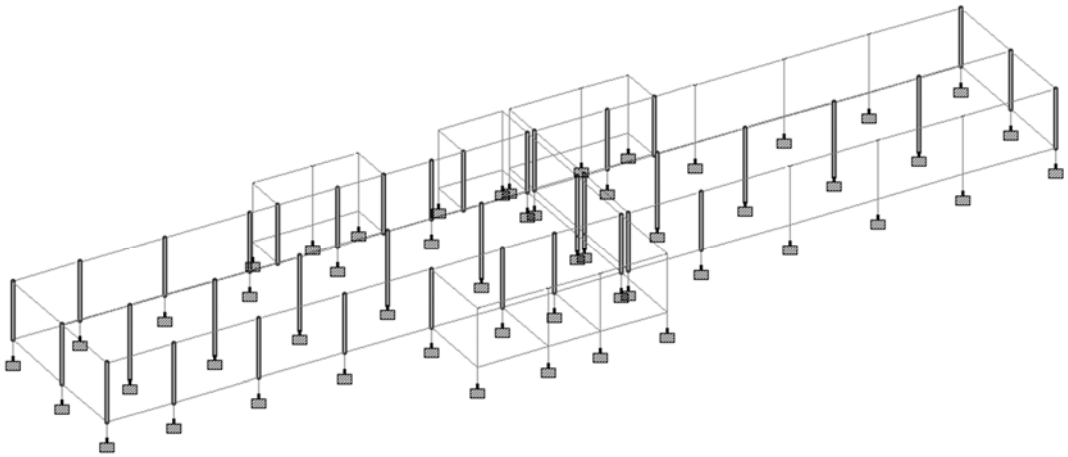
    4 - 40 MM                 2.394        54       END      0.650
  (PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
  TIE BAR NUMBER    10 SPACING 320.00 MM
```

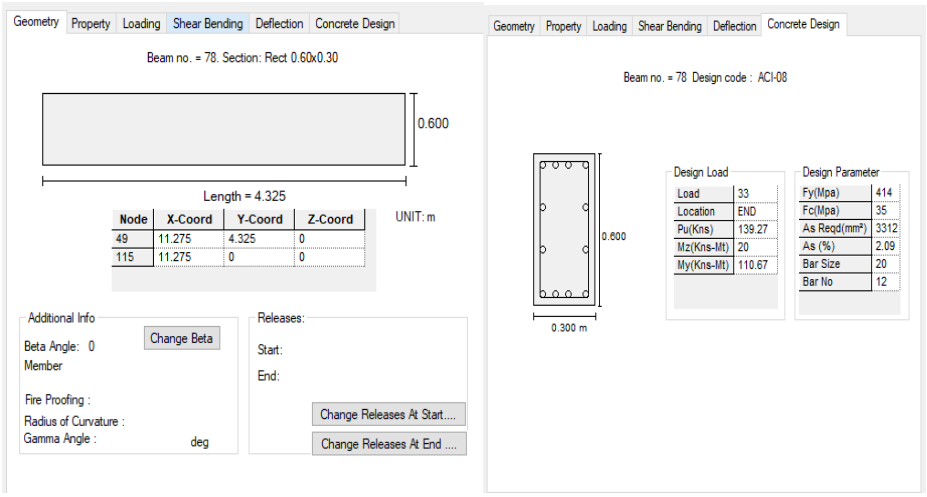

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12.2. Design of Column -(C1)





Column Design C1

Provide equivalent 8 No's-Dia 25 equivalent Vertical Reinforcement in the Column C1.

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12.2.1. Staad Output For Column C1

```
      COLUMN  NO.    78  DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 413.7  FC -  35.0 MPA,  RECT SIZE - 300.0 X 600.0 MMS, TIED
        AREA OF STEEL REQUIRED =  3312.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
-------------------------------------------------------------

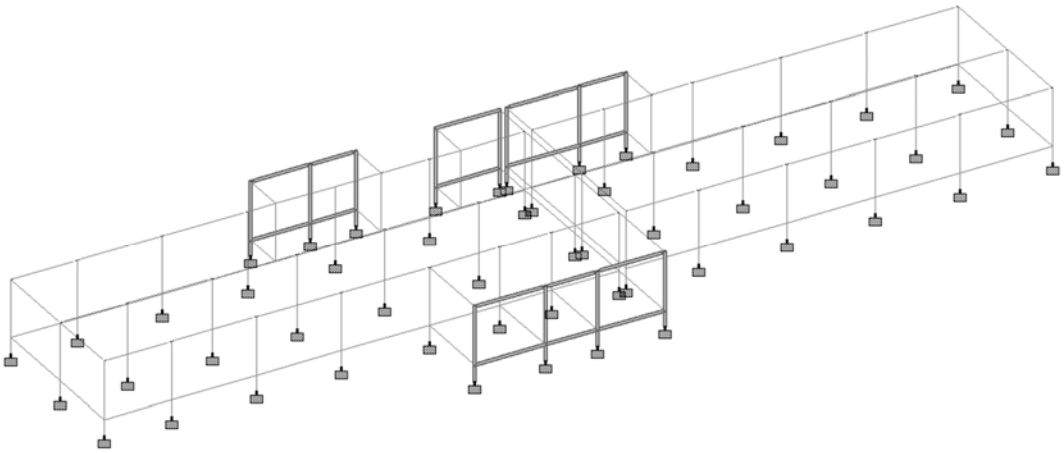
 12 - 20 MM               2.094       33      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 300.00 MM
```

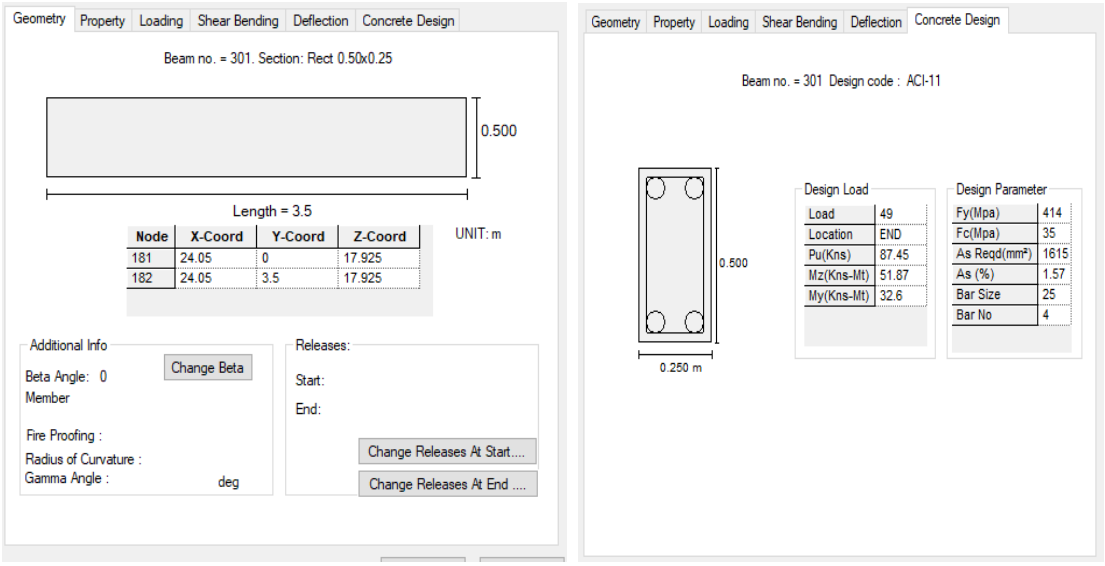

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12.3. Design of Column -(C2)





Column Design C2

Provide equivalent 8 No's-Dia 20 equivalent Vertical Reinforcement in the Column C2.

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12.3.1. Staad Output For Column C2

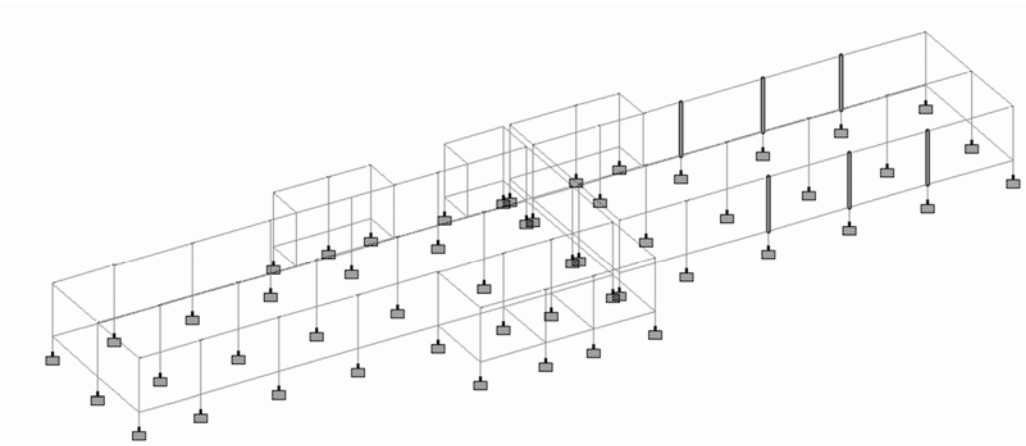
```
        ACI 318-11   COLUMN  NO.   301   DESIGN RESULTS
        ==============================================


FY - 413.7  FC -  35.0 MPA,  RECT SIZE - 250.0 X 500.0 MMS, TIED
        AREA OF STEEL REQUIRED =  1615.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
-------------------------------------------------------------

  4 - 25 MM                1.571       49      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 250.00 MM
```

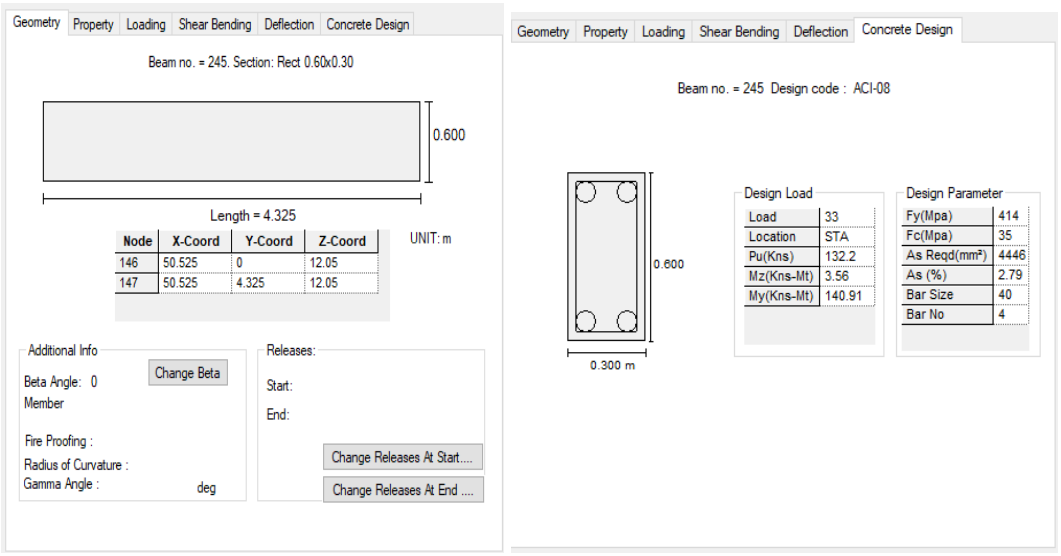
12.4. Design of Column -(C3)



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Column Design C3

Provide equivalent 10 No's-Dia 25 equivalent Vertical Reinforcement in the Column Neck.

12.4.1. Staad Output For Column C3

```
      COLUMN   NO.    245   DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 413.7   FC -   35.0 MPA,   RECT SIZE - 300.0 X 600.0 MMS, TIED
         AREA OF STEEL REQUIRED =   4446.0   SQ. MM

BAR CONFIGURATION        REINF PCT.    LOAD    LOCATION    PHI
-------------------------------------------------------------

  4 - 40 MM                 2.793        33       STA      0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    10 SPACING 300.00 MM
```


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13. CANTILEVER SLAB DESIGN

DESIGN OF CANTILEVER SLAB

Geometrical Properties			Material Properties		
Span to center of support "sp_1"	0.8	met	Weight of reinforced concrete "γ_r	24	kN/m^3
Clear Span	0.6	met	Strength of concrete "f'_c"	35	N/mm^2
Clear span + depth of slab "sp_2"	0.75	met	Strength of steel "f_y"	420	N/mm^2
Max. eff span le = min (sp_1;sp_2)	0.8	met	Clear cover to concrete("c")	20	mm
Width of Slab "b"	1000	mm	Flexure strength reduction factor	0.9	
Thickness of slab "t"	150	mm	Shear strength reduction factor	0.75	
Eff. Depth "d"	125		Modulus of Elasticity of steel "Es"	2E+05	N/mm^2
Loading			Modulus of Elasticity of concrete	27806	N/mm^2
Self Weight of slab	3.6	kN/m^2	Modular Ratio "n = Es/Ec"	7.19	
Floor Finish	3.5	kN/m^2	**Provided Reinforcement**		
Wall load	0	kN/m^2	Top layer Nos.	6 Φ 10	mm
Total dead Load "U_{dl}"	7.1	kN/m^2	Bottom layer Nos.	0 Φ 12	mm
Live load	1	kN/m^2	Area of Top steel "$A_{s\ t}$"	468	mm^2
Total Ultimate Load "$Uu_{,tot}$"$_{(1.2D+1.6L)}$	10.12	kN/m^2	Area of Bottom steel "$A_{s\ b}$"	0	mm^2
Moment due to Overhang Point Load(0.125*0.15*24*0.6)	0.27	kNm	"d_b' = 10 mm		
Calculations					
Shear force Vu = U*le	**8.096**	**kN**			
Neg. Moment Mu = U*le^2/2	**3.51**	**kNm**			

Check for Section capacity								
Depth of comp. block "a" = $A_{s\ t}$*f_y/(0.85*f'_c*b)	6.61	mm						
Section capacity ΦMn = Φ*$A_{s\ t}$*f_y*(d-a/2)	**21.94**	**kNm**	**O.K**	>	**3.51**	**kNm**	(Mu)	
Shear requirements								
Shear capacity ΦVc = Φ*0.17*√f'c*b*d	**94.29**	**kN**	**O.K**	>	**8.096**	**kN**	(Vu)	
Serviceability requirements								
d' = c+d_b/2	26	mm						
B = b/(n*$A_{s\ t}$)	0.297							
r = (n-1)*$A_{s\ b}$/(n*$A_{s\ t}$)	0							
k_d = (√(2*d*B(1+r*d'/d)+(1+r)^2)-(1+r))/B	29.01	mm						
I_{cr} = bkd^3/3+n*$A_{s\ t}$*(d-k_d)^2+(n-1)*$A_{s\ b}$*(k_d-d'	3.915E+07	mm^4						
Calculate Gross moment of inertia								
Area of slab A = t*b	2E+05	mm^2						
Depth of neutral axis "y"	75	mm						
Distance of top fiber from N.A "y_t"	75	mm						
Gross moment of inertia "I_g"	2.813E+08	mm^4						
Allowable stress fr = 0.62√f'c	3.67	mm^2						
Cracking moment M_{cr} = fr*I_g/y_t	13.75	kNm						

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Service moments

Moment due to dead load "M_{dl}"	2.54	kNm	
Moment due to live load "Mll"	0.32	kNm	
Total Moment Ms = $M_{dl}+M_{ll}$	2.86	kNm	
Deflection due to dead + live loads			
R(dl+ll) = M_{cr}/Ms	1.000		
I_e(dl+ll) = $R^3*I_g+(1-R^3)*I_{cr}$	2.813E+08	mm^4	
δ = K*5/48*Ms*le^2/(Ec*I_e)	0.08	mm	**K = 3.2 (conservative)**
Deflection due to dead load			
R(dl) = M_{cr}/Md	5.411		
I_e(dl) = $R^3*I_g+(1-R^3)*I_{cr}$	3.839E+10	mm^4	
δ = K*5/48*Md*le^2/(Ec*I_e)	0.00	mm	

Deflection due to live load

Live load deflection = dl+ll deflection - dl deflection 0.08 mm

Allowable immediate deflection due to Live load = per ACI 318 Tab 9.5b I_e/360 **2.22 mm > 0.08 mm O.K**

Long term deflection

% of liveload sustained "SL"	50	%
M_{sus} = $M_{dl}+SL*M_{ll}$	2.70	kNm
R(sus) = M_{cr}/Msus	5.091	
I_e(sus) = $R^3*I_g+(1-R^3)*I_{cr}$	3.198E+10	
δ = K*5/48*Msus*le^2/(Ec*I_e)	0.00	mm

Magnification factor for creep and shrinkage

$\lambda_\Delta = \xi / (1+50*\rho')$ 2.000

Where:-

ξ = 2 for 5 years or more cl.9.5.2.5 ACI 318

$\rho' = A_{s\ b}/(b*d)$ 0

Total deflection = λ_Δ*sus deflection + ll deflection 0.079 mm

Allowable deflection = I_e/240 3.333 mm > 0.079 mm O.K

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14. FOUNDATION DESIGN

Foundations have been designed based on the Soil Report Recommendations. The minimum Embedment Depth of Foundations has been considered as 1.2m & applicable SBC of 350 Kpa. (Refer Doc-MD-216-3000-EG-CV-RPT-1008 for Details)

TABLE 4: RECOMMENDED NET ALLOWABLE BEARING CAPACITIES

PRPJECT: MANSOURAH - MANSSARAH GOLD PROJECT
CLIENT: L & T CONSTRUCTION

Area	Borehole No.	Proposed Structures	Foundation Type	Depth of Foundation below FGL	Footing Size (B x L)	Net Allowable Bearing Capacity Based on		Recommended Net Allowable Bearing Capacity	Estimated Total Settlement	Vertical Modulus of Subgrade Reaction (Ks)	Horizontal Modulus of Subgrade Reaction (Kh)
						Settlement	Shear Failure				
				(m)	(m)	(kN/m^2)	(kN/m^2)	(kN/m^2)	(mm)	($kN/m^2/m$)	(kN/m^3)
Village Area	BH-21	Village	Isolated Square	1.20	1.0 x 1.0	1000	350	350	25	42,000	32,200
				1.50		1100	500	500	25	60,000	38,900
				1.20	2.0 x 2.0	700	580	580	25	69,600	37,500
				1.50		850	750	750	25	90,000	44,200
				1.20	3.0 x 3.0	600	750	600	25	72,000	42,800
				1.50		700	930	700	25	84,000	49,500
				1.20	4.0 x 4.0	550	910	550	25	66,000	48,100
				1.50		625	1090	625	25	75,000	54,800
				1.20	5.0 x 5.0	475	1060	475	25	57,000	53,400
				1.50		550	1240	550	25	66,000	60,200

Refer Attachment- II for Foundation Design calculations

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REFERENCES

1.)	BUILDING PLANS	MD-216-6000-EG-AR-GAD-3135
2.)	OVERALL LAYOUT PLAN	MD-216-6000-EG-CV-GAD-1535
3.)	LAYOUT & DETAILS OF FOUNDATION & COLUMN.	MD-216-6000-EG-CV-DED-1498
4.)	LAYOUT & DETAILS OF GB & GRADE SLAB	MD-216-6000-EG-CV-DED-1569
5.)	LAYOUT & DETAILS OF ROOF BEAM ROOF SLAB	MD-216-6000-EG-CV-DED-1512
6.)	SCHEDULES OF FDN, GB, ROOF BEAMS & WALL DETAILS	MD-216-6000-EG-CV-DED-1501
7.)	GEOTECHNICAL REPORT	MD-216-3000-EG-CV-RPT-1008
8.)	STRUCTURAL DESIGN CRITERIA	MD-216-0000-EG-ST-CRT-0001

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ATTACHMENT 1 STAAD INPUT DATA

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STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 18-Oct-19
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
42 0 0 0; 43 0 4.325 0; 44 0 -1.2 0; 46 4.975 4.325 0; 47 4.975 -1.2 0;
49 11.275 4.325 0; 50 11.275 -1.2 0; 52 17.575 4.325 0; 53 17.575 -1.2 0;
55 24.05 4.325 0; 56 24.05 -1.2 0; 63 0 0 6.375; 64 0 4.325 6.375;
65 0 -1.2 6.375; 67 4.975 4.325 6.375; 68 4.975 -1.2 6.375;
70 11.275 4.325 6.375; 71 11.275 -1.2 6.375; 73 17.575 4.325 6.375;
74 17.575 -1.2 6.375; 76 24.05 4.325 6.375; 77 24.05 -1.2 6.375; 84 0 0 12.05;
85 0 4.325 12.05; 86 0 -1.2 12.05; 87 4.975 0 12.05; 88 4.975 4.325 12.05;
89 4.975 -1.2 12.05; 90 11.275 0 12.05; 91 11.275 4.325 12.05;
92 11.275 -1.2 12.05; 93 17.575 0 12.05; 94 17.575 4.325 12.05;
95 17.575 -1.2 12.05; 96 24.05 0 12.05; 97 24.05 4.325 12.05;
98 24.05 -1.2 12.05; 114 4.975 0 0; 115 11.275 0 0; 116 17.575 0 0;
117 24.05 0 0; 118 31.025 4.325 0; 119 31.025 -1.2 0; 120 31.025 4.325 6.375;
121 31.025 -1.2 6.375; 125 31.025 0 0; 126 38.1 4.325 0; 127 38.1 -1.2 0;
128 38.1 4.325 6.375; 129 38.1 -1.2 6.375; 130 38.1 0 12.05;
131 38.1 4.325 12.05; 132 38.1 -1.2 12.05; 133 38.1 0 0; 134 43.975 4.325 0;
135 43.975 -1.2 0; 136 43.975 4.325 6.375; 137 43.975 -1.2 6.375;
138 43.975 0 12.05; 139 43.975 4.325 12.05; 140 43.975 -1.2 12.05;
141 43.975 0 0; 142 50.525 4.325 0; 143 50.525 -1.2 0; 144 50.525 4.325 6.375;
145 50.525 -1.2 6.375; 146 50.525 0 12.05; 147 50.525 4.325 12.05;
148 50.525 -1.2 12.05; 149 50.525 0 0; 150 57.075 4.325 0; 151 57.075 -1.2 0;
152 57.075 4.325 6.375; 153 57.075 -1.2 6.375; 154 57.075 0 12.05;
155 57.075 4.325 12.05; 156 57.075 -1.2 12.05; 157 57.075 0 0;
158 63.375 4.325 0; 159 63.375 -1.2 0; 160 63.375 4.325 6.375;
161 63.375 -1.2 6.375; 162 63.375 0 12.05; 163 63.375 4.325 12.05;
164 63.375 -1.2 12.05; 165 63.375 0 0; 166 70.175 4.325 0; 167 70.175 -1.2 0;
168 70.175 4.325 6.375; 169 70.175 -1.2 6.375; 170 70.175 0 12.05;
171 70.175 4.325 12.05; 172 70.175 -1.2 12.05; 173 70.175 0 0;
174 70.175 0 6.375; 175 29.275 0 12.05; 176 29.275 4.325 12.05;
177 29.275 -1.2 12.05; 178 33.125 0 12.05; 179 33.125 4.325 12.05;
180 33.125 -1.2 12.05; 181 24.05 0 17.925; 182 24.05 4.325 17.925;
183 24.05 -1.2 17.925; 184 38.1 0 17.925; 185 38.1 4.325 17.925;
186 38.1 -1.2 17.925; 187 29.275 0 17.925; 188 29.275 4.325 17.925;
189 29.275 -1.2 17.925; 190 33.125 0 17.925; 191 33.125 4.325 17.925;
192 33.125 -1.2 17.925; 193 24.05 4.325 -3.25; 194 24.05 -1.2 -3.25;
195 24.05 0 -3.25; 196 19.65 4.325 0; 197 19.65 0 0; 198 19.65 4.325 -3.25;
199 19.65 -1.2 -3.25; 200 19.65 0 -3.25; 201 27.45 4.325 0; 202 27.45 0 0;
203 27.45 4.325 -3.25; 204 27.45 -1.2 -3.25; 205 27.45 0 -3.25; 210 44.05 0 0;
213 38.62 4.325 -3.25; 216 38.62 0 -3.25; 217 38.62 -1.2 -3.25;
218 47.45 4.325 -3.25; 219 47.45 0 -3.25; 220 47.45 4.325 0; 221 47.45 0 0;
222 47.45 -1.2 -3.25; 225 43.975 4.325 -3.25; 226 43.975 0 -3.25;
227 43.975 -1.2 -3.25; 228 38.62 4.325 0; 229 38.62 0 0; 230 38.62 4.325 6.375;
231 38.62 -1.2 6.375; 232 38.62 0 12.05; 233 38.62 4.325 12.05;
234 38.62 -1.2 12.05; 235 38.62 -1.2 0; 236 38.1 0 6.375; 237 38.62 0 6.375;
238 38.1 4.325 -3.25; 239 38.1 0 -3.25; 240 38.1 -1.2 -3.25; 241 33.35 4.325 0;
242 33.35 4.325 -3.25; 243 33.35 0 0; 244 33.35 0 -3.25; 245 33.35 -1.2 -3.25;
MEMBER INCIDENCES
70 42 43; 71 42 44; 73 43 46; 74 46 114; 77 46 49; 78 49 115; 81 49 52;
82 52 116; 85 52 196; 86 55 117; 96 42 63; 97 43 64; 110 63 64; 111 63 65;
113 64 67; 114 67 68; 117 67 70; 118 70 71; 121 70 73; 122 73 74; 125 73 76;
126 76 77; 136 63 84; 137 64 85; 150 84 85; 151 84 86; 152 84 87; 153 85 88;
154 87 88; 155 87 89; 156 87 90; 157 88 91; 158 90 91; 159 90 92; 160 90 93;
161 91 94; 162 93 94; 163 93 95; 164 93 96; 165 94 97; 166 96 97; 167 96 98;

190 114 47; 192 115 50; 194 116 53; 196 117 56; 201 42 114; 202 114 115;
203 115 116; 204 116 197; 205 55 201; 207 76 120; 212 117 202; 213 118 125;
214 120 121; 217 125 119; 218 118 241; 219 120 128; 222 125 243; 223 126 133;
224 128 236; 225 130 131; 226 130 132; 227 133 127; 233 134 141; 234 136 137;
235 138 139; 236 138 140; 237 141 135; 238 134 220; 239 136 144; 240 138 146;
241 139 147; 242 141 210; 243 142 149; 244 144 145; 245 146 147; 246 146 148;
247 149 143; 248 142 150; 249 144 152; 250 146 154; 251 147 155; 252 149 157;
253 150 157; 254 152 153; 255 154 155; 256 154 156; 257 157 151; 258 150 158;
259 152 160; 260 154 162; 261 155 163; 262 157 165; 263 158 165; 264 160 161;
265 162 163; 266 162 164; 267 165 159; 268 158 166; 269 160 168; 270 162 170;
271 163 171; 272 165 173; 273 166 173; 274 168 174; 275 170 171; 276 170 172;
277 173 167; 278 166 168; 279 168 171; 280 174 169; 281 173 174; 282 174 170;
283 96 175; 284 97 176; 285 175 176; 286 175 177; 287 175 178; 288 176 179;
289 178 179; 290 178 180; 291 179 131; 293 96 181; 294 97 182; 295 130 184;
296 131 185; 297 175 187; 299 178 190; 301 181 182; 302 181 183; 303 184 185;
304 184 186; 305 187 188; 306 187 189; 307 190 191; 308 190 192; 309 182 188;
310 188 191; 311 191 185; 312 181 187; 313 187 190; 317 193 195; 318 195 194;
319 196 55; 320 197 117; 321 193 198; 322 195 200; 323 196 198; 324 197 200;
325 198 200; 326 200 199; 327 201 118; 328 202 125; 329 193 203; 330 195 205;
331 201 203; 332 202 205; 333 203 205; 334 205 204; 341 210 221; 345 225 218;
347 228 213; 348 229 141; 349 229 216; 350 213 216; 351 216 217; 354 220 142;
355 220 218; 356 221 149; 357 221 219; 358 218 219; 359 219 222; 364 225 213;
365 225 226; 366 226 227; 369 226 219; 370 216 226; 394 178 130; 398 190 184;
401 228 229; 402 230 136; 403 230 237; 404 232 138; 405 233 139; 406 232 233;
407 232 234; 408 229 235; 409 126 128; 410 128 131; 411 228 230; 412 230 233;
413 196 197; 414 201 202; 416 220 221; 417 236 129; 418 130 236; 419 236 133;
420 237 231; 421 229 237; 422 237 232; 423 126 238; 424 133 239; 425 238 239;
426 239 240; 427 241 126; 428 241 242; 429 243 133; 430 243 244; 431 242 244;
432 244 245; 433 244 239; 434 242 238; 435 241 243; 436 228 134;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 28000
G 9.28139e+006
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
70 74 78 82 86 110 111 114 118 122 126 150 154 158 162 166 213 214 -
223 TO 225 233 TO 235 243 TO 245 253 TO 255 263 TO 265 273 TO 275 280 285 -
289 401 403 406 413 414 416 417 420 435 PRIS YD 0.6 ZD 0.3
MEMBER PROPERTY AMERICAN
113 117 121 125 207 219 239 249 259 269 309 TO 311 402 409 TO 411 -
412 PRIS YD 0.55 ZD 0.25
MEMBER PROPERTY AMERICAN
73 77 81 85 96 97 136 137 152 153 156 157 160 161 164 165 201 TO 205 212 218 -
222 238 240 TO 242 248 250 TO 252 258 260 TO 262 268 270 TO 272 278 279 281 -
282 TO 284 287 288 291 293 TO 297 299 301 TO 308 312 313 317 TO 322 -
324 TO 330 332 TO 334 341 345 348 TO 351 354 356 TO 359 364 TO 366 369 370 -
394 398 404 405 418 419 421 422 424 TO 427 429 TO 434 -
436 PRIS YD 0.5 ZD 0.25
MEMBER PROPERTY AMERICAN
323 331 347 355 423 428 PRIS YD 0.35 ZD 0.25
MEMBER PROPERTY AMERICAN
71 151 155 159 163 167 190 192 194 196 217 226 227 236 237 246 247 256 257 -
266 267 276 277 286 290 407 408 PRIS YD 0.6 ZD 0.35

```
CONSTANTS
BETA 90 MEMB 110 111 150 151 274 280
MATERIAL CONCRETE ALL
SUPPORTS
44 47 50 53 56 65 68 71 74 77 86 89 92 95 98 119 121 127 129 132 135 137 140 -
143 145 148 151 153 156 159 161 164 167 169 172 177 180 183 186 189 192 194 -
199 204 217 222 227 231 234 235 240 245 FIXED
******************************************************************************
MEMBER OFFSET
113 117 121 125 207 219 239 249 259 269 402 START 0.3 0 0
113 117 121 125 207 219 239 249 259 269 402 END -0.3 0 0
73 77 81 153 157 161 165 218 241 248 251 258 261 268 271 284 288 291 -
309 TO 311 327 329 345 354 405 436 START 0.25 0 0
73 77 81 153 157 161 165 241 248 251 258 261 268 271 284 288 291 309 TO 311 -
327 329 345 354 405 427 436 END -0.25 0 0
97 137 278 279 START 0 0 0.125
97 137 278 279 END 0 0 -0.125
321 364 START -0.25 0 0
321 364 END 0.25 0 0
MEMBER CRACKED
70 71 74 78 82 86 110 111 114 118 122 126 150 151 154 155 158 159 162 163 -
166 167 190 192 194 196 213 214 217 223 TO 227 233 TO 237 243 TO 247 253 -
254 TO 257 263 TO 267 273 TO 277 280 285 286 289 290 301 TO 308 317 318 325 -
326 333 334 350 351 358 359 365 366 401 403 406 TO 408 413 414 416 417 420 -
425 426 431 432 435 REDUCTION RIX 0.700000 RIY 0.700000 RIZ 0.700000
73 77 81 85 96 97 113 117 121 125 136 137 152 153 156 157 160 161 164 165 -
201 TO 205 207 212 218 219 222 238 TO 242 248 TO 252 258 TO 262 268 TO 272 -
278 279 281 TO 284 287 288 291 293 TO 297 299 309 TO 313 319 TO 324 -
327 TO 332 341 345 347 TO 349 354 TO 357 364 369 370 394 398 402 404 405 -
409 TO 412 418 419 421 TO 424 427 TO 430 433 434 -
436 REDUCTION RIX 0.350000 RIY 0.350000 RIZ 0.350000
DEFINE IBC 2003
SDS 0.106 SD1 0.032 S1 0.02 IE 1 RX 2.5 RZ 2.5 SCLASS 4
SELFWEIGHT 1
MEMBER WEIGHT
357 369 370 394 398 404 418 419 421 422 429 430 433 UNI 12
309 TO 311 UNI 19.2
321 329 345 364 434 UNI 9.45
113 117 121 125 207 219 239 249 259 269 402 UNI 38.56
284 288 291 UNI 37.36
85 205 238 319 427 436 UNI 29.85
153 157 161 165 241 251 261 271 405 UNI 18.16
73 77 81 218 248 258 268 327 354 UNI 20.4
DEFINE WIND LOAD
TYPE 1 WIND X WINDWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.000 M 35.000 M 25.000 M 2.000 0.010 0 -
0 0 1 0 1.030 1.000 1.000 0.850 0 -
1 0 0 0.850 0.800 -0.550
!> END GENERATED DATA BLOCK
INT 1.59699 1.59699 HEIG  0  4.572
TYPE 2 WIND X LEEWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.500 M 75.000 M 25.000 M 2.000 0.010 1 -
0 0 1 0 1.030 1.000 1.000 0.850 0 -
1 0 0 0.850 -0.200 0.550
!> END GENERATED DATA BLOCK
```

```
INT -0.934826 -0.934826 HEIG  0  4.5
TYPE 3 SIDEWARD X
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.500 M 35.000 M 25.000 M 2.000 0.010 2 -
0 0 1 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.875 -0.700 0.550
!> END GENERATED DATA BLOCK
INT -1.50938 -1.50938 HEIG  0  4.5
TYPE 4 WIND Z WINDWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.000 M 25.000 M 75.000 M 2.000 0.010 0 -
0 0 0 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.841 0.800 -0.550
!> END GENERATED DATA BLOCK
INT 1.58753 1.58753 HEIG  0  4.572
TYPE 5 WIND Z LEEWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.500 M 25.000 M 75.000 M 2.000 0.010 1 -
0 0 0 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.841 -0.500 0.550
!> END GENERATED DATA BLOCK
INT -1.25982 -1.25982 HEIG  0  4.5
TYPE 6 SIDEWARD Z
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.500 M 25.000 M 75.000 M 2.000 0.010 2 -
0 0 0 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.841 -0.700 0.550
!> END GENERATED DATA BLOCK
INT -1.47811 -1.47811 HEIG  0  4.5
LOAD 1 LOADTYPE Seismic  TITLE EQK LOAD Z
IBC LOAD Z 1
PERFORM ANALYSIS
CHANGE
LOAD 2 LOADTYPE Seismic  TITLE EQK LOAD X
IBC LOAD X 1
PERFORM ANALYSIS
CHANGE
LOAD 3 LOADTYPE Dead  TITLE DL
SELFWEIGHT Y -1
MEMBER LOAD
97 137 294 323 347 423 428 UMOM GZ 1
153 157 161 165 241 251 261 271 309 TO 311 405 UMOM GX 1
278 279 296 331 355 UMOM GZ -1
73 77 81 85 218 248 258 268 321 327 329 345 354 364 427 434 UMOM GX -1
96 136 152 156 160 164 201 TO 204 212 222 240 242 250 252 260 262 270 272 -
281 TO 283 287 293 295 297 299 312 313 320 322 324 328 330 332 341 348 356 -
357 369 370 394 398 404 418 419 421 422 429 430 433 UNI GY -14
309 TO 311 UNI GY -19.2
321 329 345 364 434 UNI GY -9.45
113 117 121 125 207 219 239 249 259 269 402 UNI GY -38.56
284 288 291 UNI GY -37.36
85 205 238 319 427 436 UNI GY -29.85
153 157 161 165 241 251 261 271 405 UNI GY -18.16
73 77 81 218 248 258 268 327 354 UNI GY -20.4
LOAD 4 LOADTYPE Live  TITLE LL
```

MEMBER LOAD
97 137 UMOM GZ 0.18
153 157 161 165 UMOM GX 0.18
309 TO 311 UNI GY -1.5
321 329 345 364 434 UNI GY -0.8
113 117 121 125 207 219 239 249 259 269 402 UNI GY -3
284 288 291 UNI GY -2.92
205 238 319 427 436 UNI GY -2.4
153 157 161 165 241 251 261 271 405 UNI GY -1.412
73 77 81 85 218 248 258 268 327 354 UNI GY -1.59
LOAD 5 LOADTYPE Wind TITLE WL+X
WIND LOAD X 1 TYPE 1 YR 0 4.5
WIND LOAD -X 1 TYPE 2 YR 0 4.5
WIND LOAD -Z 1 TYPE 3 YR 0 4.5
WIND LOAD -Z -1 TYPE 3 YR 0 4.5
WIND LOAD X 1 TYPE 1 XR 33.35 33.4 YR 0 4.5
WIND LOAD -X 1 TYPE 2 XR 27.4 27.45 YR 0 4.5
LOAD 6 LOADTYPE Wind TITLE WL+Z
WIND LOAD Z 1 TYPE 4 XR 0 75 YR 0 4.5
WIND LOAD -Z 1 TYPE 5 XR 0 75 YR 0 4.5
WIND LOAD -X 1 TYPE 6 XR 0 75 YR 0 4.5
WIND LOAD -X -1 TYPE 6 XR 0 75 YR 0 4.5
WIND LOAD -X 1 TYPE 6 XR 27.4 27.45 YR 0 4.5
WIND LOAD -X -1 TYPE 6 XR 33.35 33.4 YR 0 4.5
LOAD 7 LOADTYPE Wind TITLE WL-X
WIND LOAD X -1 TYPE 1 YR 0 4.5
WIND LOAD -X -1 TYPE 2 YR 0 4.5
WIND LOAD -Z 1 TYPE 3 YR 0 4.5
WIND LOAD -Z -1 TYPE 3 YR 0 4.5
WIND LOAD X -1 TYPE 1 XR 27.45 27.45 YR 0 4.5
WIND LOAD -X -1 TYPE 2 XR 33.35 33.4 YR 0 4.5
LOAD 8 LOADTYPE Wind TITLE WL-Z
WIND LOAD Z -1 TYPE 4 YR 0 4.5
WIND LOAD -Z -1 TYPE 5 YR 0 4.5
WIND LOAD -X 1 TYPE 6 YR 0 4.5
WIND LOAD -X -1 TYPE 6 YR 0 4.5
WIND LOAD -X 1 TYPE 6 XR 27.45 27.45 YR 0 4.5
WIND LOAD -X -1 TYPE 6 XR 33.35 33.4 YR 0 4.5
LOAD 9 LOADTYPE Temperature TITLE TEMP
TEMPERATURE LOAD
70 73 74 77 78 81 85 97 110 113 117 121 125 137 150 153 154 157 158 161 162 -
165 205 207 218 219 235 238 239 241 243 245 248 249 251 253 255 258 259 261 -
263 265 268 269 271 273 TO 275 278 279 284 288 291 294 296 309 TO 311 317 -
319 321 323 325 327 329 331 333 345 347 350 354 355 358 364 365 402 405 409 -
410 TO 412 423 425 427 428 431 434 436 TEMP 25
LOAD COMB 10 SLS: (DL)
3 1.0
LOAD COMB 11 SLS: (DL+T)
3 1.0 9 1.0
LOAD COMB 12 SLS: (D+LR)
3 1.0 4 1.0
LOAD COMB 13 SLS: (D+0.75LR)
3 1.0 4 0.75
LOAD COMB 14 SLS: (D+0.7EZ)
3 1.0 1 0.7
LOAD COMB 15 SLS: (D+0.7EX)
3 1.0 2 0.7
LOAD COMB 16 SLS: (D+0.75WX1+0.75LR)
3 1.0 5 0.75 4 0.75

LOAD COMB 17 SLS: (D+0.75WZ1+0.75LR)
3 1.0 6 0.75 4 0.75
LOAD COMB 18 SLS: (0.6D+WX1)
3 0.6 5 1.0
LOAD COMB 19 SLS: (0.6D+WZ1)
3 0.6 6 1.0
LOAD COMB 20 SLS: (0.6D+0.7EX)
3 0.6 2 0.7
LOAD COMB 21 SLS: (0.6D+0.7EZ)
3 0.6 1 0.7
LOAD COMB 22 SLS: (D+WX1)
3 1.0 5 1.0
LOAD COMB 23 SLS: (D+WZ1)
3 1.0 6 1.0
LOAD COMB 24 ULS: (1.4 DL)
3 1.4
LOAD COMB 25 ULS: (1.4D+1.4T+0.5LR)
3 1.4 4 0.5 9 1.4
LOAD COMB 26 ULS: (1.4D+1.4T+0.5LR)
3 1.4 4 0.5 9 1.4
LOAD COMB 27 ULS: (1.2D+0.80WX1+1.6LR)
3 1.2 5 0.8 4 1.6
LOAD COMB 28 ULS: (1.2D+1.6WZ1+0.5LR)
3 1.2 6 1.6 4 0.5
LOAD COMB 29 ULS: (1.2D+1.6WX1+0.5LR)
3 1.2 4 0.5 5 1.6
LOAD COMB 30 ULS: (1.2D+1.6WZ1+0.5LR)
3 1.2 4 0.5 6 1.6
LOAD COMB 31 ULS: (1.2D+EX)
3 1.2 2 1.0
LOAD COMB 32 ULS: (1.2D+EZ)
3 1.2 1 1.0
LOAD COMB 33 ULS: (0.90D+1.6WX1)
3 0.9 5 1.6
LOAD COMB 34 ULS: (0.90D+1.6WZ1)
3 0.9 6 1.6
LOAD COMB 35 ULS: (0.90D+EZ)
3 0.9 1 1.0
LOAD COMB 36 ULS: (0.90D+EX)
3 0.9 2 1.0
LOAD COMB 39 SLS: (D+0.75WX2+0.75LR)
3 1.0 7 0.75 4 0.75
LOAD COMB 40 SLS: (D+0.75WZ2+0.75LR)
3 1.0 8 0.75 4 0.75
LOAD COMB 41 SLS: (0.6D+WX2)
3 0.6 7 1.0
LOAD COMB 42 SLS: (0.6D+WZ2)
3 0.6 8 1.0
LOAD COMB 45 SLS: (D+WX2)
3 1.0 7 1.0
LOAD COMB 46 SLS: (D+WZ2)
3 1.0 8 1.0
LOAD COMB 47 ULS: (1.2D+0.80WX2+1.6LR)
3 1.2 7 0.8 4 1.6
LOAD COMB 48 ULS: (1.2D+1.6WZ2+0.5LR)
3 1.2 8 1.6 4 0.5
LOAD COMB 49 ULS: (1.2D+1.6WX2+0.5LR)
3 1.2 4 0.5 7 1.6
LOAD COMB 50 ULS: (1.2D+1.6WZ2+0.5LR)

```
3 1.2 4 0.5 8 1.6
LOAD COMB 53 ULS: (0.90D+1.6WX2)
3 0.9 7 1.6
LOAD COMB 54 ULS: (0.90D+1.6WZ2)
3 0.9 8 1.6
PERFORM ANALYSIS
START CONCRETE DESIGN
CODE ACI
FC 35000 ALL
CLB 0.04 ALL
CLS 0.04 ALL
CLT 0.04 ALL
MINSEC 10 ALL
MINMAIN 20 MEMB 70 71 74 78 82 85 86 110 111 113 114 117 118 121 122 125 126 -
150 151 154 155 158 159 162 163 166 167 190 192 194 196 213 214 217 -
224 TO 227 233 TO 237 243 TO 247 253 TO 257 263 TO 267 273 TO 277 280 285 -
286 289 290 301 TO 308 317 TO 319 321 325 326 329 333 334 345 350 358 359 -
364 TO 366 403 406 TO 408 417 420 425 431 434 436
SFACE 0.5 MEMB 73 77 81 85 153 157 161 165 205 218 238 241 248 251 258 261 -
268 271 284 288 291 309 TO 311 319 327 354 405 427 436
EFACE 0.5 MEMB 73 77 81 85 153 157 161 165 205 218 238 241 248 251 258 261 -
268 271 284 288 291 309 TO 311 319 327 354 405 427 436
SFACE 0.55 MEMB 113 117 121 125 207 219 239 249 259 269 402
EFACE 0.55 MEMB 113 117 121 125 207 219 239 249 259 269 402
DESIGN COLUMN 70 71 74 78 82 86 110 111 114 118 122 126 150 151 154 155 158 -
159 162 163 166 167 190 192 194 196 213 214 217 223 TO 227 233 TO 237 243 -
244 TO 247 253 TO 257 263 TO 267 273 TO 277 280 285 286 289 290 301 TO 308 -
317 318 325 326 333 334 350 351 358 359 365 366 401 403 406 TO 408 413 414 -
416 417 420 425 426 431 432 435
DESIGN BEAM 73 77 81 85 96 97 113 117 121 125 136 137 152 153 156 157 160 -
161 164 165 201 TO 205 207 212 218 219 222 238 TO 242 248 TO 252 258 TO 262 -
268 TO 272 278 279 281 TO 284 287 288 291 293 TO 297 299 309 TO 313 -
319 TO 324 327 TO 332 341 345 347 TO 349 354 TO 357 364 369 370 394 398 402 -
404 405 409 TO 412 418 419 421 TO 424 427 TO 430 433 434 436
END CONCRETE DESIGN
FINISH
FINISH
```

ATTACHMENT 2 FOUNDATION DESIGN CALCULATIONS

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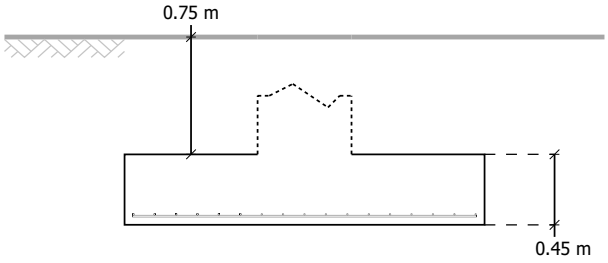
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Isolated Footing Design(ACI 318-08)

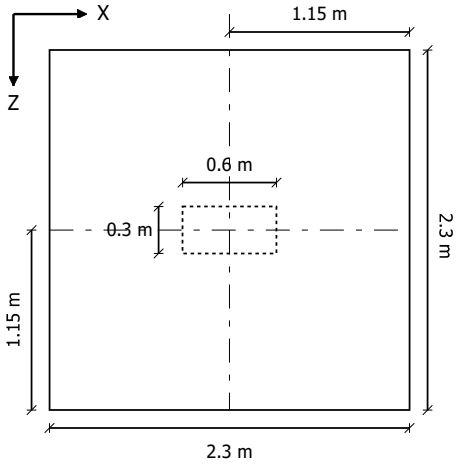
Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
153	1	2.300m	2.300m	0.450m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
153	17 - ϕ12	17 - ϕ12	N/A	N/A	N/A	N/A

FOUNDATION-F1



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 450.000mm
Footing Length - X (Fl) : 2300.000mm
Footing Width - Z (Fw) : 2300.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.600m

Column Width - Z (B_{col}) : 0.300m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 415.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #12
Pedestal Maximum Bar Size : #22
Minimum Bar Spacing : 100.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 750.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.530
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 2.300m

Initial Width (W_o) = 2.300m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
11	SLS: (DL+T)	1.00	1.00	1.00
12	SLS: (D+LR)	1.00	1.00	1.00
13	SLS: (D+0.75LR)	1.00	1.00	1.00
14	SLS: (D+0.7EZ)	1.00	1.00	1.00
15	SLS: (D+0.7EX)	1.00	1.00	1.00
16	SLS: (D+0.75WX1+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ1+0.75LR)	1.00	1.00	1.00
18	SLS: (0.6D+WX1)	1.00	1.00	1.00
19	SLS: (0.6D+WZ1)	1.00	1.00	1.00
20	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
21	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
22	SLS: (D+WX1)	1.00	1.00	1.00
23	SLS: (D+WZ1)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
39	SLS: (D+0.75WX2+0.75LR)	1.00	1.00	1.00
40	SLS: (D+0.75WZ2+0.75LR)	1.00	1.00	1.00
41	SLS: (0.6D+WX2)	1.00	1.00	1.00
42	SLS: (0.6D+WZ2)	1.00	1.00	1.00
45	SLS: (D+WX2)	1.00	1.00	1.00
46	SLS: (D+WZ2)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
24	ULS: (1.4 DL)	1.00	1.00	1.00
25	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+0.80WX1+1.6LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+1.6WX1+0.5LR)	1.00	1.00	1.00
30	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
31	ULS: (1.2D+EX)	1.00	1.00	1.00
32	ULS: (1.2D+EZ)	1.00	1.00	1.00
33	ULS: (0.90D+1.6WX1)	1.00	1.00	1.00
34	ULS: (0.90D+1.6WZ1)	1.00	1.00	1.00
35	ULS: (0.90D+EZ)	1.00	1.00	1.00
36	ULS: (0.90D+EX)	1.00	1.00	1.00
47	ULS: (1.2D+0.80WX2+1.6LR)	1.00	1.00	1.00
48	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
49	ULS: (1.2D+1.6WX2+0.5LR)	1.00	1.00	1.00
50	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
53	ULS: (0.90D+1.6WX2)	1.00	1.00	1.00
54	ULS: (0.90D+1.6WZ2)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
11	257.572	9.448	-0.006	-0.029	-25.008
12	276.873	5.455	-0.001	-0.022	-12.443
13	272.633	5.361	-0.001	-0.022	-12.229
14	259.912	5.121	7.520	37.889	-11.710
15	259.592	12.379	-0.002	-0.028	-33.749
16	272.517	6.906	-0.008	-0.095	-16.906
17	272.369	8.818	-0.051	0.003	-22.696
18	155.794	5.109	-0.009	-0.110	-13.188
19	155.596	7.657	-0.067	0.020	-20.908
20	155.627	10.347	-0.002	-0.019	-29.113

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
21	155.947	3.089	7.521	37.897	-7.075
22	259.759	7.141	-0.010	-0.119	-17.823
23	259.561	9.690	-0.067	0.011	-25.543
26	369.081	13.413	-0.008	-0.041	-35.440
39	272.937	1.726	-0.006	-0.083	-1.225
40	272.398	8.033	0.047	-0.046	-20.314
41	156.353	-1.798	-0.007	-0.095	7.720
42	155.635	6.611	0.064	-0.045	-17.732
45	260.318	0.234	-0.008	-0.104	3.085
46	259.600	8.644	0.064	-0.053	-22.367

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
24	363.878	7.114	-0.002	-0.031	-16.223
25	369.081	13.413	-0.008	-0.041	-35.440
26	369.081	13.413	-0.008	-0.041	-35.440
27	338.908	8.343	-0.009	-0.104	-20.263
28	319.813	13.658	-0.107	0.027	-36.662
29	320.129	9.580	-0.016	-0.181	-24.310
30	319.813	13.658	-0.107	0.027	-36.662
31	311.437	16.523	-0.003	-0.035	-45.564
32	311.894	6.155	10.744	54.132	-14.080
33	233.676	7.869	-0.015	-0.175	-20.406
34	233.359	11.947	-0.107	0.033	-32.758
35	233.920	4.630	10.744	54.138	-10.604
36	233.463	14.999	-0.003	-0.028	-42.087
47	339.356	2.818	-0.007	-0.092	-3.536
48	319.875	11.984	0.103	-0.077	-31.581
49	321.024	-1.471	-0.012	-0.157	9.143
50	319.875	11.984	0.103	-0.077	-31.581
53	234.570	-3.182	-0.012	-0.150	13.047
54	233.421	10.273	0.103	-0.070	-27.677

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 5.290m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 1.624m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 2.300 m Governing Load Case : # 11

Width (W_2) = 2.300 m Governing Load Case : # 11

Depth (D_2) = 0.450 m Governing Load Case : # 32

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 5.290 m^2

Final Soil Height = 0.750 m

Footing Self Weight = 57.130 kN

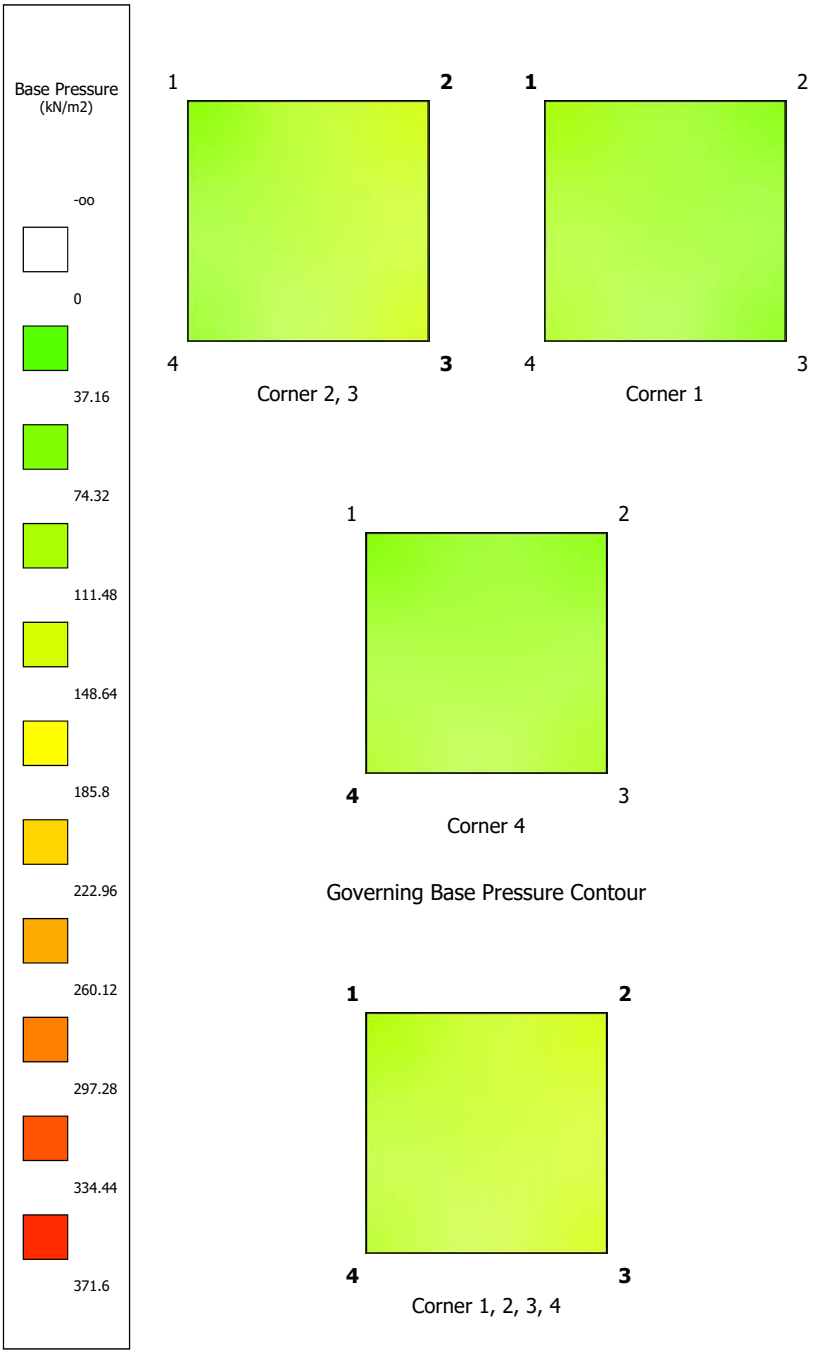
Gross Soil Bearing Capacity = 371.60 kN/m2

Soil Weight On Top Of Footing = 68.983 kN

Pressures at 4 Corners

Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner



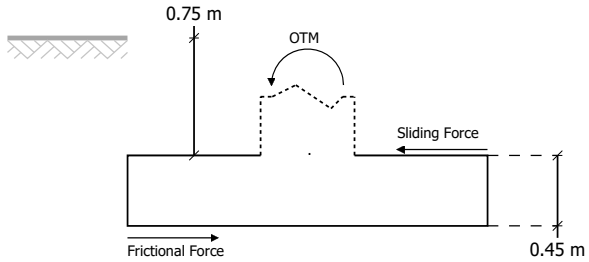
Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
45	**74.5713**	71.6328	71.5272	74.4657	0.000
26	73.1779	**114.0848**	114.0407	73.1338	0.000
26	73.1779	114.0848	**114.0407**	73.1338	0.000
14	45.7077	59.5302	100.2372	**86.4147**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
45	**74.5713**	71.6328	71.5272	74.4657
26	73.1779	**114.0848**	114.0407	73.1338
26	73.1779	114.0848	**114.0407**	73.1338
14	45.7077	59.5302	100.2372	**86.4147**

Stability Check



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
11	21.524	36397.484	30.440	1.500	13952.833	15.080	1.500
12	39.155	158797.147	55.373	1.500	20173.142	31.107	1.500
13	39.418	158540.843	55.746	1.500	20051.694	31.318	1.500
14	39.949	27.205	31.800	1.500	10.807	31.675	1.500
15	16.513	88303.770	23.353	1.500	15403.416	11.281	1.500
16	30.591	26852.299	43.262	1.500	4679.260	22.905	1.500
17	23.951	4157.549	33.871	1.500	22755.724	17.186	1.500
18	29.246	15742.329	41.360	1.500	2857.416	20.933	1.500
19	19.499	2237.423	27.574	1.500	32804.252	13.302	1.500
20	14.432	83141.304	20.409	1.500	16413.494	9.594	1.500
21	48.397	19.877	26.002	1.500	7.908	38.318	1.500
22	28.638	20430.757	40.501	1.500	3617.291	21.094	1.500
23	21.095	3039.508	29.833	1.500	23447.166	14.832	1.500
26	19.566	33446.211	27.671	1.500	12789.360	13.730	1.500
39	122.515	34062.840	173.261	1.500	5343.435	229.248	1.500
40	26.292	4448.412	37.182	1.500	18799.703	19.152	1.500
41	83.257	20566.948	117.742	1.500	3336.058	38.085	1.500
42	22.587	2322.080	31.941	1.500	20880.529	15.647	1.500
45	874.091	26267.571	1235.467	1.500	4170.337	149.156	1.500
46	23.651	3204.795	33.446	1.500	18055.263	16.893	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 20

Governing Disturbing Force : 10.347kN

Governing Restoring Force : 149.322kN

Minimum Sliding Ratio for the Critical Load Case : 14.432

Critical Load Case for Overturning about X-Direction : 21
Governing Overturning Moment : 41.282kNm
Governing Resisting Moment : 326.458kNm
Minimum Overturning Ratio for the Critical Load Case : 7.908

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 21
Governing Disturbing Force : 7.521kN
Governing Restoring Force : 149.491kN
Minimum Sliding Ratio for the Critical Load Case : 19.877
Critical Load Case for Overturning about Z-Direction : 20
Governing Overturning Moment : -33.769kNm
Governing Resisting Moment : 323.994kNm
Minimum Overturning Ratio for the Critical Load Case : 9.594

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

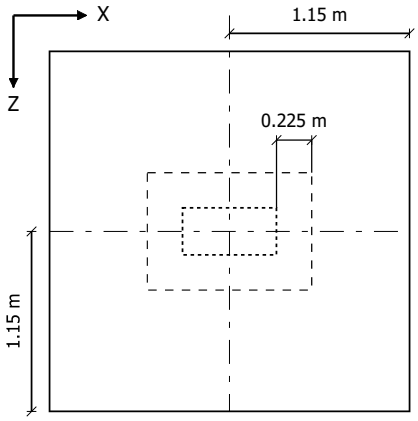
Critical Load Case for Sliding along Resultant Direction : 20
Governing Disturbing Force : 10.347kN
Governing Restoring Force : 211.173kN
Minimum Sliding Ratio for the Critical Load Case : 20.409

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.450m
Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.394m
For rectangular column, β_c = B_{col} / D_{col} = 2.000

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 430.618kN, Load Case # 25

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 3.376m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2613.679kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 4357.164kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2613.679kN

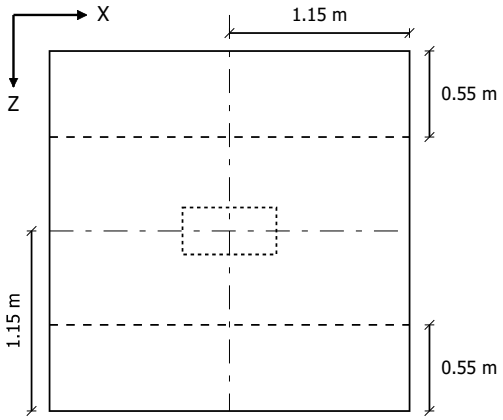
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1960.259kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 890.323kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{zd}$ = 0.606m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

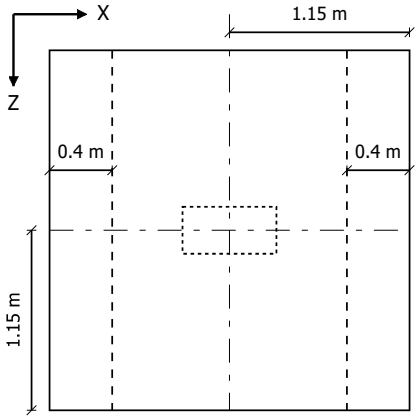
From above calculations, 0.75 X V_c = 667.742 kN

Critical load case for V_{ux} is # 32 $V_{ux} = V_{ux}|_{x=D_x}$ = 145.257 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 890.323 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.456 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

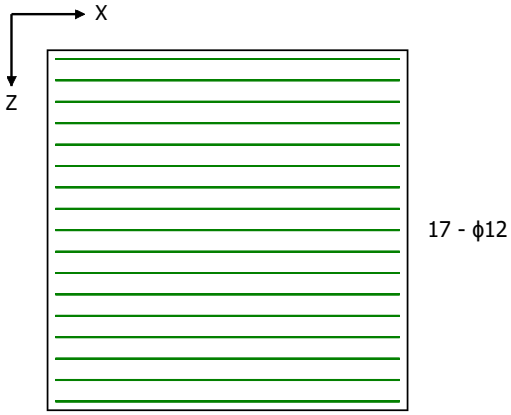
From above calculations, 0.75 X V_c = 667.742 kN

Critical load case for V_{uz} is # 25 $V_{uz} = V_{uz}|_{z=D_z} =$ 115.376 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 25

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.394 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03374
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02530
From ACI Cl. 7.12.2, ρ_{min} =		0.00178
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	13.950

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	1.450 m
Ultimate moment,	$M_u\|_{z=D_z}$ =	90.600 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	100.667 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00068
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00060
Since	$\rho = \rho_{min}$	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1846.023 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 100.000mm

Selected spacing (S) = 136.750mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

According to ACI 318 Clause No- 10.6.4 1st

Max spacing for Cracking Consideration = 254.793mm

Safe for Cracking Aspect.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 135mm o.c.

Required development length for bars =	$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f'_c}}$ =	=0.406	m
Available development length for bars, D_L =	$0.5 \times (L - D_{col}) \cdot - C_{cover}$ =	0.800	m
Try bar size # 12	Area of one bar =	113.097	mm2
Number of bars required, N_{bar} =	$\frac{A_s}{A_{bar}}$ =	17	

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	1922.641	mm2
d_{eff} =	D - C_{cover} - 0.5 X (dia. of one bar) =	0.394	m
Reinforcement ratio, ρ =	$\frac{A_{s_total}}{(d_{eff} \times W)}$ =	0.00212	

From ACI Cl.7.6.1, minimum req'd clear distance between bars

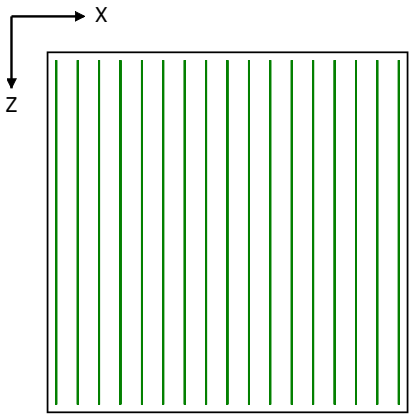
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 100.000mm

Provided Steel Area / Required Steel Area = 1.042

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



17 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 32

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =	0.382	m
Factor β_1 from ACI Cl.10.2.7.3 =	0.796	

From ACI Cl. 10.3.2, ρ_{bal} = $0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$ 0.03374

From ACI Cl. 10.3.3, ρ_{max} = $0.75 \times \rho_{bal}$ = 0.02530

From ACI Cl.7.12.2, ρ_{min} = 0.00185

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{\left(0.85 \times F_c'\right)}$ = 13.950

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 1.300 m

Ultimate moment, $M_u\big|_{x=D_x}$ = 118.982 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 132.202 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00096

(Based on gross depth) ρ x d_{eff} / Depth = 0.00081

Since ρ= ρ_{min} ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 1917.814 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 100.000mm

Selected spacing (S) = 136.750mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

According to ACI 318 Clause No- 10.6.4 3rd

Max spacing for Cracking Consideration = 254.793mm

Safe for Cracking Aspect.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 135mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.406 m

Available development length for bars, D_L = $0.5 \times \left(L - D_{col}\right) - C_{cover}$ = 0.950 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 17

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1922.641 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.382 m

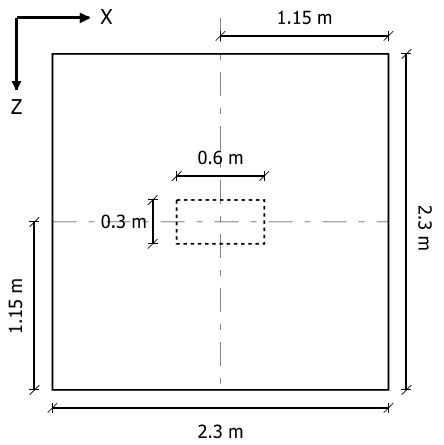
Reinforcement ratio, ρ = $\frac{A_{s_total}}{\left(d_{eff} \times W\right)}$ = 0.00219

From ACI Cl.7.6.1, minimum req'd clear distance between bars

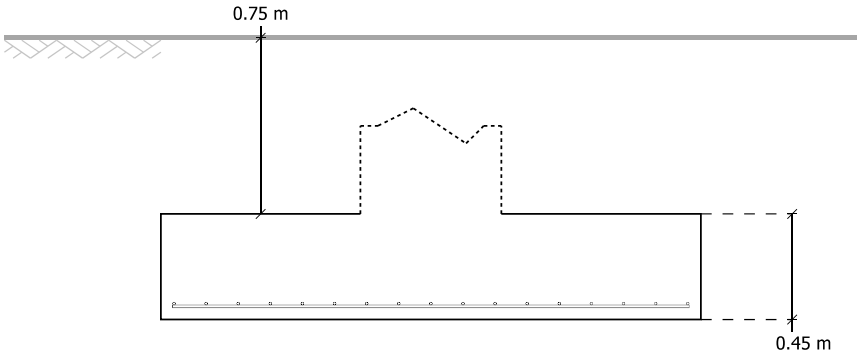
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 100.000mm

Provided Steel Area / Required Steel Area = 1.003

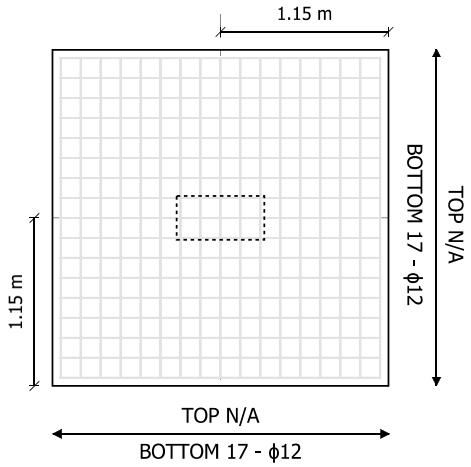
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

	Dimensions (m)			Reinforcement			
Foundation	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
	2.3 m	2.3 m	0.45 m	N/A	N/A	17 - ϕ12	17 - ϕ12
Pedestal	l	b	h	Main Reinforcement		Trans Reinforcement	
	N/A	N/A	N/A	N/A		N/A	

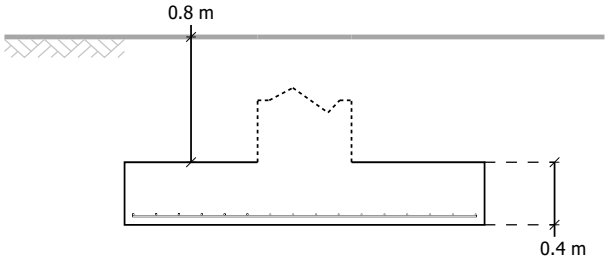
Print Calculation Sheet

Isolated Footing Design(ACI 318-08)

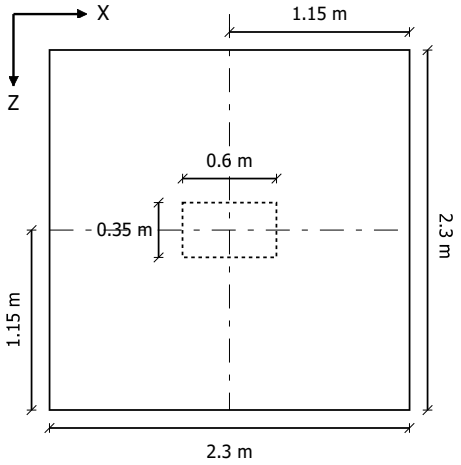
Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
148	1	2.300m	2.300m	0.400m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
148	16 - ϕ12	16 - ϕ12	N/A	N/A	N/A	N/A

FOUNDATION-F2



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 2300.000mm
Footing Width - Z (Fw) : 2300.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.600m

Column Width - Z (B_{col}) : 0.350m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 415.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #12
Pedestal Maximum Bar Size : #22
Minimum Bar Spacing : 100.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 800.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.530
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 2.300m

Initial Width (W_o) = 2.300m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
11	SLS: (DL+T)	1.00	1.00	1.00
12	SLS: (D+LR)	1.00	1.00	1.00
13	SLS: (D+0.75LR)	1.00	1.00	1.00
14	SLS: (D+0.7EZ)	1.00	1.00	1.00
15	SLS: (D+0.7EX)	1.00	1.00	1.00
16	SLS: (D+0.75WX1+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ1+0.75LR)	1.00	1.00	1.00
18	SLS: (0.6D+WX1)	1.00	1.00	1.00
19	SLS: (0.6D+WZ1)	1.00	1.00	1.00
20	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
21	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
22	SLS: (D+WX1)	1.00	1.00	1.00
23	SLS: (D+WZ1)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
39	SLS: (D+0.75WX2+0.75LR)	1.00	1.00	1.00
40	SLS: (D+0.75WZ2+0.75LR)	1.00	1.00	1.00
41	SLS: (0.6D+WX2)	1.00	1.00	1.00
42	SLS: (0.6D+WZ2)	1.00	1.00	1.00
45	SLS: (D+WX2)	1.00	1.00	1.00
46	SLS: (D+WZ2)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
24	ULS: (1.4 DL)	1.00	1.00	1.00
25	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+0.80WX1+1.6LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+1.6WX1+0.5LR)	1.00	1.00	1.00
30	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
31	ULS: (1.2D+EX)	1.00	1.00	1.00
32	ULS: (1.2D+EZ)	1.00	1.00	1.00
33	ULS: (0.90D+1.6WX1)	1.00	1.00	1.00
34	ULS: (0.90D+1.6WZ1)	1.00	1.00	1.00
35	ULS: (0.90D+EZ)	1.00	1.00	1.00
36	ULS: (0.90D+EX)	1.00	1.00	1.00
47	ULS: (1.2D+0.80WX2+1.6LR)	1.00	1.00	1.00
48	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
49	ULS: (1.2D+1.6WX2+0.5LR)	1.00	1.00	1.00
50	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
53	ULS: (0.90D+1.6WX2)	1.00	1.00	1.00
54	ULS: (0.90D+1.6WZ2)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
11	260.478	0.783	-0.083	4.891	5.346
12	273.168	1.024	-0.071	4.555	-0.206
13	271.011	1.061	-0.071	4.554	-0.245
14	264.533	1.153	4.107	24.841	-0.323
15	264.465	5.403	-0.061	4.632	-8.734
16	270.965	2.167	31.552	100.519	-2.206
17	270.954	2.760	26.294	85.128	-3.237
18	158.664	2.177	42.121	130.684	-2.830
19	158.649	2.969	35.111	110.162	-4.206
20	158.647	4.934	-0.033	2.812	-8.590

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
21	158.716	0.685	4.136	23.020	-0.179
22	264.481	2.646	42.093	132.504	-2.974
23	264.466	3.438	35.082	111.983	-4.349
26	368.982	1.021	-0.117	6.849	7.561
39	271.045	-0.871	31.544	100.463	3.177
40	270.980	2.912	-33.293	-96.785	-3.542
41	158.770	-1.873	42.111	130.609	4.346
42	158.684	3.172	-44.338	-132.388	-4.612
45	264.587	-1.404	42.082	132.430	4.203
46	264.501	3.641	-44.367	-130.568	-4.756

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
24	370.360	1.641	-0.100	6.372	-0.503
25	368.982	1.021	-0.117	6.849	7.561
26	368.982	1.021	-0.117	6.849	7.561
27	331.202	2.349	33.646	107.830	-2.278
28	321.641	4.958	56.160	177.354	-6.739
29	321.665	3.691	67.377	210.188	-4.538
30	321.641	4.958	56.160	177.354	-6.739
31	317.340	7.450	-0.071	5.577	-12.395
32	317.438	1.380	5.884	34.447	-0.379
33	237.990	3.414	67.398	208.821	-4.506
34	237.966	4.680	56.181	175.986	-6.708
35	238.075	1.028	5.905	33.081	-0.271
36	237.977	7.099	-0.050	4.212	-12.287
47	331.286	-0.891	33.638	107.770	3.463
48	321.696	5.282	-70.958	-210.727	-7.389
49	321.835	-2.789	67.360	210.069	6.944
50	321.696	5.282	-70.958	-210.727	-7.389
53	238.160	-3.067	67.381	208.701	6.976
54	238.021	5.004	-70.937	-212.094	-7.358

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 5.290m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 2.130m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 2.300 m Governing Load Case : # 11

Width (W_2) = 2.300 m Governing Load Case : # 11

Depth (D_2) = 0.400 m Governing Load Case : # 48

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 5.290 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 50.782 kN

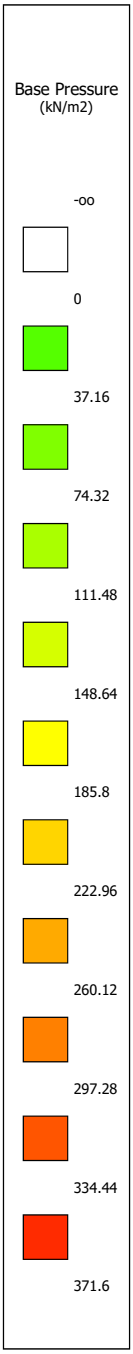
Gross Soil Bearing Capacity = 371.60 kN/m2

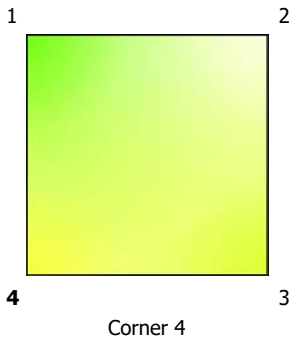
Soil Weight On Top Of Footing = 73.149 kN

Pressures at 4 Corners

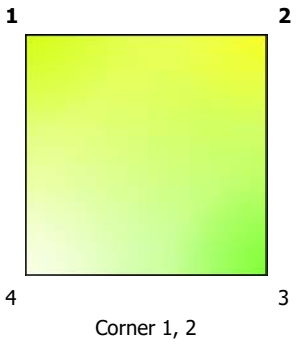
Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner

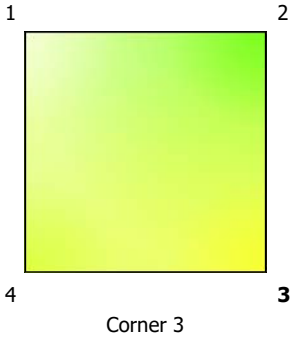




Corner 4

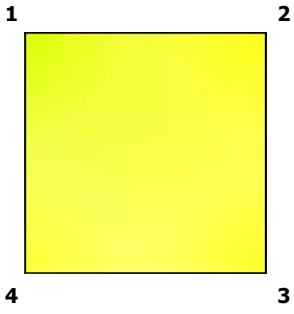


Corner 1, 2



Corner 3

Governing Base Pressure Contour



Corner 1, 2, 3, 4

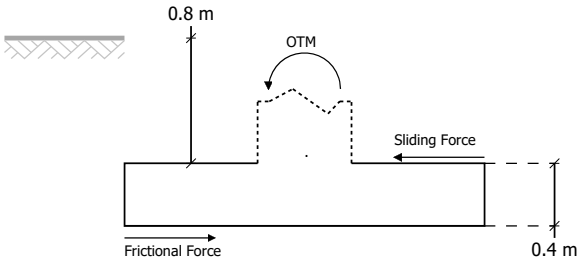
Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
46	**143.5049**	149.6316	3.3504	-2.7762	0.000
46	143.5049	**149.6316**	3.3504	-2.7762	0.000
22	-2.2115	1.7654	**149.0594**	145.0825	0.000
45	2.1855	-2.5135	144.7025	**149.4015**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
46	**143.5134**	149.6404	0.0000	0.0000
46	143.5134	**149.6404**	0.0000	0.0000
22	0.0000	0.0000	**149.0722**	145.0952
45	0.0000	0.0000	144.7183	**149.4174**

Stability Check



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
11	260.334	2443.184	366.095	1.500	91.518	87.934	1.500
12	205.526	2955.852	289.957	1.500	101.430	742.075	1.500
13	197.267	2935.226	278.349	1.500	100.903	679.609	1.500
14	178.490	50.127	68.250	1.500	16.961	570.296	1.500
15	38.101	3362.463	53.879	1.500	97.470	41.042	1.500
16	96.601	6.633	9.359	1.500	4.036	147.980	1.500
17	75.821	7.960	11.195	1.500	4.774	104.712	1.500
18	68.786	3.556	5.022	1.500	2.219	87.944	1.500
19	50.444	4.266	6.011	1.500	2.636	60.344	1.500
20	30.355	4599.438	42.928	1.500	116.994	30.809	1.500
21	218.833	36.220	50.535	1.500	13.273	718.595	1.500
22	77.790	4.891	6.903	1.500	3.007	110.900	1.500
23	59.876	5.868	8.259	1.500	3.564	78.110	1.500
26	255.750	2241.957	359.355	1.500	83.695	79.326	1.500
39	240.322	6.636	9.382	1.500	4.039	128.990	1.500
40	71.870	6.287	8.857	1.500	4.147	96.589	1.500
41	80.003	3.558	5.027	1.500	2.221	63.899	1.500
42	47.227	3.378	4.765	1.500	2.181	55.351	1.500
45	146.675	4.893	6.916	1.500	3.010	93.883	1.500
46	56.549	4.640	6.540	1.500	3.028	71.990	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 20

Governing Disturbing Force : 4.934kN

Governing Restoring Force : 149.767kN

Minimum Sliding Ratio for the Critical Load Case : 30.355

Critical Load Case for Overturning about X-Direction : 42
Governing Overturning Moment : -150.123kNm
Governing Resisting Moment : 327.465kNm
Minimum Overturning Ratio for the Critical Load Case : 2.181

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 42
Governing Disturbing Force : -44.338kN
Governing Restoring Force : 149.786kN
Minimum Sliding Ratio for the Critical Load Case : 3.378
Critical Load Case for Overturning about Z-Direction : 20
Governing Overturning Moment : -10.564kNm
Governing Resisting Moment : 325.457kNm
Minimum Overturning Ratio for the Critical Load Case : 30.809

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

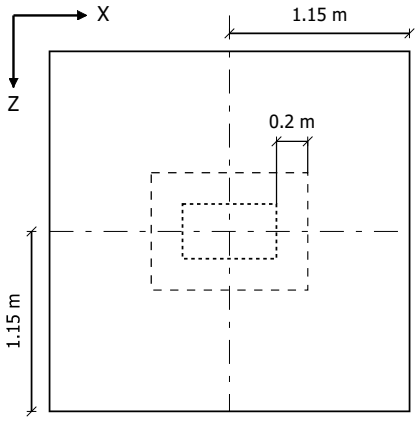
Critical Load Case for Sliding along Resultant Direction : 42
Governing Disturbing Force : 44.451kN
Governing Restoring Force : 211.830kN
Minimum Sliding Ratio for the Critical Load Case : 4.765

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m

For rectangular column, β_c = B_{col} / D_{col} = 1.714

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 499.589kN, Load Case # 54

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 3.276m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2398.933kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 3432.454kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2214.399kN

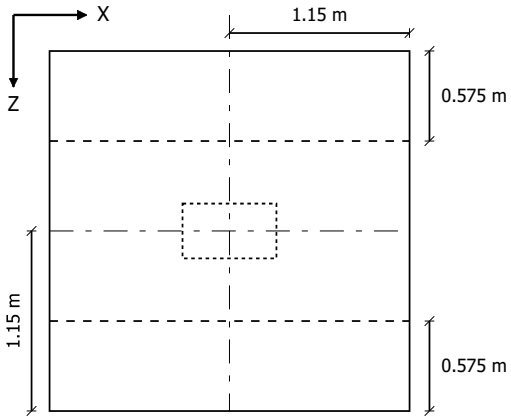
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1660.799kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 777.338kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{zd}$ = 0.631m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

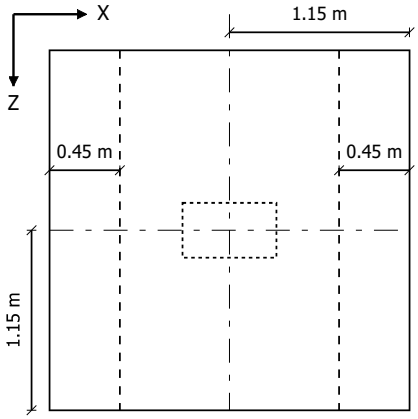
From above calculations, 0.75 X V_c = 583.003 kN

Critical load case for V_{ux} is # 48 $V_{ux} = V_{ux}|_{x=D_x}$ = 246.439 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 777.338 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.506 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

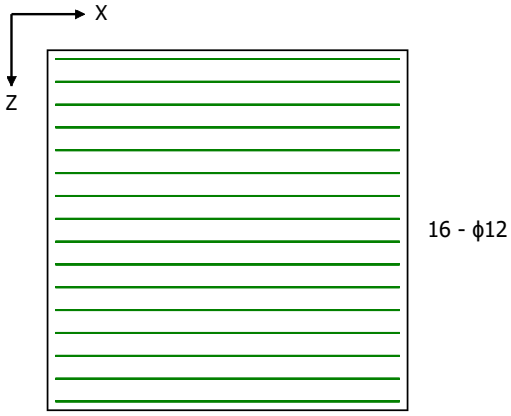
From above calculations, 0.75 X V_c = 583.003 kN

Critical load case for V_{uz} is # 25 $V_{uz} = V_{uz}|_{z=D_z} =$ 111.643 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 48

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03374
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02530
From ACI Cl. 7.12.2, ρ_{min} =		0.00173
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	13.950

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	1.450 m
Ultimate moment,	$M_u\|_{z=D_z}$ =	79.709 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	88.565 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00079
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00068
Since	$\rho = \rho_{min}$	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1593.594 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 100.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

According to ACI 318 Clause No- 10.6.4 1st

Max spacing for Cracking Consideration = 254.793mm

Safe for Cracking Aspect.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f'_c}}$ = =0.406 m

Available development length for bars, D_L = $0.5 \times (L - D_{col}) \cdot - C_{cover}$ = 0.800 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 16

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1809.545 mm2

d_{eff} = D - C_{cover} - 0.5 X (dia. of one bar) = 0.344 m

Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00229

From ACI Cl.7.6.1, minimum req'd clear distance between bars

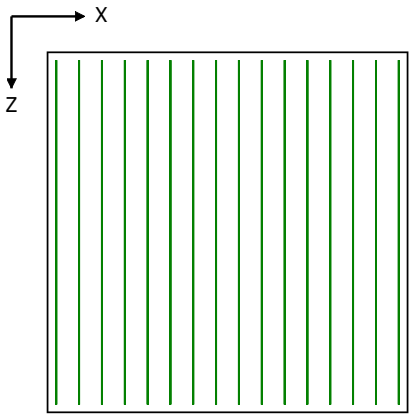
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 100.000mm

Provided Steel Area / Required Steel Area = 1.136

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



16 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 29

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} = 0.332 m

Factor β_1 from ACI Cl.10.2.7.3 = 0.796

From ACI Cl. 10.3.2, ρ_{bal} = $0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$ = 0.03374

From ACI Cl. 10.3.3, ρ_{max} = $0.75 \times \rho_{bal}$ = 0.02530

From ACI Cl.7.12.2, ρ_{min} = 0.00181

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{\left(0.85 \times F_c'\right)}$ = 13.950

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 1.325 m

Ultimate moment, $M_u\big|_{x=D_x}$ = 198.590 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 220.656 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00213

(Based on gross depth) ρ x d_{eff} / Depth = 0.00177

Since ρ= ρ_{min} ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 1665.442 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 100.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

According to ACI 318 Clause No- 10.6.4 3rd

Max spacing for Cracking Consideration = 254.793mm

Safe for Cracking Aspect.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.406 m

Available development length for bars, D_L = $0.5 \times \left(L - D_{col}\right) - C_{cover}$ = 0.925 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 16

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1809.545 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.332 m

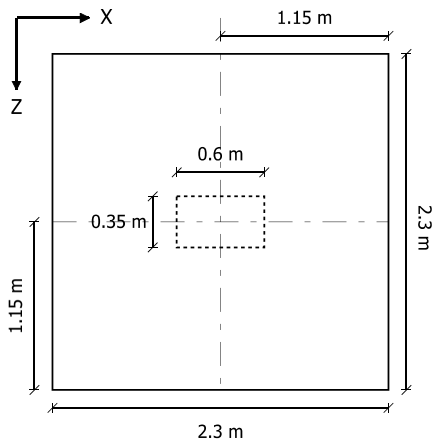
Reinforcement ratio, ρ = $\frac{A_{s_total}}{\left(d_{eff} \times W\right)}$ = 0.00237

From ACI Cl.7.6.1, minimum req'd clear distance between bars

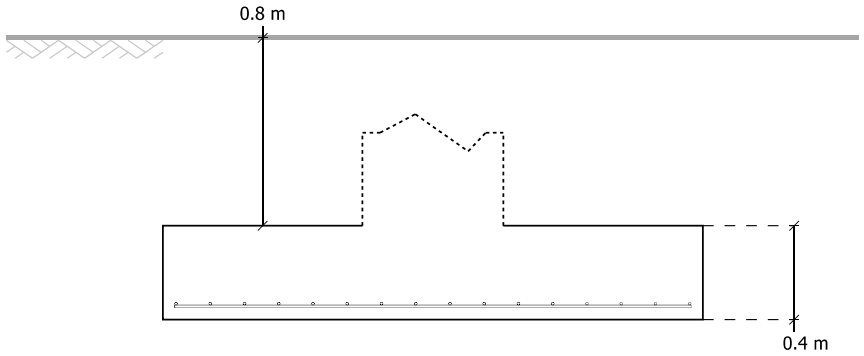
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 100.000mm

Provided Steel Area / Required Steel Area = 1.087

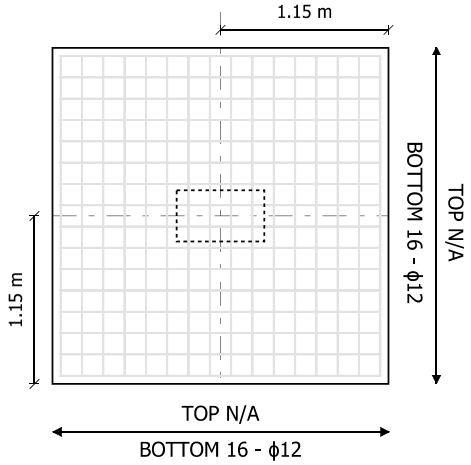
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

	Dimensions (m)			Reinforcement			
Foundation	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
	2.3 m	2.3 m	0.4 m	N/A	N/A	16 - ϕ12	16 - ϕ12
Pedestal	l	b	h	Main Reinforcement		Trans Reinforcement	
	N/A	N/A	N/A	N/A		N/A	

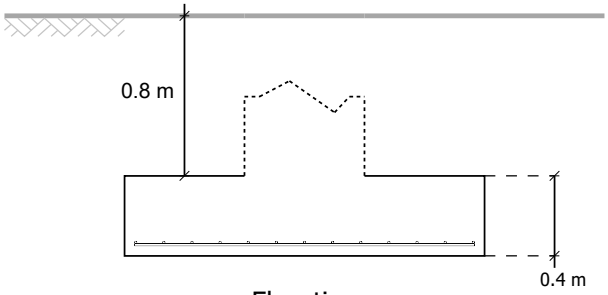
Print Calculation Sheet

Isolated Footing Design(ACI 318-08)

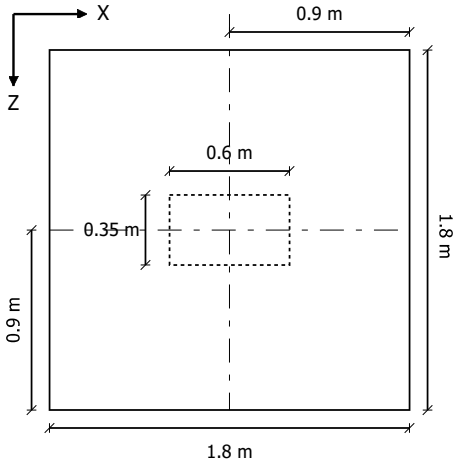
Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
167	1	1.800m	1.800m	0.400m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
167	13 - ϕ12	13 - ϕ12	N/A	N/A	N/A	N/A

FOUNDATION-F3



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 1800.000mm
Footing Width - Z (Fw) : 1800.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.600m

Column Width - Z (B_{col}) : 0.350m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 415.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #12
Pedestal Maximum Bar Size : #22
Minimum Bar Spacing : 100.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 800.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.530
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 1.800m

Initial Width (W_o) = 1.800m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
11	SLS: (DL+T)	1.00	1.00	1.00
12	SLS: (D+LR)	1.00	1.00	1.00
13	SLS: (D+0.75LR)	1.00	1.00	1.00
14	SLS: (D+0.7EZ)	1.00	1.00	1.00
15	SLS: (D+0.7EX)	1.00	1.00	1.00
16	SLS: (D+0.75WX1+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ1+0.75LR)	1.00	1.00	1.00
18	SLS: (0.6D+WX1)	1.00	1.00	1.00
19	SLS: (0.6D+WZ1)	1.00	1.00	1.00
20	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
21	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
22	SLS: (D+WX1)	1.00	1.00	1.00
23	SLS: (D+WZ1)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
39	SLS: (D+0.75WX2+0.75LR)	1.00	1.00	1.00
40	SLS: (D+0.75WZ2+0.75LR)	1.00	1.00	1.00
41	SLS: (0.6D+WX2)	1.00	1.00	1.00
42	SLS: (0.6D+WZ2)	1.00	1.00	1.00
45	SLS: (D+WX2)	1.00	1.00	1.00
46	SLS: (D+WZ2)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
24	ULS: (1.4 DL)	1.00	1.00	1.00
25	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+0.80WX1+1.6LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+1.6WX1+0.5LR)	1.00	1.00	1.00
30	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
31	ULS: (1.2D+EX)	1.00	1.00	1.00
32	ULS: (1.2D+EZ)	1.00	1.00	1.00
33	ULS: (0.90D+1.6WX1)	1.00	1.00	1.00
34	ULS: (0.90D+1.6WZ1)	1.00	1.00	1.00
35	ULS: (0.90D+EZ)	1.00	1.00	1.00
36	ULS: (0.90D+EX)	1.00	1.00	1.00
47	ULS: (1.2D+0.80WX2+1.6LR)	1.00	1.00	1.00
48	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
49	ULS: (1.2D+1.6WX2+0.5LR)	1.00	1.00	1.00
50	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
53	ULS: (0.90D+1.6WX2)	1.00	1.00	1.00
54	ULS: (0.90D+1.6WZ2)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
11	220.679	16.879	-33.915	-9.553	-27.332
12	214.436	22.655	-38.683	-10.519	-3.986
13	213.272	22.792	-38.683	-10.518	-3.842
14	204.721	22.954	-34.312	-3.767	-2.849
15	211.822	26.110	-38.683	-10.513	-11.240
16	216.178	27.218	-38.784	-11.187	-11.232
17	205.059	26.822	-27.516	3.926	-8.649
18	129.743	19.823	-23.343	-7.202	-11.899
19	114.917	19.295	-8.320	12.949	-8.454
20	127.909	16.829	-23.211	-6.307	-9.877

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
21	120.809	13.673	-18.839	0.439	-1.485
22	213.655	29.104	-38.816	-11.408	-13.262
23	198.829	28.576	-23.793	8.743	-9.818
26	311.278	23.357	-47.482	-13.375	-38.554
39	214.309	18.473	-38.787	-11.197	1.335
40	223.186	28.713	-49.823	-24.805	-13.112
41	127.251	8.163	-23.348	-7.215	4.858
42	139.087	21.817	-38.062	-25.359	-14.405
45	211.163	17.445	-38.820	-11.422	3.494
46	222.999	31.098	-53.534	-29.566	-15.769

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
24	293.693	32.485	-54.154	-14.723	-4.772
25	311.278	23.357	-47.482	-13.375	-38.554
26	311.278	23.357	-47.482	-13.375	-38.554
27	262.285	31.688	-46.528	-13.336	-12.897
28	236.543	36.167	-22.597	18.194	-14.634
29	260.263	37.012	-46.634	-14.048	-20.145
30	236.543	36.167	-22.597	18.194	-14.634
31	254.652	31.997	-46.420	-12.615	-15.279
32	244.509	27.488	-40.175	-2.978	-3.290
33	195.001	30.325	-35.028	-10.892	-18.834
34	171.281	29.480	-10.992	21.350	-13.323
35	181.575	20.527	-28.571	0.177	-2.268
36	191.718	25.036	-34.816	-9.460	-14.256
47	260.292	22.360	-46.532	-13.347	0.508
48	275.214	40.202	-70.184	-43.100	-24.156
49	256.277	18.356	-46.641	-14.069	6.665
50	275.214	40.202	-70.184	-43.100	-24.156
53	191.015	11.669	-35.036	-10.913	7.977
54	209.952	33.515	-58.578	-39.944	-22.844

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 3.240m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 1.759m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.800 m Governing Load Case : # 11

Width (W_2) = 1.800 m Governing Load Case : # 11

Depth (D_2) = 0.400 m Governing Load Case : # 50

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 3.240 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 31.103 kN

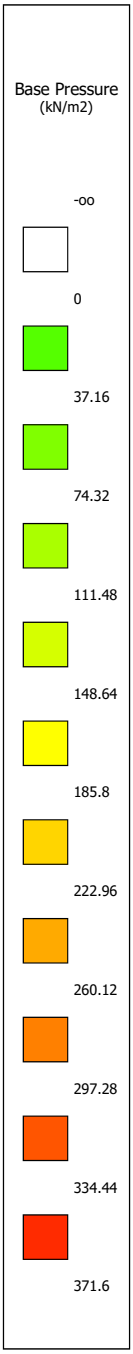
Gross Soil Bearing Capacity = 371.60 kN/m2

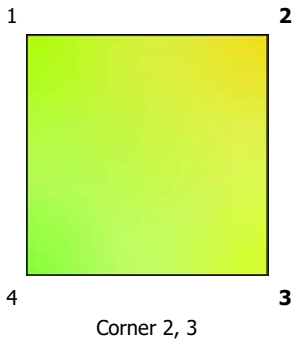
Soil Weight On Top Of Footing = 43.630 kN

Pressures at 4 Corners

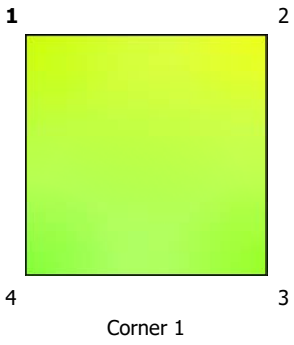
Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner

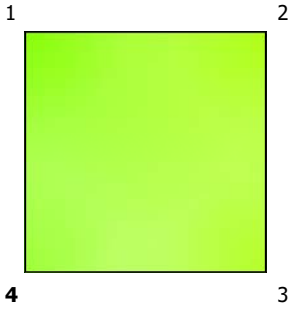




Corner 2, 3

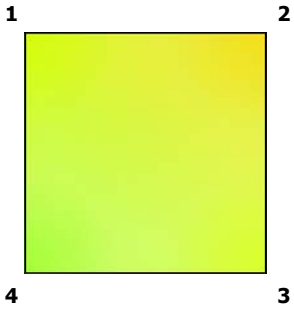


Corner 1



Corner 4

Governing Base Pressure Contour



Corner 1, 2, 3, 4

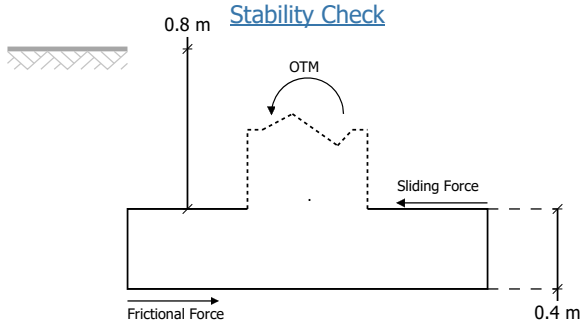
Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
46	**115.3203**	173.3626	68.4651	10.4228	0.000
26	103.1627	**201.7165**	135.1162	36.5625	0.000
26	103.1627	201.7165	**135.1162**	36.5625	0.000
23	63.3687	107.0902	105.4972	**61.7757**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
46	**115.3203**	173.3626	68.4651	10.4228
26	103.1627	**201.7165**	135.1162	36.5625
26	103.1627	201.7165	**135.1162**	36.5625
23	63.3687	107.0902	105.4972	**61.7757**

Stability Check



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
11	9.276	4.617	5.845	1.500	11.584	7.812	1.500
12	6.765	3.962	4.835	1.500	10.087	19.975	1.500
13	6.697	3.946	4.808	1.500	10.047	20.032	1.500
14	6.452	4.317	5.074	1.500	14.489	20.938	1.500
15	5.817	3.926	4.602	1.500	9.998	11.911	1.500
16	5.665	3.975	4.602	1.500	9.878	11.854	1.500
17	5.529	5.389	5.458	1.500	35.839	13.015	1.500
18	5.467	4.643	5.005	1.500	11.243	9.301	1.500
19	5.209	12.081	6.765	1.500	17.940	10.578	1.500
20	6.382	4.627	5.298	1.500	11.821	11.004	1.500
21	7.580	5.501	6.296	1.500	25.072	25.362	1.500
22	5.252	3.938	4.455	1.500	9.708	10.437	1.500
23	5.074	6.094	5.514	1.500	320.507	11.605	1.500
26	8.759	4.309	5.468	1.500	10.793	7.261	1.500
39	8.293	3.950	5.043	1.500	9.811	43.034	1.500
40	5.499	3.169	3.883	1.500	6.037	10.916	1.500
41	13.114	4.585	6.121	1.500	11.098	114.408	1.500
42	5.194	2.977	3.653	1.500	4.789	8.336	1.500
45	8.686	3.903	5.035	1.500	9.619	73.971	1.500
46	5.074	2.948	3.605	1.500	5.294	9.513	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 23

Governing Disturbing Force : 28.576kN

Governing Restoring Force : 144.988kN

Minimum Sliding Ratio for the Critical Load Case : 5.074

Critical Load Case for Overturning about X-Direction : 42
Governing Overturning Moment : -40.584kNm
Governing Resisting Moment : 194.362kNm
Minimum Overturning Ratio for the Critical Load Case : 4.789

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 46
Governing Disturbing Force : -53.534kN
Governing Restoring Force : 157.798kN
Minimum Sliding Ratio for the Critical Load Case : 2.948
Critical Load Case for Overturning about Z-Direction : 26
Governing Overturning Moment : -47.896kNm
Governing Resisting Moment : 347.793kNm
Minimum Overturning Ratio for the Critical Load Case : 7.261

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

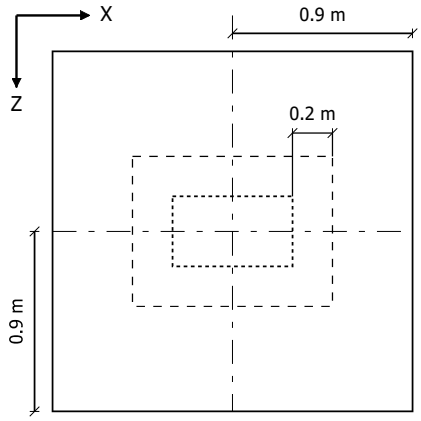
Critical Load Case for Sliding along Resultant Direction : 46
Governing Disturbing Force : 61.911kN
Governing Restoring Force : 223.160kN
Minimum Sliding Ratio for the Critical Load Case : 3.605

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m
Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m
For rectangular column, β_c = B_{col} / D_{col} = 1.714

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 307.959kN, Load Case # 25

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 3.276m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2398.933kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 3432.454kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2214.399kN

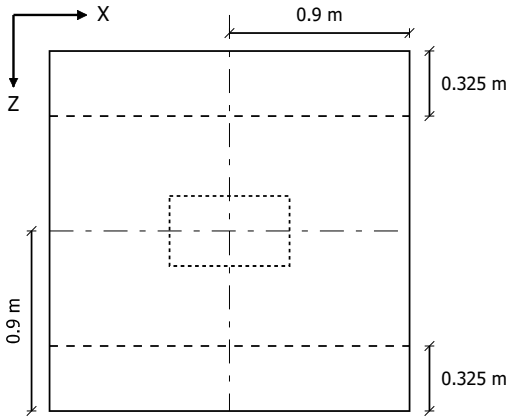
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1660.799kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 608.351kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{zd}$ = 0.381m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

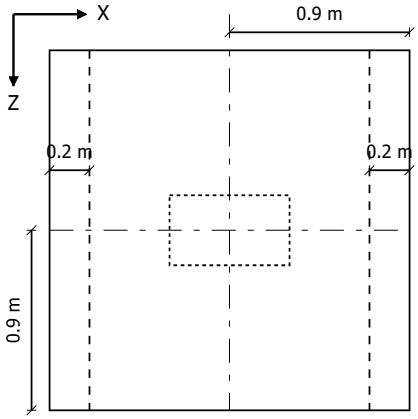
From above calculations, 0.75 X V_c = 456.264 kN

Critical load case for V_{ux} is # 50 $V_{ux} = V_{ux}|_{x=D_x}$ = 113.660 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 608.351 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.256 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

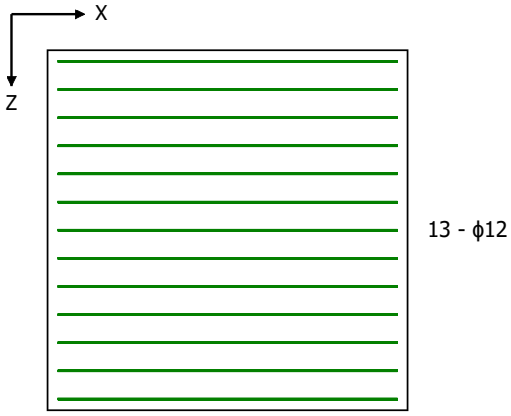
From above calculations, 0.75 X V_c = 456.264 kN

Critical load case for V_{uz} is # 25 $V_{uz} = V_{uz}|_{z=D_z} =$ 74.377 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 25

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03374
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02530
From ACI Cl. 7.12.2, ρ_{min} =		0.00185
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	13.950

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	1.200 m
Ultimate moment,	$M_u\|_{z=D_z}$ =	51.041 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	56.712 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00064
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00055
Since	$\rho = \rho_{min}$	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1333.463 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 100.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

According to ACI 318 Clause No- 10.6.4 1st

Max spacing for Cracking Consideration = 254.793mm

Safe for Cracking Aspect.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f'_c}}$ = =0.406 m

Available development length for bars, D_L = $0.5 \times (L - D_{col}) \cdot - C_{cover}$ = 0.550 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 13

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1470.255 mm2

d_{eff} = D - C_{cover} - 0.5 X (dia. of one bar) = 0.344 m

Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00237

From ACI Cl.7.6.1, minimum req'd clear distance between bars

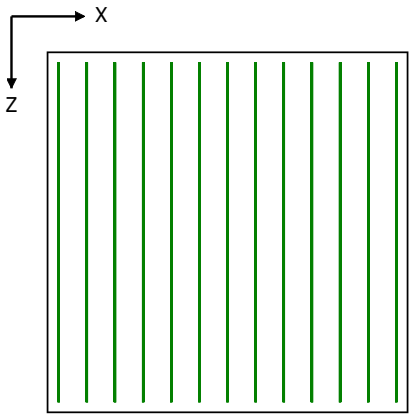
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 100.000mm

Provided Steel Area / Required Steel Area = 1.103

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



13 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 50

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} = 0.332 m

Factor β_1 from ACI Cl.10.2.7.3 = 0.796

From ACI Cl. 10.3.2, ρ_{bal} = $0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$ 0.03374

From ACI Cl. 10.3.3, ρ_{max} = $0.75 \times \rho_{bal}$ = 0.02530

From ACI Cl.7.12.2, ρ_{min} = 0.00191

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{\left(0.85 \times F_c'\right)}$ = 13.950

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 0.725 m

Ultimate moment, $M_u\big|_{x=D_x}$ = 76.424 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 84.916 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00104

(Based on gross depth) ρ x d_{eff} / Depth = 0.00086

Since ρ= ρ_{min} ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 1371.976 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 100.000mm

Selected spacing (S) = 140.667mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

According to ACI 318 Clause No- 10.6.4 3rd

Max spacing for Cracking Consideration = 254.793mm

Safe for Cracking Aspect.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 140mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.406 m

Available development length for bars, D_L = $0.5 \times \left(L - D_{col}\right) - C_{cover}$ = 0.675 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 13

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1470.255 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.332 m

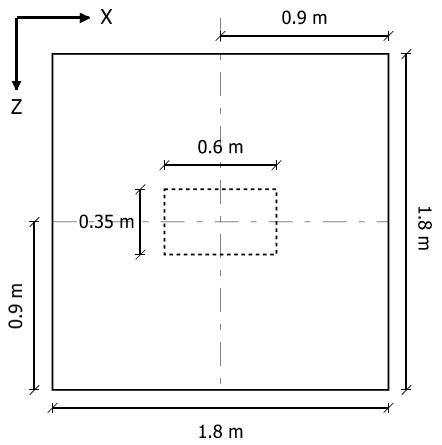
Reinforcement ratio, ρ = $\frac{A_{s_total}}{\left(d_{eff} \times W\right)}$ = 0.00246

From ACI Cl.7.6.1, minimum req'd clear distance between bars

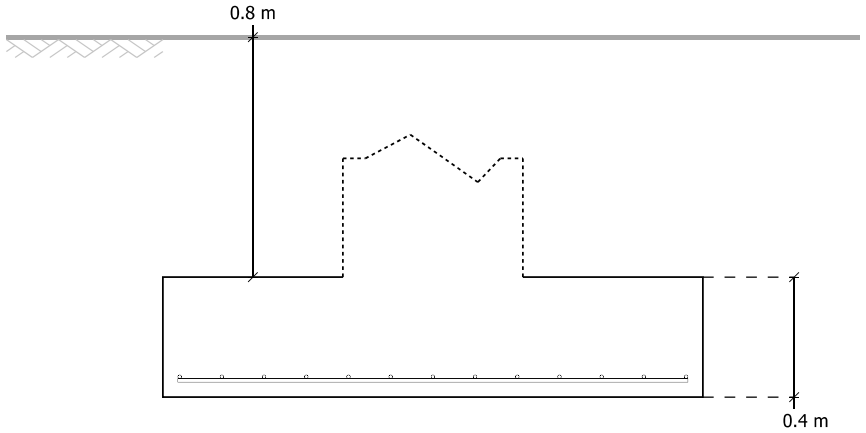
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 100.000mm

Provided Steel Area / Required Steel Area = 1.072

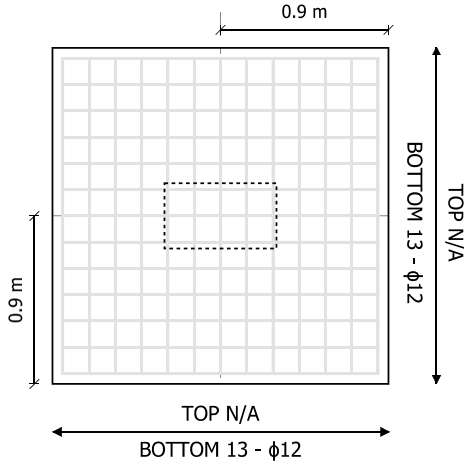
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	1.8 m	1.8 m	0.4 m	N/A	N/A	13 - ϕ12	13 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

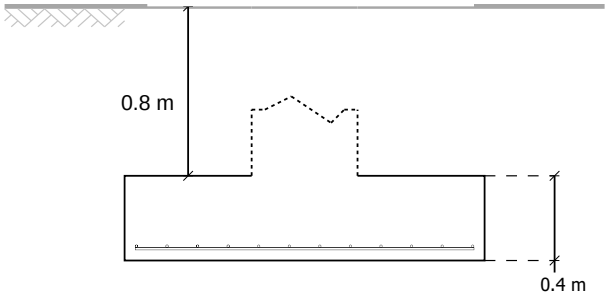
Print Calculation Sheet

Isolated Footing Design(ACI 318-08)

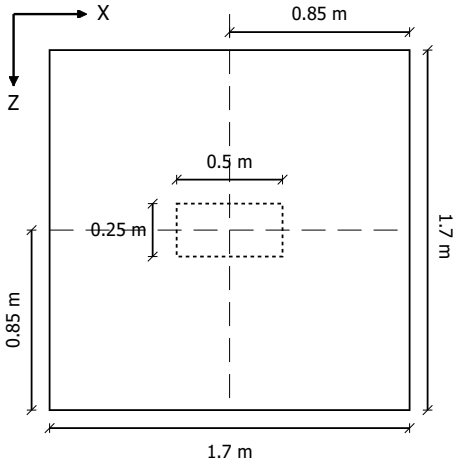
Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
245	1	1.700m	1.700m	0.400m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
245	12 - ϕ12	12 - ϕ12	N/A	N/A	N/A	N/A

FOUNDATION-F4



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 1700.000mm
Footing Width - Z (Fw) : 1700.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.500m

Column Width - Z (B_{col}) : 0.250m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 415.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #12
Pedestal Maximum Bar Size : #22
Minimum Bar Spacing : 100.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 800.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.530
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 1.700m

Initial Width (W_o) = 1.700m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
11	SLS: (DL+T)	1.00	1.00	1.00
12	SLS: (D+LR)	1.00	1.00	1.00
13	SLS: (D+0.75LR)	1.00	1.00	1.00
14	SLS: (D+0.7EZ)	1.00	1.00	1.00
15	SLS: (D+0.7EX)	1.00	1.00	1.00
16	SLS: (D+0.75WX1+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ1+0.75LR)	1.00	1.00	1.00
18	SLS: (0.6D+WX1)	1.00	1.00	1.00
19	SLS: (0.6D+WZ1)	1.00	1.00	1.00
20	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
21	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
22	SLS: (D+WX1)	1.00	1.00	1.00
23	SLS: (D+WZ1)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
39	SLS: (D+0.75WX2+0.75LR)	1.00	1.00	1.00
40	SLS: (D+0.75WZ2+0.75LR)	1.00	1.00	1.00
41	SLS: (0.6D+WX2)	1.00	1.00	1.00
42	SLS: (0.6D+WZ2)	1.00	1.00	1.00
45	SLS: (D+WX2)	1.00	1.00	1.00
46	SLS: (D+WZ2)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
24	ULS: (1.4 DL)	1.00	1.00	1.00
25	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+0.80WX1+1.6LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+1.6WX1+0.5LR)	1.00	1.00	1.00
30	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
31	ULS: (1.2D+EX)	1.00	1.00	1.00
32	ULS: (1.2D+EZ)	1.00	1.00	1.00
33	ULS: (0.90D+1.6WX1)	1.00	1.00	1.00
34	ULS: (0.90D+1.6WZ1)	1.00	1.00	1.00
35	ULS: (0.90D+EZ)	1.00	1.00	1.00
36	ULS: (0.90D+EX)	1.00	1.00	1.00
47	ULS: (1.2D+0.80WX2+1.6LR)	1.00	1.00	1.00
48	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
49	ULS: (1.2D+1.6WX2+0.5LR)	1.00	1.00	1.00
50	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
53	ULS: (0.90D+1.6WX2)	1.00	1.00	1.00
54	ULS: (0.90D+1.6WZ2)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
11	128.100	-13.820	-6.797	4.847	1.615
12	131.349	-18.487	-6.229	5.233	5.840
13	130.777	-18.568	-6.184	5.204	5.855
14	121.178	-18.684	-0.544	9.444	5.645
15	127.212	-17.188	-6.076	5.299	3.383
16	138.838	-12.268	-21.496	-6.095	-2.324
17	118.193	-23.708	11.574	18.469	12.314
18	88.185	-2.886	-24.045	-11.995	-7.365
19	60.658	-18.139	20.048	20.757	12.152
20	75.588	-9.664	-3.657	3.252	1.023

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
21	69.554	-11.160	1.875	7.397	3.285
22	139.809	-10.409	-26.464	-9.949	-5.005
23	112.282	-25.663	17.629	22.804	14.512
26	180.485	-19.187	-9.606	6.845	2.231
39	143.082	-18.641	-21.142	-6.064	5.414
40	145.414	-21.086	-20.183	-5.396	8.653
41	93.842	-11.383	-23.572	-11.954	2.952
42	96.953	-14.643	-22.295	-11.063	7.270
45	145.466	-18.907	-25.991	-9.907	5.313
46	148.577	-22.166	-24.714	-9.017	9.631

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
24	180.684	-26.332	-8.467	7.164	8.261
25	180.485	-19.187	-9.606	6.845	2.231
26	180.485	-19.187	-9.606	6.845	2.231
27	167.134	-15.336	-23.880	-5.726	-1.740
28	129.171	-33.376	30.535	34.497	20.829
29	173.214	-8.970	-40.014	-17.906	-10.398
30	129.171	-33.376	30.535	34.497	20.829
31	152.233	-20.255	-7.297	6.400	3.485
32	143.613	-22.392	0.605	12.321	6.715
33	133.352	-3.489	-38.109	-19.499	-12.138
34	89.309	-27.894	32.440	32.904	19.089
35	104.895	-16.749	2.420	10.786	4.945
36	113.515	-14.612	-5.483	4.865	1.715
47	171.660	-22.134	-23.502	-5.693	6.514
48	187.243	-27.782	-37.214	-16.415	13.019
49	182.267	-22.567	-39.258	-17.840	6.110
50	187.243	-27.782	-37.214	-16.415	13.019
53	142.404	-17.085	-37.353	-19.433	4.370
54	147.380	-22.300	-35.309	-18.009	11.279

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 2.890m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 1.003m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.700 m Governing Load Case : # 11

Width (W_2) = 1.700 m Governing Load Case : # 11

Depth (D_2) = 0.400 m Governing Load Case : # 50

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 2.890 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 27.743 kN

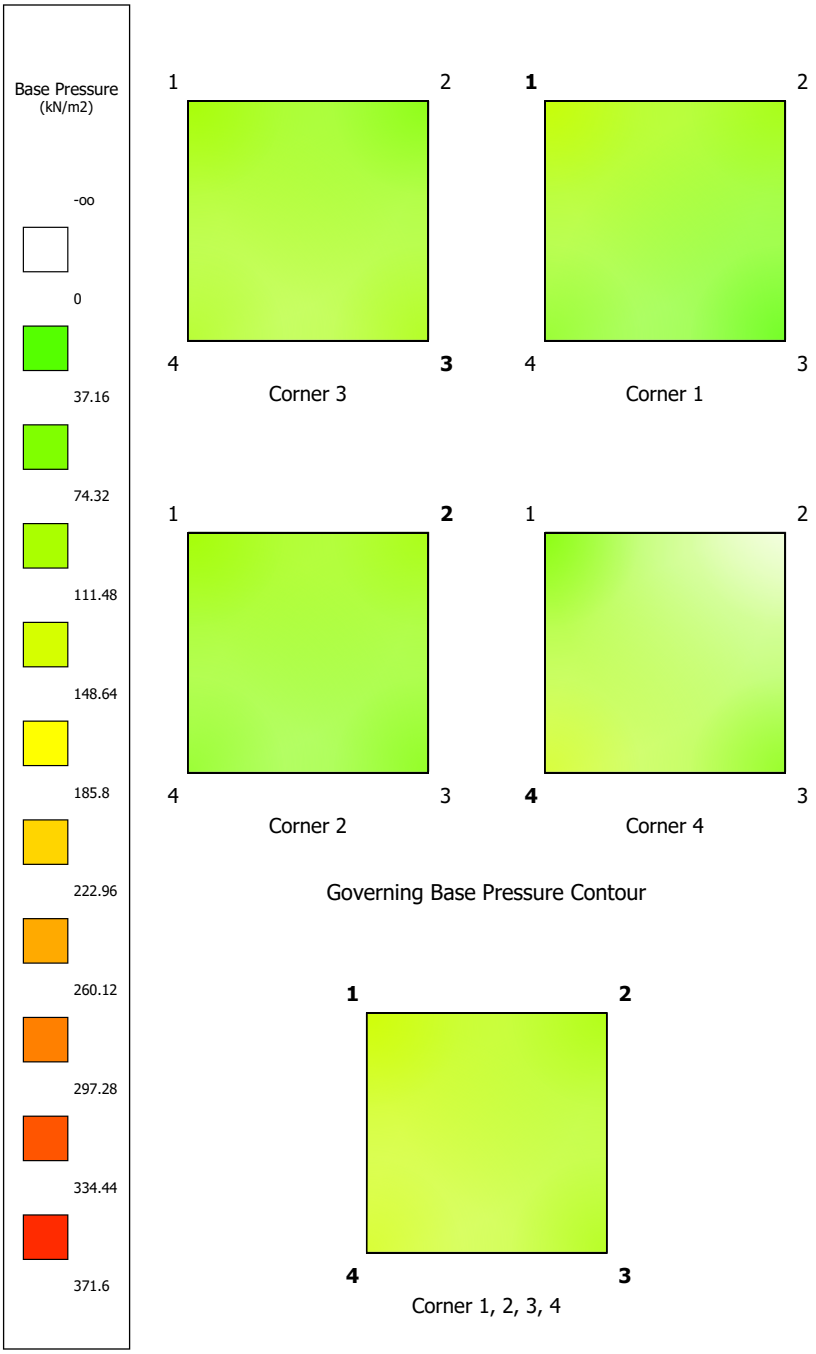
Gross Soil Bearing Capacity = 371.60 kN/m2

Soil Weight On Top Of Footing = 39.815 kN

Pressures at 4 Corners

Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner

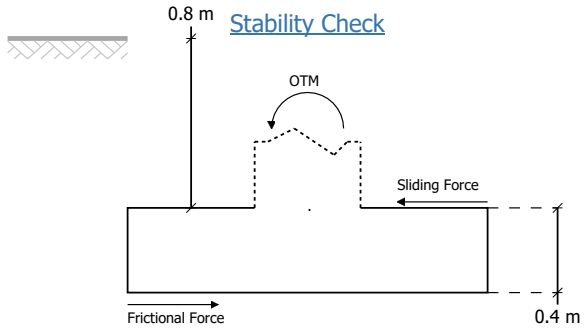


Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
46	**120.4618**	75.2816	29.1120	74.2922	0.000
22	95.8036	**97.8576**	47.7024	45.6485	0.000
26	94.2592	70.0640	**77.3967**	101.5920	0.000
23	56.0264	-4.4927	68.4299	**128.9490**	0.005

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
46	**120.4618**	75.2816	29.1120	74.2922
22	95.8036	**97.8576**	47.7024	45.6485
26	94.2592	70.0640	**77.3967**	101.5920
23	56.0282	0.0000	68.4325	**128.9700**



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
11	7.504	15.257	9.522	1.500	78.665	23.282	1.500
12	5.702	16.923	7.642	1.500	62.091	12.774	1.500
13	5.661	16.998	7.596	1.500	62.162	12.692	1.500
14	5.354	183.885	7.568	1.500	17.513	12.229	1.500
15	6.006	16.990	8.008	1.500	58.110	16.138	1.500
16	8.917	5.089	6.250	1.500	12.018	67.907	1.500
17	4.153	8.506	5.277	1.500	6.885	7.243	1.500
18	28.604	3.433	4.820	1.500	6.178	21.315	1.500
19	3.746	3.390	3.555	1.500	3.827	5.615	1.500
20	7.850	20.748	10.384	1.500	68.627	24.888	1.500
21	6.511	38.751	9.081	1.500	14.446	15.041	1.500
22	10.558	4.153	5.466	1.500	8.640	209.601	1.500
23	3.714	5.407	4.329	1.500	5.159	6.169	1.500
26	6.852	13.685	8.665	1.500	70.611	21.284	1.500
39	5.989	5.281	5.601	1.500	12.409	13.911	1.500
40	5.353	5.592	5.469	1.500	13.525	10.594	1.500
41	7.515	3.629	4.621	1.500	6.470	18.278	1.500
42	5.954	3.911	4.623	1.500	7.056	10.652	1.500
45	5.972	4.344	4.968	1.500	8.974	14.063	1.500
46	5.168	4.635	4.880	1.500	9.780	9.932	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 23

Governing Disturbing Force : -25.663kN

Governing Restoring Force : 95.315kN

Minimum Sliding Ratio for the Critical Load Case : 3.714

Critical Load Case for Overturning about X-Direction : 19
Governing Overturning Moment : 28.776kNm
Governing Resisting Moment : 110.128kNm
Minimum Overturning Ratio for the Critical Load Case : 3.827

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 19
Governing Disturbing Force : 20.048kN
Governing Restoring Force : 67.954kN
Minimum Sliding Ratio for the Critical Load Case : 3.390
Critical Load Case for Overturning about Z-Direction : 19
Governing Overturning Moment : 19.407kNm
Governing Resisting Moment : 108.981kNm
Minimum Overturning Ratio for the Critical Load Case : 5.615

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

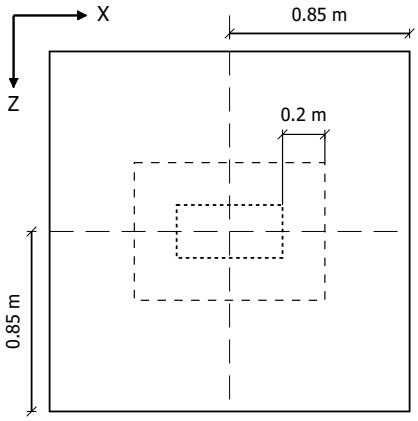
Critical Load Case for Sliding along Resultant Direction : 19
Governing Disturbing Force : 27.036kN
Governing Restoring Force : 96.102kN
Minimum Sliding Ratio for the Critical Load Case : 3.555

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m
Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m
For rectangular column, β_c = B_{col} / D_{col} = 2.000

Effective depth, d_{eff}, increased until 0.75XV_c $\geq$ Punching Shear Force

Punching Shear Force, Vu = 210.600kN, Load Case # 48

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 2.876m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 1944.021kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 3297.265kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 1944.021kN

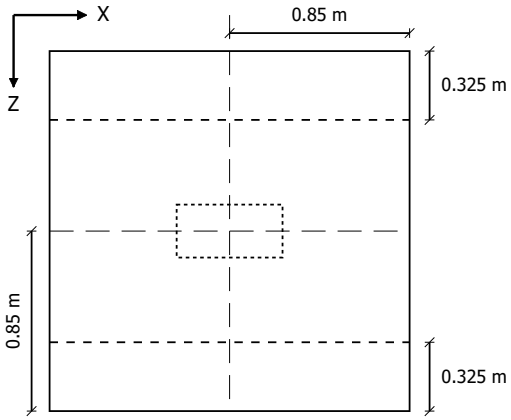
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1458.016kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 574.554kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{zd}$ = 0.381m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

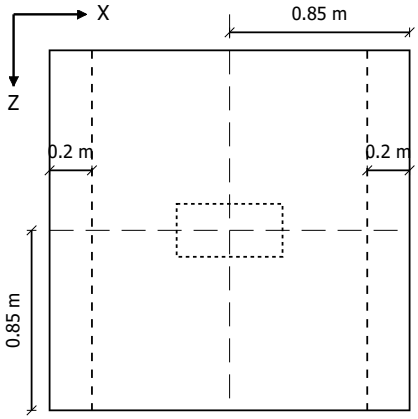
From above calculations, 0.75 X V_c = 430.916 kN

Critical load case for V_{ux} is # 49 $V_{ux} = V_{ux}|_{x=D_x}$ = 76.577 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 574.554 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.256 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

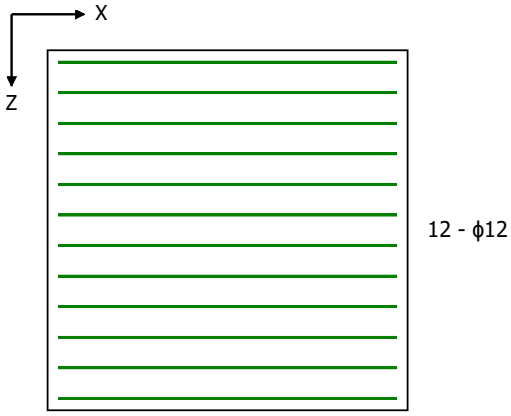
From above calculations, 0.75 X V_c = 430.916 kN

Critical load case for V_{uz} is # 50 $V_{uz} = V_{uz}|_{z=D_z} =$ 49.264 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 50

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03374
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02530
From ACI Cl. 7.12.2, ρ_{min} =		0.00185
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	13.950

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	0.600 m
Ultimate moment,	$M_u\|_{z=D_z}$ =	33.873 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	37.636 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00045
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00039
Since	ρ= ρ_{min}	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1259.382 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 100.000mm

Selected spacing (S) = 144.364mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

According to ACI 318 Clause No- 10.6.4 1st

Max spacing for Cracking Consideration = 254.793mm

Safe for Cracking Aspect.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 140mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.406 m

Available development length for bars, D_L = $0.5 \times (L - D_{col}) - C_{cover}$ = 0.550 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 12

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1357.159 mm2

d_{eff} = D - C_{cover} - 0.5 X (dia. of one bar) = 0.344 m

Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00232

From ACI Cl.7.6.1, minimum req'd clear distance between bars

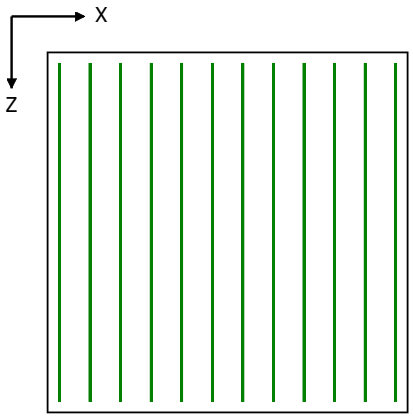
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 100.000mm

Provided Steel Area / Required Steel Area = 1.078

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



12 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 49

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} = 0.332 m

Factor β_1 from ACI Cl.10.2.7.3 = 0.796

From ACI Cl. 10.3.2, ρ_{bal} = $0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$ = 0.03374

From ACI Cl. 10.3.3, ρ_{max} = $0.75 \times \rho_{bal}$ = 0.02530

From ACI Cl.7.12.2, ρ_{min} = 0.00191

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{\left(0.85 \times F_c'\right)}$ = 13.950

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 0.725 m

Ultimate moment, $M_u\big|_{x=D_x}$ = 51.715 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 57.462 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00074

(Based on gross depth) ρ x d_{eff} / Depth = 0.00062

Since ρ= ρ_{min} ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 1295.755 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 100.000mm

Selected spacing (S) = 144.364mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

According to ACI 318 Clause No- 10.6.4 3rd

Max spacing for Cracking Consideration = 254.793mm

Safe for Cracking Aspect.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 140mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.406 m

Available development length for bars, D_L = $0.5 \times \left(L - D_{col}\right) - C_{cover}$ = 0.675 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 12

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1357.159 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.332 m

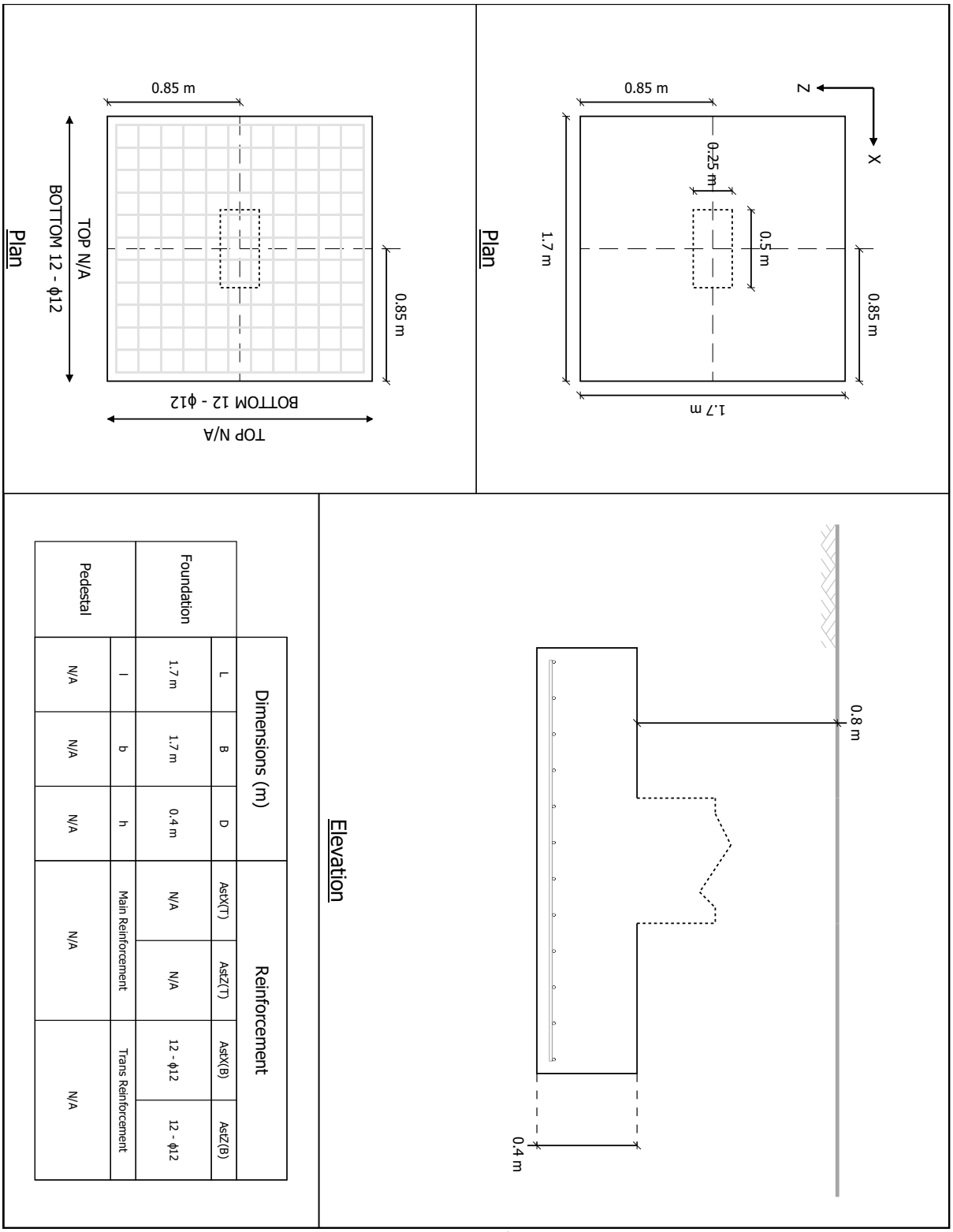
Reinforcement ratio, ρ = $\frac{A_{s_total}}{\left(d_{eff} \times W\right)}$ = 0.00240

From ACI Cl.7.6.1, minimum req'd clear distance between bars

C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 100.000mm

Provided Steel Area / Required Steel Area = 1.047

Check to see if width is sufficient to accomodate bars



Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	1.7 m	1.7 m	0.4 m	N/A	N/A	12 - ϕ12	12 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

Print Calculation Sheet

COMBINED FOUNDATION DESIGN (ACI 318-08)

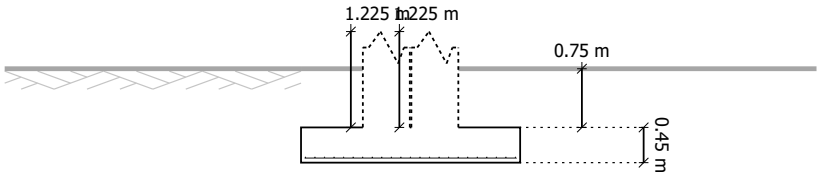
Result Summary

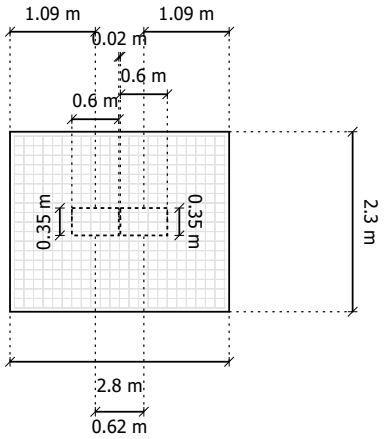
Footing No.	Left Overhang (m)	Right Overhang (m)	Length (m)	Width (m)	Thickness (m)
1	1.090	1.090	2.800	2.300	0.450

Footing No.	Footing Reinforcement			
-	Main Steel Top	Main Steel Bottom	Secondary Steel Top	Secondary Steel Bottom
1	-	19 - ϕ12	-	23 - ϕ12

FOUNDATION-CF1

Input Data





Geometry of Footing

For Column 127

Column Dimensions

Column Shape : Rectangular

Column Length - X (D_{col}) : 0.600m

Column Width - Z (B_{col}) : 0.350m

Pedestal

Include Pedestal? No

Pedestal Shape : N/A

Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Eccentricity

Column Offset in Transverse Direction : 0.000m

For Column 235

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.600m
Column Width - Z (B_{col}) : 0.350m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Eccentricity

Column Offset in Transverse Direction : 0.000m

Length of left overhang : 1.090m
Length of right overhang : 1.090m
Is the length of left overhang fixed? No
Is the length of right overhang fixed? No
Maximum length of Footing (Lo) : 450.000mm
Width of Footing (Wo) : 2.300m
Thickness of Footing (Do) : 450.000mm

Cover and Soil Properties

Pedestal Clear Cover : 50.000mm
Footing Clear Cover : 50.000mm
Unit Weight of soil : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 0.750m
Type of Depth : Fixed Top
Depth of Water Table : 24.000m

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Compressive Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 415.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Minimum Pedestal Bar Size : #12
Maximum Pedestal Bar Size : #25
Minimum Bar Spacing : 100.000mm

Maximum Bar Spacing : 150.000mm

Design Calculations

Footing Size Calculations

Gross Soil Bearing Capacity = 371.60kN/m^2
Reduction of force due to buoyancy = 0.000kN

Area from initial length and width, A_o = L_o X W_o = 6.440m^2
Min. area required from bearing pressure, A_{min} = P / q_{max} = 2.256m^2

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final footing dimensions are:

Length of footing, L : 2.800m
Width of footing, W : 2.300m
Depth of footing, Do : 0.450m
Area, A : 6.440m^2
Length of left overhang, $L_{left_overhang}$: 1.090m
Length of right overhang, $L_{right_overhang}$: 1.090m
Footing self weight : 69.550kN
Soi weight on top of footing : 81.267kN

Load Combinations

Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
10	SLS: (DL)	1.00	1.00	1.00
11	SLS: (DL+T)	1.00	1.00	1.00
12	SLS: (D+LR)	1.00	1.00	1.00
13	SLS: (D+0.75LR)	1.00	1.00	1.00
14	SLS: (D+0.7EZ)	1.00	1.00	1.00
15	SLS: (D+0.7EX)	1.00	1.00	1.00
16	SLS: (D+0.75WX1+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ1+0.75LR)	1.00	1.00	1.00
18	SLS: (0.6D+WX1)	1.00	1.00	1.00
19	SLS: (0.6D+WZ1)	1.00	1.00	1.00
20	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
21	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
22	SLS: (D+WX1)	1.00	1.00	1.00
23	SLS: (D+WZ1)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
39	SLS: (D+0.75WX2+0.75LR)	1.00	1.00	1.00
40	SLS: (D+0.75WZ2+0.75LR)	1.00	1.00	1.00
41	SLS: (0.6D+WX2)	1.00	1.00	1.00
42	SLS: (0.6D+WZ2)	1.00	1.00	1.00
45	SLS: (D+WX2)	1.00	1.00	1.00
46	SLS: (D+WZ2)	1.00	1.00	1.00
Load Combination Number	**Load Combination Title**	**Load Combination Factor**	**Soil Bearing Factor**	**Self Weight Factor**
24	ULS: (1.4 DL)	1.00	1.00	1.00
25	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00

Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
27	ULS: (1.2D+0.80WX1+1.6LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+1.6WX1+0.5LR)	1.00	1.00	1.00
30	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
31	ULS: (1.2D+EX)	1.00	1.00	1.00
32	ULS: (1.2D+EZ)	1.00	1.00	1.00
33	ULS: (0.90D+1.6WX1)	1.00	1.00	1.00
34	ULS: (0.90D+1.6WZ1)	1.00	1.00	1.00
35	ULS: (0.90D+EZ)	1.00	1.00	1.00
36	ULS: (0.90D+EX)	1.00	1.00	1.00
47	ULS: (1.2D+0.80WX2+1.6LR)	1.00	1.00	1.00
48	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
49	ULS: (1.2D+1.6WX2+0.5LR)	1.00	1.00	1.00
50	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
53	ULS: (0.90D+1.6WX2)	1.00	1.00	1.00
54	ULS: (0.90D+1.6WZ2)	1.00	1.00	1.00

Applied Loads

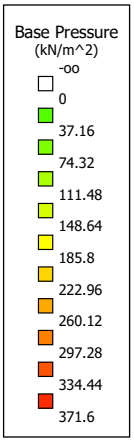
Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
-					
Column Number : 127					
10	250.839	27.960	-26.482	-6.263	-16.328
11	262.709	29.157	-24.660	-6.648	-42.678
12	257.273	28.512	-26.472	-6.229	-17.484
13	255.665	28.374	-26.475	-6.237	-17.195
14	255.452	28.833	-19.630	1.053	-17.130
15	254.924	33.160	-26.515	-6.280	-24.812
16	254.638	30.282	-26.965	-6.399	-20.510
17	261.639	28.812	-17.886	1.977	-16.738
18	149.135	19.321	-16.543	-3.974	-14.216
19	158.469	17.360	-4.438	7.195	-9.187
20	154.589	21.977	-15.922	-3.775	-18.280
21	155.116	17.649	-9.037	3.558	-10.598
22	249.471	30.504	-27.136	-6.479	-20.748
23	258.804	28.544	-15.030	4.690	-15.718
26	371.009	41.095	-34.519	-9.289	-60.328
39	249.930	21.464	-27.067	-6.513	-8.342
40	247.297	23.362	-35.221	-14.680	-10.757
41	142.857	7.563	-16.679	-4.126	2.007
42	139.346	10.094	-27.551	-15.015	-1.213
45	243.193	18.747	-27.271	-6.631	-4.524
46	239.682	21.278	-38.144	-17.521	-7.744
-					
Column Number : 235					
10	219.137	-10.012	-31.192	-10.215	5.268
11	224.193	-3.121	-28.698	-9.708	22.279
12	224.615	-9.384	-31.192	-10.208	5.774
13	223.246	-9.541	-31.192	-10.209	5.647
14	222.566	-9.920	-23.981	-2.653	5.442
15	216.111	-6.376	-31.157	-10.185	-2.941
16	221.352	-7.164	-31.378	-10.022	0.493
17	228.699	-7.903	-17.889	2.260	3.298
18	128.957	-2.838	-18.963	-5.879	-3.712
19	138.754	-3.823	-0.978	10.497	0.028
20	128.456	-2.371	-18.680	-6.099	-5.048
21	134.911	-5.915	-11.504	1.432	3.334
22	216.612	-6.842	-31.440	-9.965	-1.605
23	226.409	-7.828	-13.455	6.411	2.135
26	316.609	-4.055	-40.177	-13.588	31.444
39	223.935	-11.262	-31.331	-9.972	8.193
40	214.981	-6.582	-44.514	-22.781	-0.492

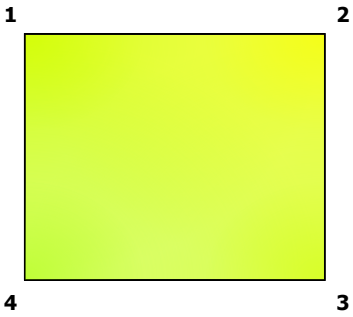
Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
41	132.401	-8.301	-18.900	-5.813	6.555
42	120.462	-2.062	-36.478	-22.891	-5.025
45	220.056	-12.306	-31.377	-9.898	8.662
46	208.117	-6.067	-48.955	-26.977	-2.918

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
-					
Column Number : 127					
24	351.175	39.144	-37.074	-8.769	-22.859
25	371.009	41.095	-34.519	-9.289	-60.328
26	371.009	41.095	-34.519	-9.289	-60.328
27	310.207	36.471	-32.287	-7.633	-24.979
28	316.968	34.762	-13.451	10.026	-19.196
29	302.034	37.899	-32.820	-7.844	-27.243
30	316.968	34.762	-13.451	10.026	-19.196
31	306.843	40.981	-31.826	-7.540	-31.713
32	307.596	34.799	-21.990	2.935	-20.739
33	223.566	29.235	-24.880	-5.982	-21.766
34	238.499	26.098	-5.511	11.888	-13.719
35	232.345	26.411	-14.046	4.814	-15.840
36	231.591	32.593	-23.881	-5.661	-26.814
47	305.184	27.065	-32.395	-7.755	-12.000
48	286.372	23.136	-50.432	-25.510	-6.437
49	291.990	19.087	-33.037	-8.087	-1.286
50	286.372	23.136	-50.432	-25.510	-6.437
53	213.521	10.423	-25.097	-6.225	4.191
54	207.903	14.472	-42.493	-23.649	-0.961
-					
Column Number : 235					
24	306.792	-14.017	-43.669	-14.300	7.375
25	316.609	-4.055	-40.177	-13.588	31.444
26	316.609	-4.055	-40.177	-13.588	31.444
27	269.709	-8.474	-37.629	-12.047	1.633
28	277.338	-8.206	-9.051	14.347	1.562
29	261.663	-6.629	-37.828	-11.855	-4.422
30	277.338	-8.206	-9.051	14.347	1.562
31	258.641	-6.820	-37.381	-12.215	-5.405
32	267.863	-11.883	-27.129	-1.456	6.570
33	193.183	-3.939	-28.470	-8.794	-6.255
34	208.858	-5.516	0.307	17.408	-0.271
35	202.122	-8.880	-17.771	1.609	4.989
36	192.900	-3.816	-28.023	-9.150	-6.985
47	272.464	-12.845	-37.579	-11.994	9.846
48	248.072	-5.388	-65.851	-39.074	-6.523
49	267.173	-15.371	-37.726	-11.748	12.005
50	248.072	-5.388	-65.851	-39.074	-6.523
53	198.693	-12.681	-28.369	-8.687	10.172
54	179.592	-2.698	-56.493	-36.013	-8.357

Pressures at 4 Corners

Governing Base Pressure Contour for Each Corner





Corner 1, 2, 3, 4

Load Case	Pressure at corner 1 (q_1) (kN/m^2)	Pressure at corner 2 (q_2) (kN/m^2)	Pressure at corner 3 (q_3) (kN/m^2)	Pressure at corner 4 (q_4) (kN/m^2)	Area of footing in uplift (A_u) (sq. m)
26	**143.5291**	162.6206	116.8545	97.7629	0.000
26	143.5291	**162.6206**	116.8545	97.7629	0.000
26	143.5291	162.6206	**116.8545**	97.7629	0.000
26	143.5291	162.6206	116.8545	**97.7629**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m^2)	Pressure at corner 2 (q_2) (kN/m^2)	Pressure at corner 3 (q_3) (kN/m^2)	Pressure at corner 4 (q_4) (kN/m^2)
26	**143.5291**	162.6206	116.8545	97.7629
26	143.5291	**162.6206**	116.8545	97.7629
26	143.5291	162.6206	**116.8545**	97.7629
26	143.5291	162.6206	116.8545	**97.7629**

Stability Check

Load Case	Shear X (kN)	Shear Z (kN)	Resultant Shear (kN)	Resisting Sliding Force (kN)	Ratio X	Ratio Z	Resultant Ratio	Required FOS
10	17.948	-57.674	60.402	248.317	13.836	4.306	4.111	1.500
11	26.036	-53.358	59.371	255.088	9.798	4.781	4.296	1.500
12	19.128	-57.665	60.754	253.082	13.231	4.389	4.166	1.500
13	18.833	-57.667	60.664	251.891	13.375	4.368	4.152	1.500
14	18.913	-43.611	47.535	251.534	13.300	5.768	5.292	1.500
15	26.785	-57.673	63.589	248.741	9.287	4.313	3.912	1.500
16	23.118	-58.344	62.757	250.723	10.845	4.297	3.995	1.500
17	20.909	-35.776	41.437	256.462	12.266	7.169	6.189	1.500
18	16.483	-35.507	39.146	171.564	10.409	4.832	4.383	1.500
19	13.536	-5.416	14.580	179.216	13.239	33.091	12.292	1.500
20	19.605	-34.603	39.771	173.545	8.852	5.015	4.364	1.500
21	11.733	-20.541	23.656	176.338	15.029	8.584	7.454	1.500
22	23.662	-58.576	63.175	246.760	10.428	4.213	3.906	1.500
23	20.716	-28.485	35.222	254.412	12.281	8.931	7.223	1.500

Load Case	Shear X (kN)	Shear Z (kN)	Resultant Shear (kN)	Resisting Sliding Force (kN)	Ratio X	Ratio Z	Resultant Ratio	Required FOS
26	37.040	-74.696	83.376	335.374	9.054	4.490	4.022	1.500
39	10.202	-58.398	59.282	249.873	24.492	4.279	4.215	1.500
40	16.780	-79.735	81.482	245.238	14.615	3.076	3.010	1.500
41	-0.738	-35.579	35.587	170.430	230.785	4.790	4.789	1.500
42	8.032	-64.029	64.531	164.250	20.450	2.565	2.545	1.500
45	6.441	-58.649	59.001	245.626	38.137	4.188	4.163	1.500
46	15.211	-87.098	88.417	239.446	15.742	2.749	2.708	1.500

Check for stability against overturning (Moments printed against Local axis)

Load Case	Moment X (kNm)	Moment Z (kNm)	Resisting Moment X (kNm)	Resisting Moment Z (kNm)	Ratio X	Ratio Z	Required FOS
10	-42.432	-9.309	713.899	869.095	16.825	93.357	1.500
11	-40.367	-20.175	733.363	892.790	18.168	44.252	1.500
12	-42.386	-10.194	727.598	885.771	17.166	86.892	1.500
13	-42.397	-9.973	724.173	881.602	17.081	88.400	1.500
14	-21.226	-10.004	723.147	880.353	34.070	87.998	1.500
15	-42.418	-27.773	715.117	870.577	16.859	31.346	1.500
16	-42.676	-20.102	720.815	877.514	16.891	43.654	1.500
17	-11.862	-12.638	737.314	897.600	62.160	71.026	1.500
18	-25.831	-19.091	493.236	600.462	19.095	31.453	1.500
19	15.255	-9.139	515.236	627.244	33.775	68.637	1.500
20	-25.445	-24.049	498.931	607.395	19.608	25.256	1.500
21	-4.253	-6.280	506.961	617.171	119.198	98.268	1.500
22	-42.804	-22.814	709.422	863.644	16.574	37.855	1.500
23	-1.718	-12.862	731.421	890.426	425.844	69.227	1.500
26	-56.491	-28.688	964.183	1173.789	17.068	40.916	1.500
39	-42.765	3.318	718.370	874.538	16.798	263.574	1.500
40	-73.343	-8.783	705.045	858.316	9.613	97.729	1.500
41	-25.949	12.135	489.977	596.494	18.883	49.153	1.500
42	-66.720	-3.999	472.210	574.865	7.078	143.764	1.500
45	-42.922	8.412	706.162	859.676	16.453	102.200	1.500
46	-83.693	-7.722	688.396	838.047	8.225	108.522	1.500

Calculations of Footing Thickness

Footing thickness is calculated based on the ultimate load cases

Punching Shear Check

For Column 127

Critical Load case for Punching Shear Check : 25

Total Footing Depth, D_o = 0.450m

Calculated Effective Depth, d_{eff} = 0.394m

For rectangular column, β_c = B_{col} / D_{col} : 1.714

Considering the particular column as interior column, Slab Edge Factor α_s : 40.0

Effective depth, d_{eff}, increased until $0.75*V_c \geq$ Punching Shear Force

Punchng Shear Force, Vu = 294.261kN

From ACI Cl.11.11.2, b_c for column 3.476m

Equation 11-31, V_{c1} 2915.357kN

$\left(2+\frac{4}{\beta_c}\right)\times b_o\times d_{eff}\times\sqrt{1000\times F_c'}$ =

Equation 11-32, V_{c2} = $\left(\frac{\alpha_s\times d}{b_o}+2\right)\times\lambda\times\sqrt{f_c}\times b_o\times d$ = 4395.874kN

Equation 11-33, V_{c3} = $4\times b_o\times d_{eff}\times\sqrt{1000\times F_c'}$ = 2691.098kN

Punching shear strength,V_c= 0.75 x minimum of (V_{c1},V_{c2},V_{c3}) = 2018.324kN

0.75 * V_c > V_u hence, OK

For Column 235

Critical Load case for Punching Shear Check : 25

Total Footing Depth, D_o = 0.450m

Calculated Effective Depth,d_{eff} = 0.394m

For rectangular column, β_c = B_{col} / D_{col} : 1.714

Considering the particular column as interior column, Slab Edge Factor α_s : 40.0

Effective depth, d_{eff}, increased until 0.75*V_c ≥ Punching Shear Force

Punchng Shear Force, Vu = 236.735kN

From ACI Cl.11.11.2, b_c for column 3.476m

Equation 11-31, V_{c1} $\left(2+\frac{4}{\beta_c}\right)\times b_o\times d_{eff}\times\sqrt{1000\times F_c'}$ = 2915.357kN

Equation 11-32, V_{c2} = $\left(\frac{\alpha_s\times d}{b_o}+2\right)\times\lambda\times\sqrt{f_c}\times b_o\times d$ = 4395.874kN

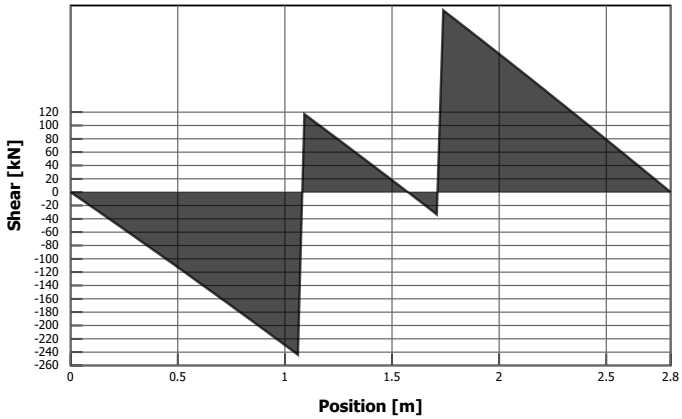
Equation 11-33, V_{c3} = $4\times b_o\times d_{eff}\times\sqrt{1000\times F_c'}$ = 2691.098kN

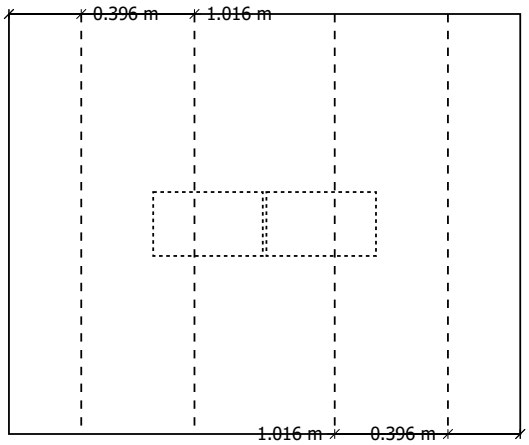
Punching shear strength,V_c= 0.75 x minimum of (V_{c1},V_{c2},V_{c3}) = 2018.324kN

0.75 * V_c > V_u hence, OK

One-Way Shear

Shear Diagram

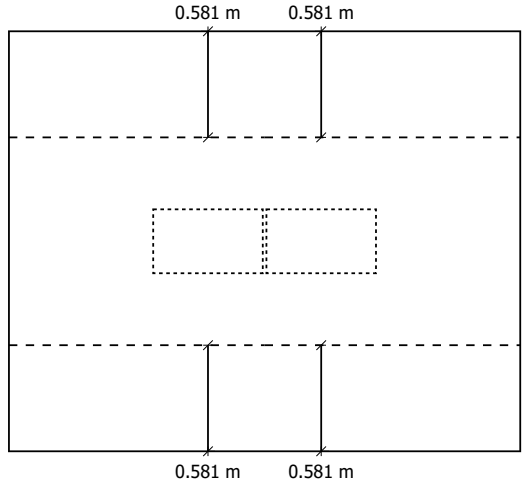




Shear Plane Parallel to Foundation Width

Critical load case for maximum shear force along the length of footing : 25

Critical Shear force,V_u	For the critical load case:	233.238kN
Point of occurance of V_u	Critical one-way shear position:	1.016m
From ACI Cl.11.2.1.1, V_c =	$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ =	890.323kN
	0.75 x V_c =	667.742kN
	Since 0.75 * V_c > V_u	hence, OK



Shear Plane Parallel to Foundation Length

Critical load case for maximum shear force along the width of footing : 25

Critical Shear force,V_u	For the critical load case:	205.759kN
Point of occurance of V_u	Critical one-way shear position:	0.581m
From ACI Cl.11.2.1.1, V_c =	$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ =	1083.872kN
	0.75 x V_c =	812.904kN
	Since 0.75 * V_c > V_u	hence, OK

Design for Flexure

Design of flexure

Moment Diagram

Bottom Reinforcement

Critical load case :	25	
Required Effective Depth :	0.394m	
β_1, from ACI Cl.10.2.7.3	max[0.85-0.05x(Fc-4),0.65]=	0.7962
From ACI Cl. 10.3.2, ρ_{bal}	$0.85 \times \beta_1 \times F_c' \times \frac{87}{[f_y \times (87 + F_y)]}$ =	0.03374
From ACI Cl. 10.3.3, ρ_{max}	$0.75 \times \rho_{bal}$ =	0.02530
From ACI Cl. 7.12.2, ρ_{min}	$\max\left(0.0018 \cdot \frac{60ksi}{F_y}, 0.0014\right)$ =	0.00200
Modular Ratio,m	$\frac{F_y}{(0.85 \times F_c')}$ =	13.9496
Ultimate Moment (Mz) :	190.585kNm	
Point of occurrence of the ultimate moment along the length of footing :	1.390m	
Nominal Moment Capacity :	211.761kNm	
Required ρ (based on effective depth) :	0.0023	
ρ x d_{eff} / Depth (based on gross depth) :	0.0020	
Area of main steel required, A_s = ρ * W * deff :	2070.000mm2	

Distribution Reinforcement

Critical load case :	25		
Critical Moment for distribution steel (Mx) :		165.5318	kNm
Nominal moment Capacity :	183.9243		kNm
Point of occurance of the critical moment along length:	1.3250		m
Required ρ (based on effective depth):	0.0024		
ρ x d_{eff} / Depth (based on gross depth) :	0.0020		
Area of distribution steel required, A_s = ρ * L * deff :	2520.001		mm2

Provided Reinforcement

Based on spacing reinforcement increment; provided reinforcement is

Main bar no. for bottom Reinforcement:	#12

Spacing of bottom reinforcement bar :	122.222	mm

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 120mm o.c.

Distribution bar no. (Bottom):	#12	
Spacing of distribution bars (Bottom):	122.727	mm

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 120mm o.c.

Based on spacing reinforcement increment; provided reinforcement is

Pedestal Design

Print Calculation Sheet

COMBINED FOUNDATION DESIGN (ACI 318-08)

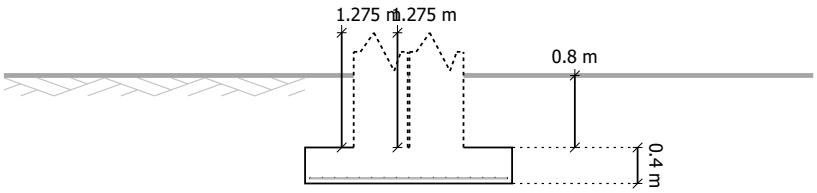
Result Summary

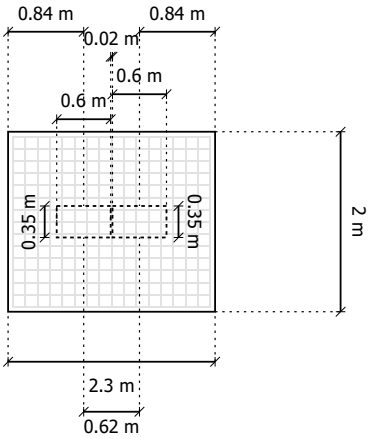
Footing No.	Left Overhang (m)	Right Overhang (m)	Length (m)	Width (m)	Thickness (m)
1	0.840	0.840	2.300	2.000	0.400

Footing No.	Footing Reinforcement			
-	Main Steel Top	Main Steel Bottom	Secondary Steel Top	Secondary Steel Bottom
1	-	15 - ϕ12	-	17 - ϕ12

FOUNDATION-CF2

Input Data





Geometry of Footing

For Column 127

Column Dimensions

Column Shape : Rectangular

Column Length - X (D_{col}) : 0.600m

Column Width - Z (B_{col}) : 0.350m

Pedestal

Include Pedestal? No

Pedestal Shape : N/A

Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Eccentricity

Column Offset in Transverse Direction : 0.000m

For Column 235

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.600m
Column Width - Z (B_{col}) : 0.350m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Eccentricity

Column Offset in Transverse Direction : 0.000m

Length of left overhang : 0.840m
Length of right overhang : 0.840m
Is the length of left overhang fixed? No
Is the length of right overhang fixed? No
Maximum length of Footing (Lo) : 400.000mm
Width of Footing (Wo) : 2.000m
Thickness of Footing (Do) : 400.000mm

Cover and Soil Properties

Pedestal Clear Cover : 50.000mm
Footing Clear Cover : 50.000mm
Unit Weight of soil : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 0.800m
Type of Depth : Fixed Top
Depth of Water Table : 24.000m

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Compressive Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 415.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Minimum Pedestal Bar Size : #12
Maximum Pedestal Bar Size : #25
Minimum Bar Spacing : 100.000mm

Maximum Bar Spacing : 150.000mm

Design Calculations

Footing Size Calculations

Gross Soil Bearing Capacity = 371.60kN/m^2
Reduction of force due to buoyancy = 0.000kN

Area from initial length and width, A_o = L_o X W_o = 4.600m^2
Min. area required from bearing pressure, A_{min} = P / q_{max} = 2.131m^2

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final footing dimensions are:

Length of footing, L : 2.300m
Width of footing, W : 2.000m
Depth of footing, Do : 0.400m
Area, A : 4.600m^2
Length of left overhang, $L_{left_overhang}$: 0.840m
Length of right overhang, $L_{right_overhang}$: 0.840m
Footing self weight : 44.158kN
Soi weight on top of footing : 60.190kN

Load Combinations

Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
10	SLS: (DL)	1.00	1.00	1.00
11	SLS: (DL+T)	1.00	1.00	1.00
12	SLS: (D+LR)	1.00	1.00	1.00
13	SLS: (D+0.75LR)	1.00	1.00	1.00
14	SLS: (D+0.7EZ)	1.00	1.00	1.00
15	SLS: (D+0.7EX)	1.00	1.00	1.00
16	SLS: (D+0.75WX1+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ1+0.75LR)	1.00	1.00	1.00
18	SLS: (0.6D+WX1)	1.00	1.00	1.00
19	SLS: (0.6D+WZ1)	1.00	1.00	1.00
20	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
21	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
22	SLS: (D+WX1)	1.00	1.00	1.00
23	SLS: (D+WZ1)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
39	SLS: (D+0.75WX2+0.75LR)	1.00	1.00	1.00
40	SLS: (D+0.75WZ2+0.75LR)	1.00	1.00	1.00
41	SLS: (0.6D+WX2)	1.00	1.00	1.00
42	SLS: (0.6D+WZ2)	1.00	1.00	1.00
45	SLS: (D+WX2)	1.00	1.00	1.00
46	SLS: (D+WZ2)	1.00	1.00	1.00
Load Combination Number	**Load Combination Title**	**Load Combination Factor**	**Soil Bearing Factor**	**Self Weight Factor**
24	ULS: (1.4 DL)	1.00	1.00	1.00
25	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00

Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
27	ULS: (1.2D+0.80WX1+1.6LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+1.6WX1+0.5LR)	1.00	1.00	1.00
30	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
31	ULS: (1.2D+EX)	1.00	1.00	1.00
32	ULS: (1.2D+EZ)	1.00	1.00	1.00
33	ULS: (0.90D+1.6WX1)	1.00	1.00	1.00
34	ULS: (0.90D+1.6WZ1)	1.00	1.00	1.00
35	ULS: (0.90D+EZ)	1.00	1.00	1.00
36	ULS: (0.90D+EX)	1.00	1.00	1.00
47	ULS: (1.2D+0.80WX2+1.6LR)	1.00	1.00	1.00
48	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
49	ULS: (1.2D+1.6WX2+0.5LR)	1.00	1.00	1.00
50	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
53	ULS: (0.90D+1.6WX2)	1.00	1.00	1.00
54	ULS: (0.90D+1.6WZ2)	1.00	1.00	1.00

Applied Loads

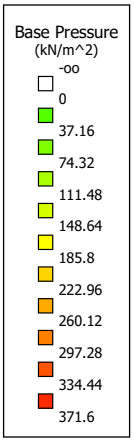
Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
-					
Column Number : 127					
10	250.839	27.960	-26.482	-6.263	-16.328
11	262.709	29.157	-24.660	-6.648	-42.678
12	257.273	28.512	-26.472	-6.229	-17.484
13	255.665	28.374	-26.475	-6.237	-17.195
14	255.452	28.833	-19.630	1.053	-17.130
15	254.924	33.160	-26.515	-6.280	-24.812
16	254.638	30.282	-26.965	-6.399	-20.510
17	261.639	28.812	-17.886	1.977	-16.738
18	149.135	19.321	-16.543	-3.974	-14.216
19	158.469	17.360	-4.438	7.195	-9.187
20	154.589	21.977	-15.922	-3.775	-18.280
21	155.116	17.649	-9.037	3.558	-10.598
22	249.471	30.504	-27.136	-6.479	-20.748
23	258.804	28.544	-15.030	4.690	-15.718
26	371.009	41.095	-34.519	-9.289	-60.328
39	249.930	21.464	-27.067	-6.513	-8.342
40	247.297	23.362	-35.221	-14.680	-10.757
41	142.857	7.563	-16.679	-4.126	2.007
42	139.346	10.094	-27.551	-15.015	-1.213
45	243.193	18.747	-27.271	-6.631	-4.524
46	239.682	21.278	-38.144	-17.521	-7.744
-					
Column Number : 235					
10	219.137	-10.012	-31.192	-10.215	5.268
11	224.193	-3.121	-28.698	-9.708	22.279
12	224.615	-9.384	-31.192	-10.208	5.774
13	223.246	-9.541	-31.192	-10.209	5.647
14	222.566	-9.920	-23.981	-2.653	5.442
15	216.111	-6.376	-31.157	-10.185	-2.941
16	221.352	-7.164	-31.378	-10.022	0.493
17	228.699	-7.903	-17.889	2.260	3.298
18	128.957	-2.838	-18.963	-5.879	-3.712
19	138.754	-3.823	-0.978	10.497	0.028
20	128.456	-2.371	-18.680	-6.099	-5.048
21	134.911	-5.915	-11.504	1.432	3.334
22	216.612	-6.842	-31.440	-9.965	-1.605
23	226.409	-7.828	-13.455	6.411	2.135
26	316.609	-4.055	-40.177	-13.588	31.444
39	223.935	-11.262	-31.331	-9.972	8.193
40	214.981	-6.582	-44.514	-22.781	-0.492

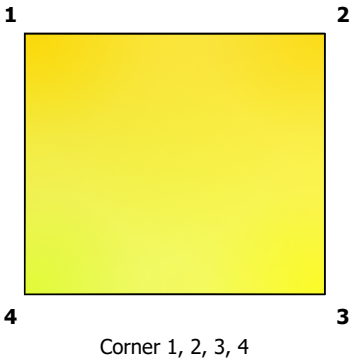
Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
41	132.401	-8.301	-18.900	-5.813	6.555
42	120.462	-2.062	-36.478	-22.891	-5.025
45	220.056	-12.306	-31.377	-9.898	8.662
46	208.117	-6.067	-48.955	-26.977	-2.918

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
-					
Column Number : 127					
24	351.175	39.144	-37.074	-8.769	-22.859
25	371.009	41.095	-34.519	-9.289	-60.328
26	371.009	41.095	-34.519	-9.289	-60.328
27	310.207	36.471	-32.287	-7.633	-24.979
28	316.968	34.762	-13.451	10.026	-19.196
29	302.034	37.899	-32.820	-7.844	-27.243
30	316.968	34.762	-13.451	10.026	-19.196
31	306.843	40.981	-31.826	-7.540	-31.713
32	307.596	34.799	-21.990	2.935	-20.739
33	223.566	29.235	-24.880	-5.982	-21.766
34	238.499	26.098	-5.511	11.888	-13.719
35	232.345	26.411	-14.046	4.814	-15.840
36	231.591	32.593	-23.881	-5.661	-26.814
47	305.184	27.065	-32.395	-7.755	-12.000
48	286.372	23.136	-50.432	-25.510	-6.437
49	291.990	19.087	-33.037	-8.087	-1.286
50	286.372	23.136	-50.432	-25.510	-6.437
53	213.521	10.423	-25.097	-6.225	4.191
54	207.903	14.472	-42.493	-23.649	-0.961
-					
Column Number : 235					
24	306.792	-14.017	-43.669	-14.300	7.375
25	316.609	-4.055	-40.177	-13.588	31.444
26	316.609	-4.055	-40.177	-13.588	31.444
27	269.709	-8.474	-37.629	-12.047	1.633
28	277.338	-8.206	-9.051	14.347	1.562
29	261.663	-6.629	-37.828	-11.855	-4.422
30	277.338	-8.206	-9.051	14.347	1.562
31	258.641	-6.820	-37.381	-12.215	-5.405
32	267.863	-11.883	-27.129	-1.456	6.570
33	193.183	-3.939	-28.470	-8.794	-6.255
34	208.858	-5.516	0.307	17.408	-0.271
35	202.122	-8.880	-17.771	1.609	4.989
36	192.900	-3.816	-28.023	-9.150	-6.985
47	272.464	-12.845	-37.579	-11.994	9.846
48	248.072	-5.388	-65.851	-39.074	-6.523
49	267.173	-15.371	-37.726	-11.748	12.005
50	248.072	-5.388	-65.851	-39.074	-6.523
53	198.693	-12.681	-28.369	-8.687	10.172
54	179.592	-2.698	-56.493	-36.013	-8.357

Pressures at 4 Corners

Governing Base Pressure Contour for Each Corner





Corner 1, 2, 3, 4

Load Case	Pressure at corner 1 (q_1) (kN/m^2)	Pressure at corner 2 (q_2) (kN/m^2)	Pressure at corner 3 (q_3) (kN/m^2)	Pressure at corner 4 (q_4) (kN/m^2)	Area of footing in uplift (A_u) (sq. m)
26	**191.3536**	221.7917	152.9797	122.5416	0.000
26	191.3536	**221.7917**	152.9797	122.5416	0.000
26	191.3536	221.7917	**152.9797**	122.5416	0.000
26	191.3536	221.7917	152.9797	**122.5416**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m^2)	Pressure at corner 2 (q_2) (kN/m^2)	Pressure at corner 3 (q_3) (kN/m^2)	Pressure at corner 4 (q_4) (kN/m^2)
26	**191.3536**	221.7917	152.9797	122.5416
26	191.3536	**221.7917**	152.9797	122.5416
26	191.3536	221.7917	**152.9797**	122.5416
26	191.3536	221.7917	152.9797	**122.5416**

Stability Check

Load Case	Shear X (kN)	Shear Z (kN)	Resultant Shear (kN)	Resisting Sliding Force (kN)	Ratio X	Ratio Z	Resultant Ratio	Required FOS
10	17.948	-57.674	60.402	229.730	12.800	3.983	3.803	1.500
11	26.036	-53.358	59.371	236.500	9.084	4.432	3.983	1.500
12	19.128	-57.665	60.754	234.495	12.259	4.067	3.860	1.500
13	18.833	-57.667	60.664	233.304	12.388	4.046	3.846	1.500
14	18.913	-43.611	47.535	232.947	12.317	5.341	4.900	1.500
15	26.785	-57.673	63.589	230.153	8.593	3.991	3.619	1.500
16	23.118	-58.344	62.757	232.135	10.041	3.979	3.699	1.500
17	20.909	-35.776	41.437	237.874	11.377	6.649	5.741	1.500
18	16.483	-35.507	39.146	152.976	9.281	4.308	3.908	1.500
19	13.536	-5.416	14.580	160.628	11.866	29.659	11.017	1.500
20	19.605	-34.603	39.771	154.957	7.904	4.478	3.896	1.500
21	11.733	-20.541	23.656	157.750	13.445	7.680	6.668	1.500
22	23.662	-58.576	63.175	228.172	9.643	3.895	3.612	1.500
23	20.716	-28.485	35.222	235.825	11.384	8.279	6.695	1.500

Load Case	Shear X (kN)	Shear Z (kN)	Resultant Shear (kN)	Resisting Sliding Force (kN)	Ratio X	Ratio Z	Resultant Ratio	Required FOS
26	37.040	-74.696	83.376	316.787	8.553	4.241	3.800	1.500
39	10.202	-58.398	59.282	231.285	22.670	3.960	3.901	1.500
40	16.780	-79.735	81.482	226.650	13.507	2.843	2.782	1.500
41	-0.738	-35.579	35.587	151.843	205.615	4.268	4.267	1.500
42	8.032	-64.029	64.531	145.663	18.136	2.275	2.257	1.500
45	6.441	-58.649	59.001	227.039	35.251	3.871	3.848	1.500
46	15.211	-87.098	88.417	220.859	14.520	2.536	2.498	1.500

Check for stability against overturning (Moments printed against Local axis)

Load Case	Moment X (kNm)	Moment Z (kNm)	Resisting Moment X (kNm)	Resisting Moment Z (kNm)	Ratio X	Ratio Z	Required FOS
10	-39.548	-8.412	574.315	660.462	14.522	78.514	1.500
11	-37.699	-18.874	591.240	679.926	15.683	36.025	1.500
12	-39.502	-9.238	586.226	674.160	14.841	72.980	1.500
13	-39.514	-9.031	583.248	670.736	14.761	74.269	1.500
14	-19.045	-9.059	582.356	669.709	30.578	73.931	1.500
15	-39.534	-26.434	575.373	661.680	14.554	25.031	1.500
16	-39.759	-18.946	580.328	667.377	14.596	35.226	1.500
17	-10.073	-11.592	594.675	683.877	59.038	58.995	1.500
18	-24.056	-18.267	382.434	439.799	15.898	24.077	1.500
19	15.526	-8.462	401.563	461.798	25.865	54.574	1.500
20	-23.715	-23.069	387.386	445.494	16.335	19.311	1.500
21	-3.226	-5.694	394.368	453.524	122.245	79.652	1.500
22	-39.875	-21.631	570.421	655.984	14.306	30.326	1.500
23	-0.293	-11.827	589.551	677.984	2009.832	57.327	1.500
26	-52.756	-26.836	791.953	910.746	15.012	33.938	1.500
39	-39.845	3.828	578.202	664.933	14.512	173.698	1.500
40	-69.357	-7.944	566.615	651.608	8.170	82.029	1.500
41	-24.170	12.099	379.599	436.540	15.706	36.082	1.500
42	-63.519	-3.597	364.150	418.773	5.733	116.420	1.500
45	-39.989	8.734	567.587	652.725	14.194	74.736	1.500
46	-79.338	-6.962	552.137	634.958	6.959	91.205	1.500

Calculations of Footing Thickness

Footing thickness is calculated based on the ultimate load cases

Punching Shear Check

For Column 127

Critical Load case for Punching Shear Check : 25

Total Footing Depth, D_o = 0.400m

Calculated Effective Depth, d_{eff} = 0.344m

For rectangular column, β_c = B_{col} / D_{col} : 1.714

Considering the particular column as interior column, Slab Edge Factor α_s : 40.0

Effective depth, d_{eff}, increased until $0.75*V_c \geq$ Punching Shear Force

Punchng Shear Force, Vu = 276.627kN

From ACI Cl.11.11.2, b_c for column 3.276m

Equation 11-31, V_{c1} 2398.933kN

$\left(2+\frac{4}{\beta_c}\right)\times b_o\times d_{eff}\times\sqrt{1000\times F_c'}$ =

Equation 11-32, V_{c2} = $\left(\frac{\alpha_s\times d}{b_o}+2\right)\times\lambda\times\sqrt{f_c}\times b_o\times d$ = 3432.454kN

Equation 11-33, V_{c3} = $4\times b_o\times d_{eff}\times\sqrt{1000\times F_c'}$ = 2214.399kN

Punching shear strength,V_c= 0.75 x minimum of (V_{c1},V_{c2},V_{c3}) = 1660.799kN

0.75 * V_c > V_u hence, OK

For Column 235

Critical Load case for Punching Shear Check : 25

Total Footing Depth, D_o = 0.400m

Calculated Effective Depth,d_{eff} = 0.344m

For rectangular column, β_c = B_{col} / D_{col} : 1.714

Considering the particular column as interior column, Slab Edge Factor α_s : 40.0

Effective depth, d_{eff}, increased until 0.75*V_c ≥ Punching Shear Force

Punchng Shear Force, Vu = 216.852kN

From ACI Cl.11.11.2, b_c for column 3.276m

Equation 11-31, V_{c1} $\left(2+\frac{4}{\beta_c}\right)\times b_o\times d_{eff}\times\sqrt{1000\times F_c'}$ = 2398.933kN

Equation 11-32, V_{c2} = $\left(\frac{\alpha_s\times d}{b_o}+2\right)\times\lambda\times\sqrt{f_c}\times b_o\times d$ = 3432.454kN

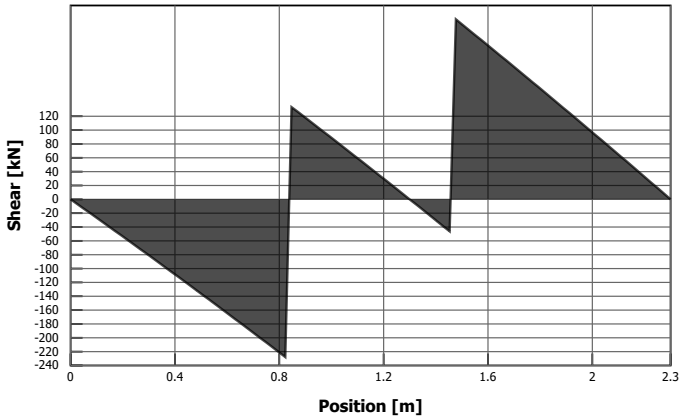
Equation 11-33, V_{c3} = $4\times b_o\times d_{eff}\times\sqrt{1000\times F_c'}$ = 2214.399kN

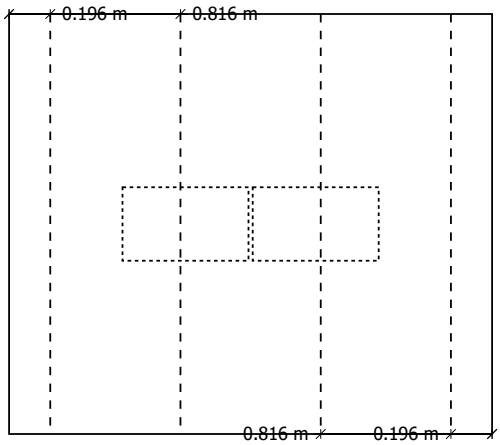
Punching shear strength,V_c= 0.75 x minimum of (V_{c1},V_{c2},V_{c3}) = 1660.799kN

0.75 * V_c > V_u hence, OK

One-Way Shear

Shear Diagram

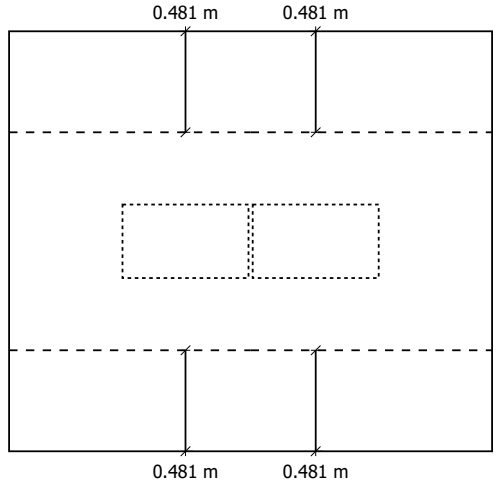




Shear Plane Parallel to Foundation Width

Critical load case for maximum shear force along the length of footing : 25

Critical Shear force,V_u	For the critical load case:	225.784kN
Point of occurance of V_u	Critical one-way shear position:	0.816m
From ACI Cl.11.2.1.1, V_c =	$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ =	675.946kN
	0.75 x V_c =	506.960kN
	Since 0.75 * V_c > V_u	hence, OK



Shear Plane Parallel to Foundation Length

Critical load case for maximum shear force along the width of footing : 25

Critical Shear force,V_u	For the critical load case:	198.875kN
Point of occurance of V_u	Critical one-way shear position:	0.481m
From ACI Cl.11.2.1.1, V_c =	$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ =	777.338kN
	0.75 x V_c =	583.004kN
	Since 0.75 * V_c > V_u	hence, OK

Design for Flexure

Design of flexure

Moment Diagram

Bottom Reinforcement

Critical load case :	25	
Required Effective Depth :	0.344m	
β_1, from ACI Cl.10.2.7.3	max[0.85-0.05x(Fc-4),0.65]=	0.7962
From ACI Cl. 10.3.2, ρ_{bal}	$0.85 \times \beta_1 \times F_c' \times \frac{87}{[f_y \times (87 + F_y)]}$ =	0.03374
From ACI Cl. 10.3.3, ρ_{max}	$0.75 \times \rho_{bal}$ =	0.02530
From ACI Cl. 7.12.2, ρ_{min}	$\max\left(0.0018 \cdot \frac{60ksi}{F_y}, 0.0014\right)$ =	0.00200
Modular Ratio,m	$\frac{F_y}{(0.85 \times F_c')}$ =	13.9496
Ultimate Moment (Mz) :	146.506kNm	
Point of occurrence of the ultimate moment along the length of footing :	1.140m	
Nominal Moment Capacity :	162.784kNm	
Required ρ (based on effective depth) :	0.0023	
ρ x d_{eff} / Depth (based on gross depth) :	0.0020	
Area of main steel required, A_s = ρ * W * deff :	1600.000mm2	

Distribution Reinforcement

Critical load case :	25		
Critical Moment for distribution steel (Mx) :		137.9842	kNm
Nominal moment Capacity :	153.3158		kNm
Point of occurance of the critical moment along length:	1.1750		m
Required ρ (based on effective depth):	0.0024		
ρ x d_{eff} / Depth (based on gross depth) :	0.0020		
Area of distribution steel required, A_s = ρ * L * deff :	1840.001		mm2

Provided Reinforcement

Based on spacing reinforcement increment; provided reinforcement is

Main bar no. for bottom Reinforcement:	#12

Spacing of bottom reinforcement bar :	135.714	mm

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 135mm o.c.

Distribution bar no. (Bottom):	#12	
Spacing of distribution bars (Bottom):	137.500	mm

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 135mm o.c.

Based on spacing reinforcement increment; provided reinforcement is

Pedestal Design

Print Calculation Sheet

ATTACHMENT 3 COMMENT RESOLUTION LOG SHEET

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Mansourah Massarah Gold Project
Comment Resolution Log Sheet / Compliance Report

PROJECT:	**MANSOURAH MASSARAH GOLD PROJECT** **PROJECT NO.: 216**
STAGE:	**4 - PROJECT EXECUTION.** **REVIEW LOG DOC. NO. & REV.:**
ACONEX REF. NO. & DATE:	BECHTEL-WTRAN-003144,19 June 2020

STATUS CODE: A - APPROVED, C - APPROVED WITH COMMENTS, R - REVISE & RESUBMIT & O - OTHERS (INFORMATION, REFERENCE, REVIEWED)

DOCUMENT NUMBER	Area / Title	Comment No.	PMC/PMT COMMENT DATE: 19 June 2020	Comment by	EPC Response DATE: 23 June,2020	Status	Remarks
			General Comments (cc)				
MD-216-6000-EG-CV-CAL-1500-A01	6000	1	Same value as indicated in this project design criteria, which is the mandatory document.	Mr.Klaus Grof Bechtel	All values are confirming to Project Design Criteria, Doc-MD-216-1000-EG-ST-CRT-1001, Clause 4.2.2.	Closed	